

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

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SCHEM, MLB, X425

08/06/2014 PROT01A

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Page	Contents	Sync	Date
1	Table of Contents	MASTER	
2	BOM Configuration	CLEAN_X305	05/30/2014
3	BOM Configuration	J15_MLB	10/31/2012
4	PD Parts	J15_MLB	10/31/2012
5	CPU DMI/PEG/FDI/RSVD	CLEAN_X305_PEG	02/18/2014
6	CPU Clock/Misc/JTAG/CFG	J15_REFERENCE	12/18/2012
7	CPU DDR3 Interfaces	J15_REFERENCE	12/18/2012
8	CPU Power	J15_REFERENCE	12/18/2012
9	CPU Ground	J15_REFERENCE	12/18/2012
10	CPU Decoupling	J15_REFERENCE	12/18/2012
11	PCH RTC/HDA/JTAG/SATA/CLK	J15_REFERENCE	12/18/2012
12	PCH DMI/FDI/PM/GFX/PCI	J15_REFERENCE	12/18/2012
13	PCH PCI-E/USB	J15_REFERENCE	12/18/2012
14	PCH GPT0/MISC/NCTF	CLEAN_X305_PEG	02/18/2014
15	PCH Power	J15_REFERENCE	12/18/2012
16	PCH Grounds	J15_REFERENCE	12/18/2012
17	PCH DECOUPLING	J15_REFERENCE	12/18/2012
18	CPU & PCH XDP	J15_MLB	10/31/2012
19	Chipset Support	J15_REFERENCE	12/18/2012
20	Project Chipset Support	J15_REFERENCE	01/14/2013
21	CPU Memory S3 Support	CLEAN_X305G	07/01/2014
22	DDR3 VREF MARGINING	CLEAN_X305	01/14/2014
23	DDR3 SDRAM Bank A (1 OF 2)	J15_MLB	10/31/2012
24	DDR3 SDRAM Bank A (2 OF 2)	J15_MLB	10/31/2012
25	DDR3 SDRAM Bank B (1 OF 2)	J15_MLB	10/31/2012
26	DDR3 SDRAM Bank B (2 OF 2)	J15_MLB	10/31/2012
27	DDR3 Termination	J15_MLB	10/31/2012
28	Thunderbolt Host (1 of 2)	T29_RR	01/14/2013
29	Thunderbolt Host (2 of 2)	T29_RR	01/14/2013
30	Thunderbolt Mobile Support	CLEAN_X305	06/24/2014
31	Thunderbolt Connector A	CLEAN_X305	06/24/2014
32	Thunderbolt Connector B	CLEAN_X305	06/24/2014
33	DDC Crossbar	J15_REFERENCE	11/16/2012
34	X87 CONNECTOR	CLEAN_X305	01/15/2014
35	SSD Connector	CLEAN_X305_PEG	02/18/2014
36	Camera 1 of 2	CLEAN_X305	06/24/2014
37	Camera 2 of 2	CLEAN_X305	06/24/2014
38	USB 3.0 CONNECTORS	J15_MLB	10/31/2012
39	KEYBOARD/TRACKPAD (1 OF 2)	CLEAN_X305_PEG	02/18/2014
40	KEYBOARD/TRACKPAD (2 OF 2)	CLEAN_X305	05/30/2014
41	SMC	CLEAN_X305	01/15/2014

Page		Contents	Sync	Date
Page 42	51	SMC Shared Support	CLEAN_X305	06/24/2014
Page 43	52	SMC Project Support	CLEAN_X305	05/30/2014
Page 44	53	SMBus Connections	CHANG_J45	11/26/2012
Page 45	54	High Side Voltage and Current Sensing	CLEAN_X305_PEG	02/18/2014
Page 46	55	Load Side Voltage and Current Sensing	CLEAN_X305_PEG	02/18/2014
Page 47	56	Debug Sensors	CLEAN_X305	01/14/2014
Page 48	58	Thermal Sensors	CHANG_J45	11/26/2012
Page 49	60	Fan Connectors	J15_MLB	10/31/2012
Page 50	61	SPI Debug Connector	CLEAN_X305_PEG	02/18/2014
Page 51	62	AUDIO:CODEC, ANALOG	JOE_J45	07/30/2013
Page 52	63	AUDIO:CODEC, DIGITAL	JOE_J45	07/30/2013
Page 53	64	AUDIO: SPEAKER AMP	JOE_J45	07/30/2013
Page 54	65	AUDIO: JACK	CLEAN_X305	06/24/2014
Page 55	66	AUDIO: JACK TRANSLATORS	CLEAN_X305	06/24/2014
Page 56	70	DC-In & Battery Connectors	CLEAN_X305_PEG	02/18/2014
Page 57	71	PBus Supply & Battery Charger	CLEAN_X305	01/15/2014
Page 58	72	CPU VR12.5 VCC Regulator IC	CLEAN_X305_PEG	02/24/2014
Page 59	73	CPU VR12.5 VCC Power Stage	J15_MLB	10/31/2012
Page 60	74	1.35V DDR3L SUPPLY	CLEAN_X305	01/15/2014
Page 61	75	5V / 3.3V Power Supply	CLEAN_X305_PEG	02/18/2014
Page 62	76	1V05V POWER SUPPLY	CLEAN_X305_PEG	02/18/2014
Page 63	77	LCD/KBD Backlight Driver	CLEAN_X305_PEG	02/19/2014
Page 64	78	Misc Power Supplies	CLEAN_X305	01/15/2014
Page 65	79	X249 Boost Power Supply	CLEAN_X305	05/30/2014
Page 66	80	Power FETS	CLEAN_X305_PEG	02/18/2014
Page 67	81	Power Control 1/ENABLE	CLEAN_X305	05/30/2014
Page 68	83	eDP Display Connector	CLEAN_X305_PEG	02/18/2014
Page 69	95	RIO Connectors	CLEAN_X305G	01/14/2014
Page 70	100	Power Aliases	CLEAN_X305	05/30/2014
Page 71	102	Signal Aliases	J15_MLB	10/31/2012
Page 72	104	Functional Test Points	J15_MLB	10/31/2012
Page 73	105	NC & No Test	J15_MLB	10/31/2012
Page 74	110	PCB Rule Definitions	SIDLE_J45	12/10/2012
Page 75	111	CPU Constraints	CLEAN_X305_PEG	02/18/2014
Page 76	112	PCH Constraints 1	SIDLE_J45	12/10/2012
Page 77	113	PCH Constraints 2	CLEAN_X305_PEG	02/18/2014
Page 78	114	Memory Constraints	SIDLE_J45	12/10/2012
Page 79	115	Thunderbolt Constraints	SIDLE_J45	12/10/2012
Page 80	116	Camera Constraints	SIDLE_J45	12/10/2012
Page 81	117	SMC Constraints	SIDLE_J45	12/10/2012
Page 82	118	Project Specific Constraints	SIDLE_J45	12/10/2012

ALIASES RESOLVED

Schematic / PCB #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
051-00330	1	SCHEM, MLB, X305	SCH	CRITICAL	
820-00138	1	PCBF, MLB, X305	PCB	CRITICAL	

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		BRANCH	
		<BRANCH>	
		PAGE	
		1 OF 118	
		SHEET	
		1 OF 82	

BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
685-00039	COMMON PARTS, MLB, X425	X425_COMMON
985-00043	DEV, MLB, X425	X425_DEV:ENG
639-00534	PCBA, MLB, CTO, 16G-HYN, X425	BASE_BOM, DEVELOPMENT_BOM, CPU_CRW:CTO, RAM:HYNIX_1866
639-00535	PCBA, MLB, CTO, 16G-MIC, X425	BASE_BOM, DEVELOPMENT_BOM, CPU_CRW:CTO, RAM:MICRON_1866

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X425 BOM Groups

BOM GROUP	BOM OPTIONS
X425_COMMON	ALTERNATE,COMMON,X425_COMMON1,X425_COMMON2,X425_PROGPARTS
X425_COMMON1	CPUMEM:S0,TBTHV:P15V,SKIP_5V3V3:AUDIBLE,CPUPEG:X8X8,S2_PWR:S0,SMC_SUSACK:YES
X425_COMMON2	EDP:YES,XDP,SSD_PWR_EN:GPIO,CAM_WAKE:NO,SAMCONN,APCLKRQ:ISOL,DDRREG_PGD:N,CRW_SPRT,WLAN_SW:SIL
X425_PVT	BKLT:PROD,SENSOR_NONPROD:N
X425_PROGPARTS	SMC_PROG:BASE,BOOTROM_PROG:PROT01A,TBTROM:PROG
X425_DEVEL:ENG	ALTERNATE,XDP_DEBUG,S6PGOOD_ISL,SENSOR_NONPROD:Y,SENSOR_NONPROD_R,BKLT:ENG,DBGLED,X249:BOOST
X425_DEVEL:DVT	ALTERNATE,XDP_DEBUG,BKLT:PROD,SENSOR_NONPROD:N,DBGLED
X425_DEVEL:PVT	XDP_DEBUG
XDP_DEBUG	XDP_CONN,XDP_PCH

Module Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
337S00057	1	CRW, SR12X, PRQ, C0, 2, 2, 47W, +3E, 1, 2, 6M, BGA	U0500	CRITICAL	CPU_CRW: BETTER
337S00058	1	CRW, SR12X, PRQ, C0, 2, 5, 47W, +3E, 1, 2, 6M, BGA	U0500	CRITICAL	CPU_CRW: BEST
337S00059	1	CRW, SR12Y, PRQ, C0, 2, 8, 47W, +3E, 1, 2, 6M, BGA	U0500	CRITICAL	CPU_CRW: CTO
337S4542	1	IC, QEW, LPT- M, HM87, C2, SR199, CRQ, FCBGA	U1100	CRITICAL	
338S1247	1	IC, T8T, FR-4C, AB, PRQ, C10, SR13C, FCBGA288	U2800	CRITICAL	
338S1264	1	IC, BCM1570BA2, S2 PCIE CMRA, 6XS, 208FCBGA	U3900	CRITICAL	
333S0700	1	IC, SDRAM, 4GBIT, DDR3L-1600, 6EMMA, 96B FBGA	U4000	CRITICAL	
333S0802	16	IC, SDRAM, 25NM, 512MK8, DDR3L-1866, 78B FBGA		CRITICAL	HYNIX_1866_S
333S0719	16	IC, SDRAM, 4GBIT, DDR3-1866, V88A, 78B, FBGA		CRITICAL	MICRON_1866_S
333S0802	32	IC, SDRAM, 25NM, 512MK8, DDR3L-1866, 78B FBGA		CRITICAL	HYNIX_1866
333S0719	32	IC, SDRAM, 4GBIT, DDR3-1866, V88A, 78B, FBGA		CRITICAL	MICRON_1866

DRAM SPD Straps

BOM GROUP	BOM OPTIONS
RAM:HYNIX_1866_S	HYNIX_1866_S, RAMCFG3:L, RAMCFG2:H, RAMCFG1:L, RAMCFG0:L
RAM:MICRON_1866_S	MICRON_1866_S, RAMCFG3:L, RAMCFG2:H, RAMCFG1:H, RAMCFG0:L
RAM:HYNIX_1866	HYNIX_1866, RAMCFG3:H, RAMCFG2:H, RAMCFG1:L, RAMCFG0:L
RAM:MICRON_1866	MICRON_1866, RAMCFG3:H, RAMCFG2:H, RAMCFG1:H, RAMCFG0:L

COMMON/DEVEL BOM

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
685-00039	1	COMMON PARTS,MLB,X425	BASE	CRITICAL	BASE_BOM
985-00043	1	DEV,MLB,X425	DEVEL	CRITICAL	DEVEL_BOM

SYNC MASTER=CLEAN X305		SYNC DATE=05/30/2014	
PAGE TITLE			
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		PAGE	2 OF 118
		SHEET	2 OF 82

Programmables - All builds

335S0915	1	IC, SERIAL SPI FLASH ROM, 4MBIT, 50MHZ, USON	U2890	CRITICAL	TBTROM: BLANK
341S00133	1	T29, FALCON RIDGE(VXXXX) PROTO 1A, X425	U2890	CRITICAL	TBTROM: PROG

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SMC

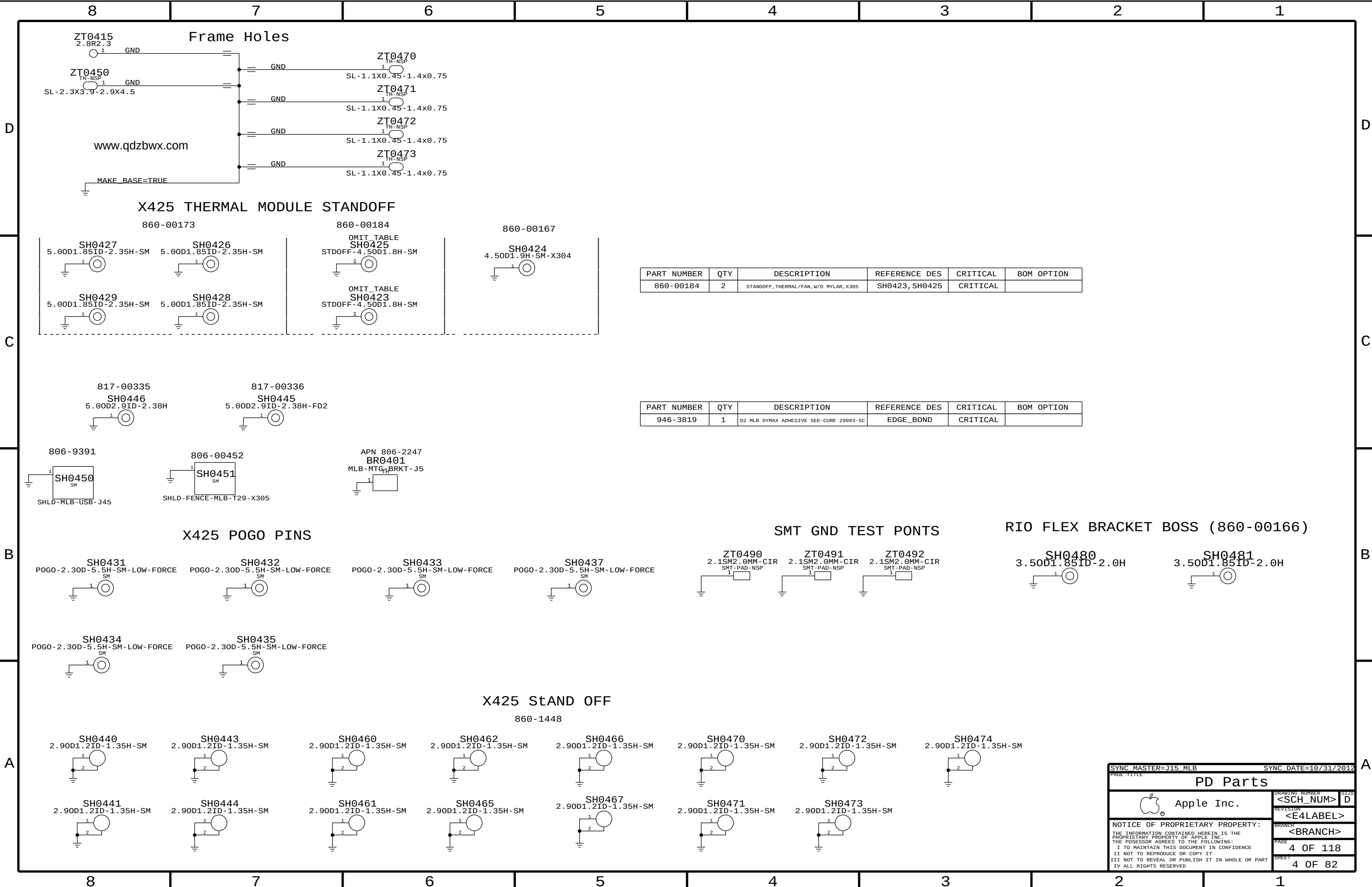
338S1214	1	IC, SMC -B1, 40MHZ/50DMIPS, SCPL FW, 1578GA	U5000	CRITICAL	SMC_PROG: BLANK
341S00125	1	IC, SMC -B1, EXT (V2. 24A31) PROTO 1A, X425	U5000	CRITICAL	SMC_PROG: BASE

EFI ROM

335S00007	1	IC, SERIAL FLASH, 64MB, 3V, WSON, 6X5MM	U6100	CRITICAL	BOOTROM_BLANK:WIN
335S00006	1	IC, SERIAL FLASH, 64MB, 3V, WSON, 6X5MM	U6100	CRITICAL	BOOTROM_BLANK:MAC
341S00131	1	IC, EFI ROM (VXXXX) PROTO 1A, X425	U6100	CRITICAL	BOOTROM_PROG:PROTO1A

Alternate Parts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S1053	376S0604		ALL	Diodes alt to Fairchild
128S0311	128S0329		ALL	NEC alt to Sanyo
138S0739	138S0706		ALL	Samsung alt to Murata
197S0481	197S0480		ALL	Epson Alt to NDK
197S0478	197S0479		ALL	NDK Alt to Epson
371S0713	371S0558		ALL	DOS alt to ST
152S0461	152S1645		ALL	Cyntec alt to Vishay
376S1080	376S0820		ALL	Diodes alt to On Semi
155S0667	155S00008		ALL	Panasonic alt to TDK
376S1217	376S0855		ALL	Toshiba alt to Diodes
376S1129	376S0855		ALL	NDP alt to Diodes
376S1089	376S1128		ALL	NDP alt to Diodes
128S0371	128S0376		ALL	Kemet alt to Sanyo
138S0803	138S0639		ALL	Samsung alt to Murata
138S0843	138S0674		ALL	Samsung alt to Murata
138S0846	138S0811		ALL	Samsung alt to Murata
127S0164	127S0162		ALL	Rohm alt to Vishay
138S0732	138S0715		ALL	Rohm alt to Vishay
128S0364	128S0264		ALL	Kemet alt to Sanyo
333S0704	333S0700		ALL	ELPIDA to HYUNEX
311S0649	311S0541		ALL	ON alt to Toshiba
376S00014	376S0761		ALL	Toshiba alt to Vishay
740S00003	740S0135		ALL	AEM alt to Tyco
377S0155	377S00011		ALL	On Semi alt to Infineon
377S0184	377S00011		ALL	Infineon alt to Infineon



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
860-00184	2	STANDOFF, THERMAL/FAN, W/O MYLAR, X385	SH0423, SH0425	CRITICAL	

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
946-3819	1	D2 MLB DYNAM ADHESIVE. SEE CURE 29993-SC	EDGE_BOND	CRITICAL	

SYNC MASTER=J15 MLB

SYNC DATE=10/31/2012

PD Parts

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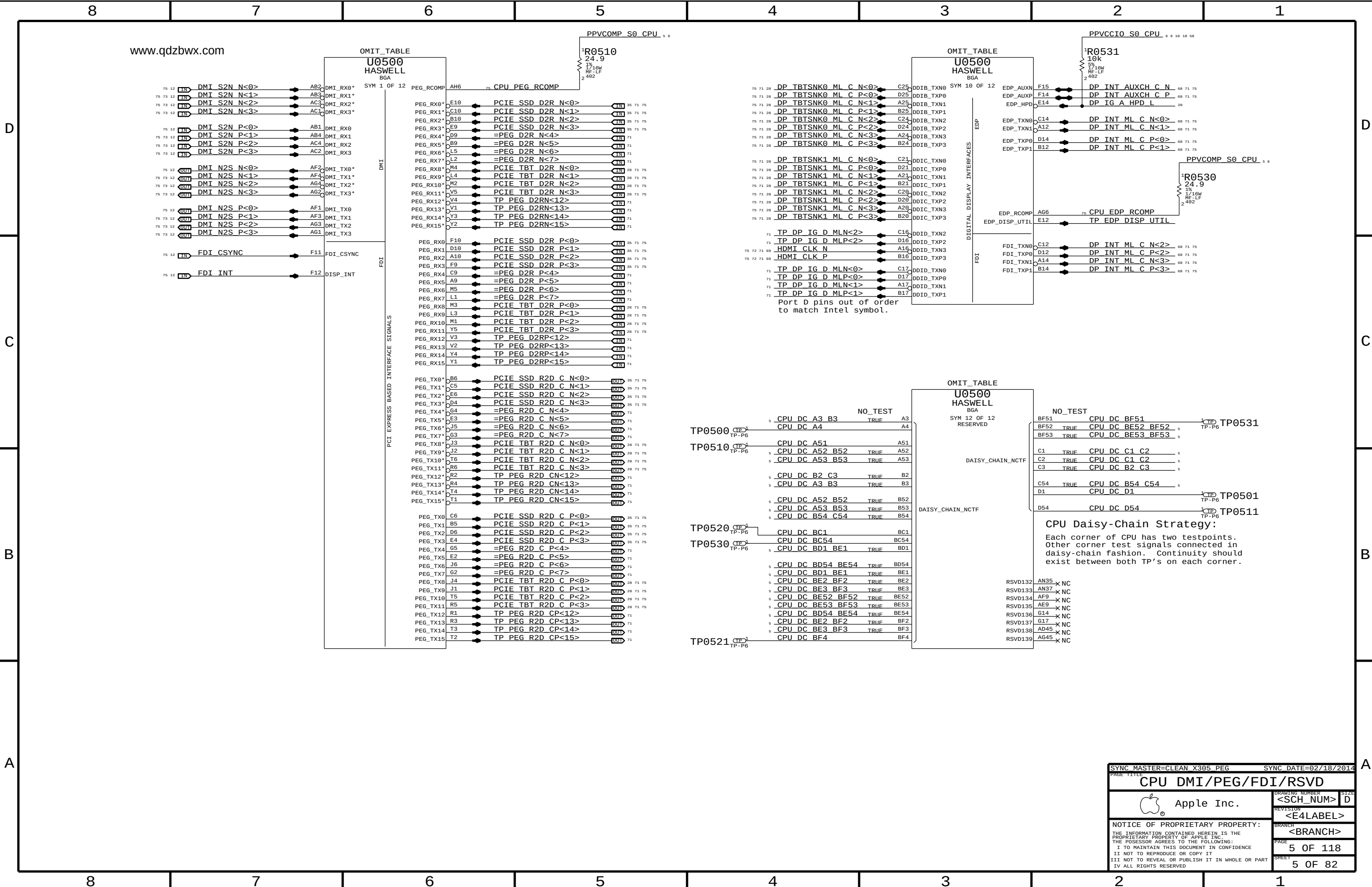
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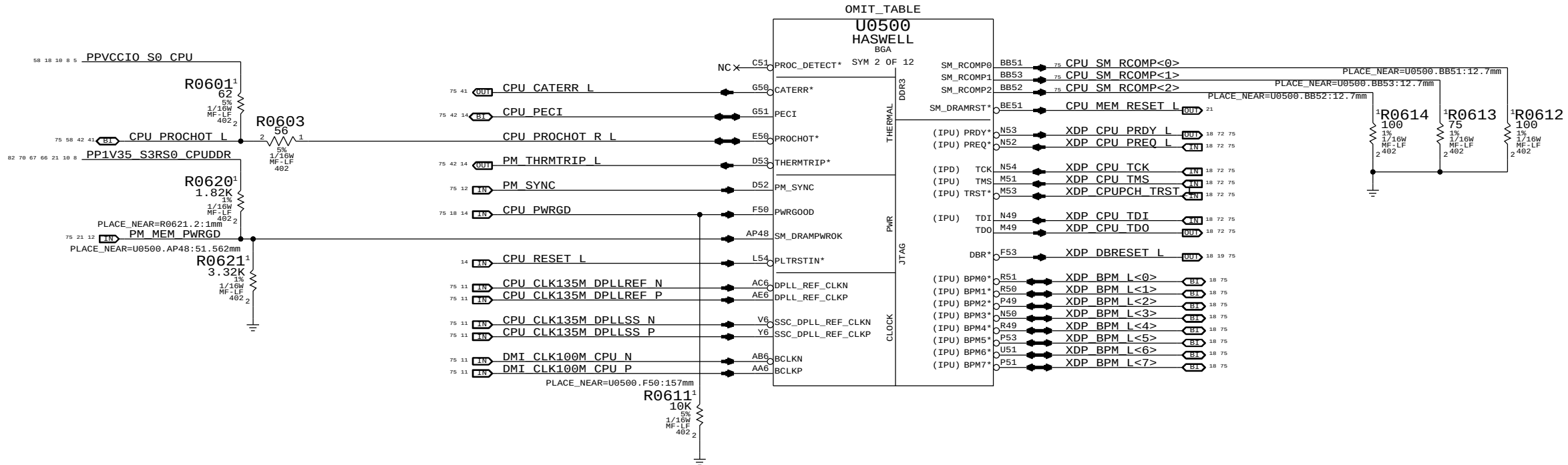
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PAGE
4 OF 118

SHEET
4 OF 82

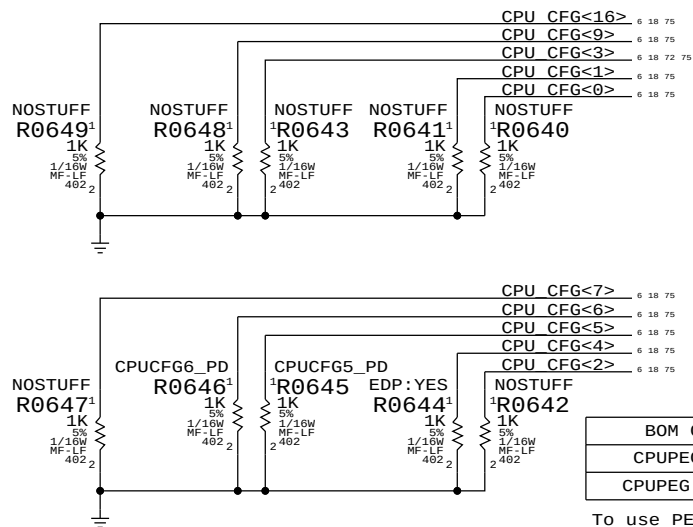


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CFG [7] :PEG DEFER TRAINING 1 = (DEFAULT) IMMEDIATELY AFTER xxRESETB 0 = WAIT FOR BIOS
CFG [6:5] :PCIE BIFURCATION 11 = 1 X16 (DEFAULT) 10 = 2 X8 01 = RSVD 00 = X8, X4, X4
CFG [4] :eDP ENABLE/DISABLE 1 = DISABLED 0 = ENABLED
CFG [3] :PCIE x4 LANE REVERSAL 1 = NORMAL OPERATION 0 = LANES REVERSED
CFG [2] :PCIE x16 LANE REVERSAL 1 = NORMAL OPERATION 0 = LANES REVERSED

These can be placed close to J1800 and only for debug access



BOM GROUP	BOM OPTIONS
CPUPEG:X8X8	CPUCFG5_PD
CPUPEG:X8X4X4	CPUCFG6_PD,CPUCFG5_PD

To use PEG X16 configuration, simply remove CPUPEG:X8X8 and CPUPEG:X8X4X4 from BOMs.

TP_CPU_RSVD_TP23	BE4	RSVD_TP23
TP_CPU_RSVD_TP24	BD3	RSVD_TP24
TP_CPU_RSVD_TP25	F6	RSVD_TP25
TP_CPU_RSVD_TP26	G6	RSVD_TP26
TP_CPU_RSVD_TP27	G21	RSVD_TP27
TP_CPU_RSVD_TP28	G24	RSVD_TP28
CPU_TESTLO_F21	F21	TESTLO_F21
	G19	VSS_G19
	F51	VSS_F51
	F52	VSS_F52
	F22	VCC_F22
TP_CPU_RSVD_TP35	L52	RSVD_TP35
TP_CPU_RSVD_TP36	L53	RSVD_TP36
TP_CPU_RSVD_TP37	L51	RSVD_TP37
TP_CPU_RSVD_TP38	F24	RSVD_TP38
TP_CPU_RSVD_TP39	F25	RSVD_TP39
CPU_TESTLO_F20	F20	TESTLO_F20

CPU_CFG<0>	AG49	CFG0 (IPU)
CPU_CFG<1>	AD49	CFG1 (IPU)
CPU_CFG<2>	AC49	CFG2 (IPU)
CPU_CFG<3>	AE49	CFG3 (IPU)
CPU_CFG<4>	Y50	CFG4 (IPU)
CPU_CFG<5>	AB49	CFG5 (IPU)
CPU_CFG<6>	V51	CFG6 (IPU)
CPU_CFG<7>	W51	CFG7 (IPU)
CPU_CFG<8>	Y49	CFG8 (IPU)
CPU_CFG<9>	Y54	CFG9 (IPU)
CPU_CFG<10>	Y53	CFG10 (IPU)
CPU_CFG<11>	W53	CFG11 (IPU)
CPU_CFG<12>	U53	CFG12 (IPU)
CPU_CFG<13>	V54	CFG13 (IPU)
CPU_CFG<14>	R53	CFG14 (IPU)
CPU_CFG<15>	R52	CFG15 (IPU)

NC X	L50	RSVD50
NC X	L49	RSVD51
NC X	E5	RSVD52

RSVD_TP1	F1	TP_CPU_RSVD_TP1
RSVD_TP2	E1	TP_CPU_RSVD_TP2
RSVD_TP3	A5	TP_CPU_RSVD_TP3
RSVD_TP4	A6	TP_CPU_RSVD_TP4

CFG_RCOMP	R54	CPU_CFG_RCOMP
(IPU) CF616	Y52	CPU_CFG<16>
CF618	V53	CPU_CFG<18>
(IPU) CF617	Y51	CPU_CFG<17>
CF619	V52	CPU_CFG<19>

RSVD92	B50	NC
RSVD93	AH49	NC
RSVD94	AM48	NC
RSVD95	AU27	NC
RSVD9	AU26	NC
RSVD10	BD4	NC
RSVD11	BC4	NC
RSVD41	AL6	NC
RSVD42	F8	NC
RSVD16	F16	NC

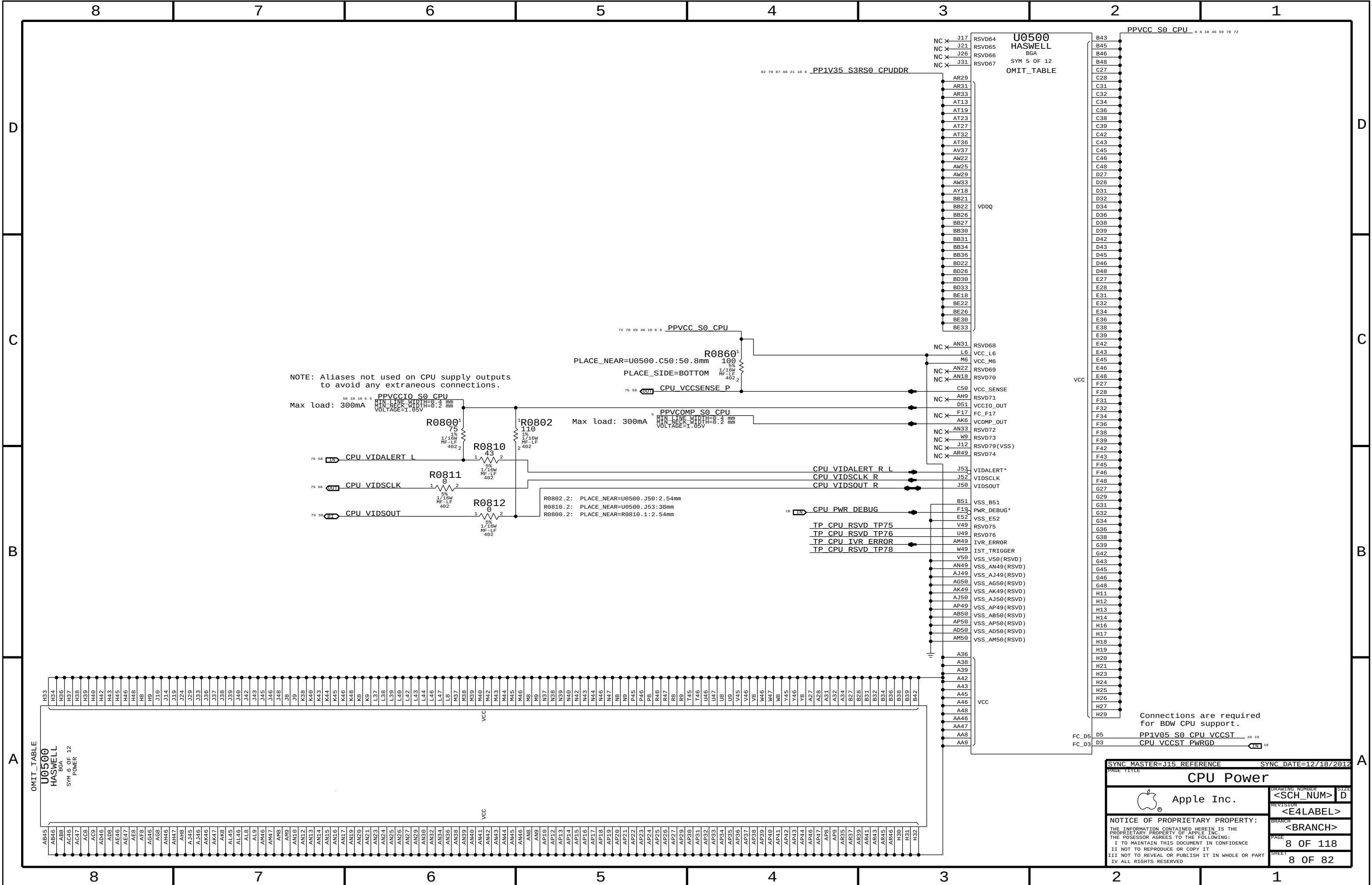
RSVD_TP17	G12	TP_CPU_RSVD_TP17
RSVD_TP18	G10	TP_CPU_RSVD_TP18

VSS_H54	H54	
VSS_H53	H53	
VSS_H51	H51	
VSS_H52	H52	

RSVD47	N51	TP_CPU_RSVD_TP47
RSVD48	G53	TP_CPU_RSVD_TP48
RSVD49	H50	TP_CPU_RSVD_TP49

SYNC MASTER=J15 REFERENCE		SYNC DATE=12/18/2012	
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NOTE: Aliases not used on CPU supply outputs to avoid any extraneous connections.

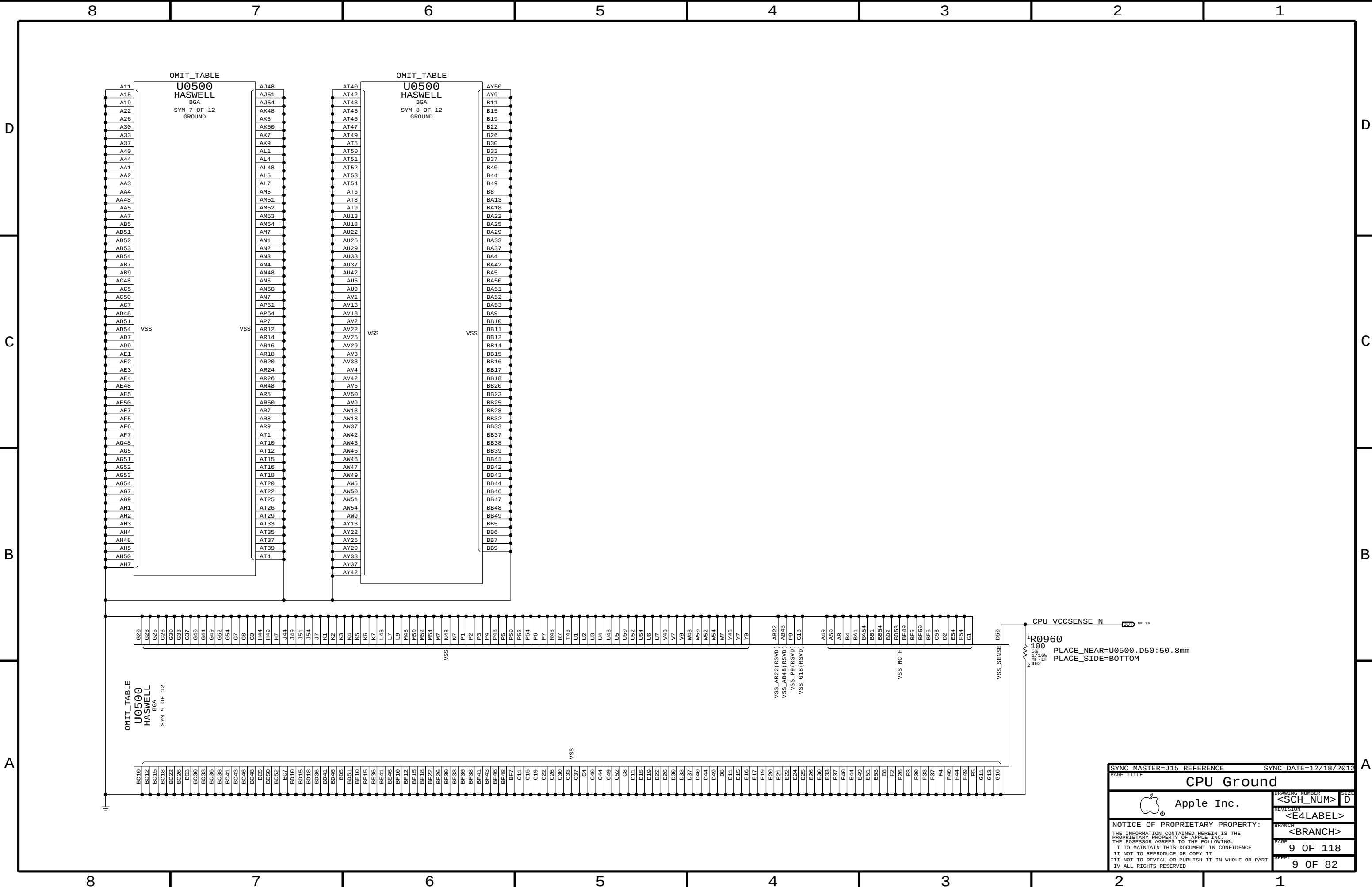
Max load: 300mA

Max load: 300mA

Connections are required for BDW CPU support.

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		<BRANCH>	
		PAGE	8 OF 118
		SHEET	8 OF 82

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SYNC_MASTER=J15 REFERENCE

SYNC_DATE=12/18/2012

CPU Ground

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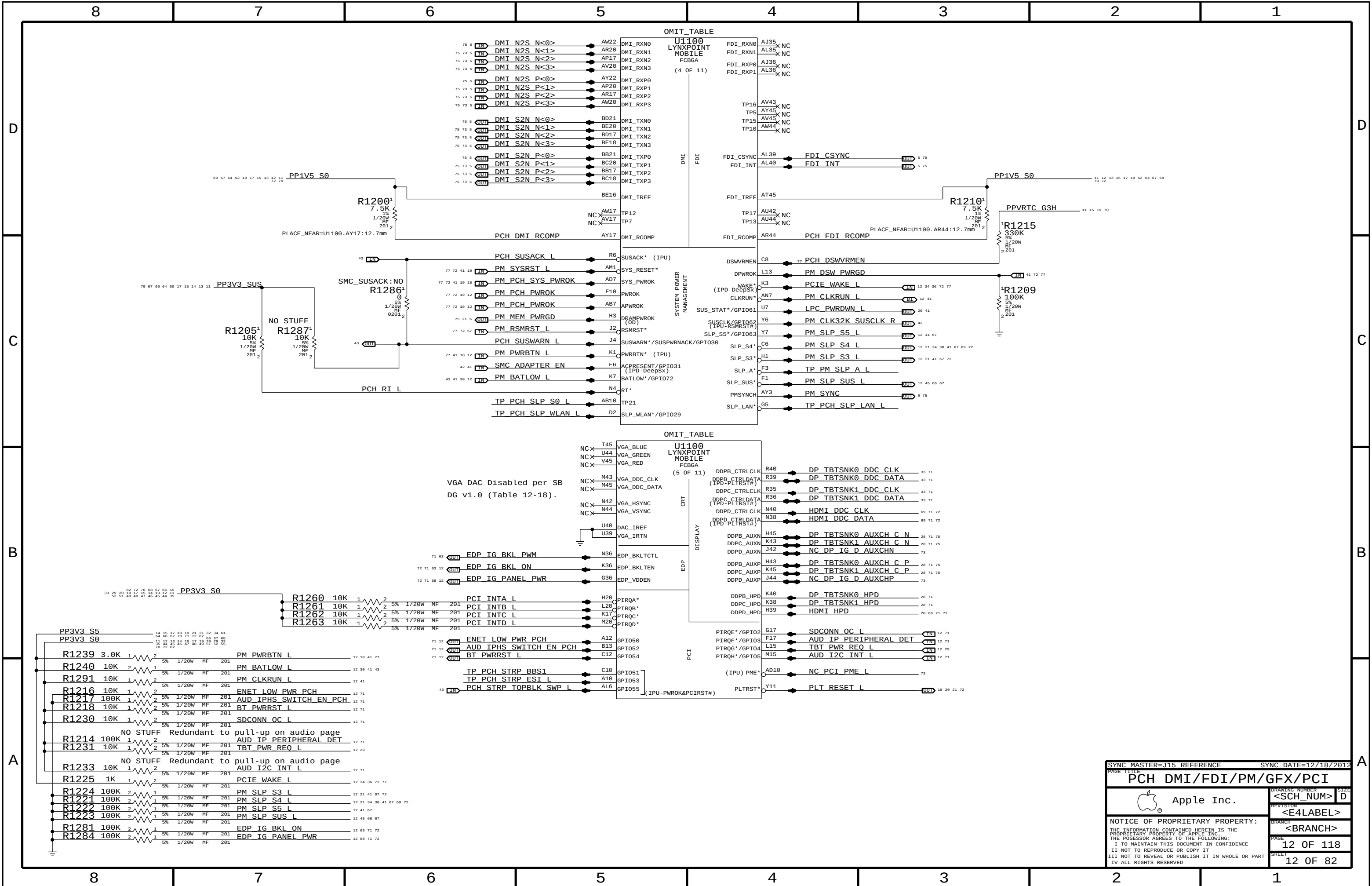
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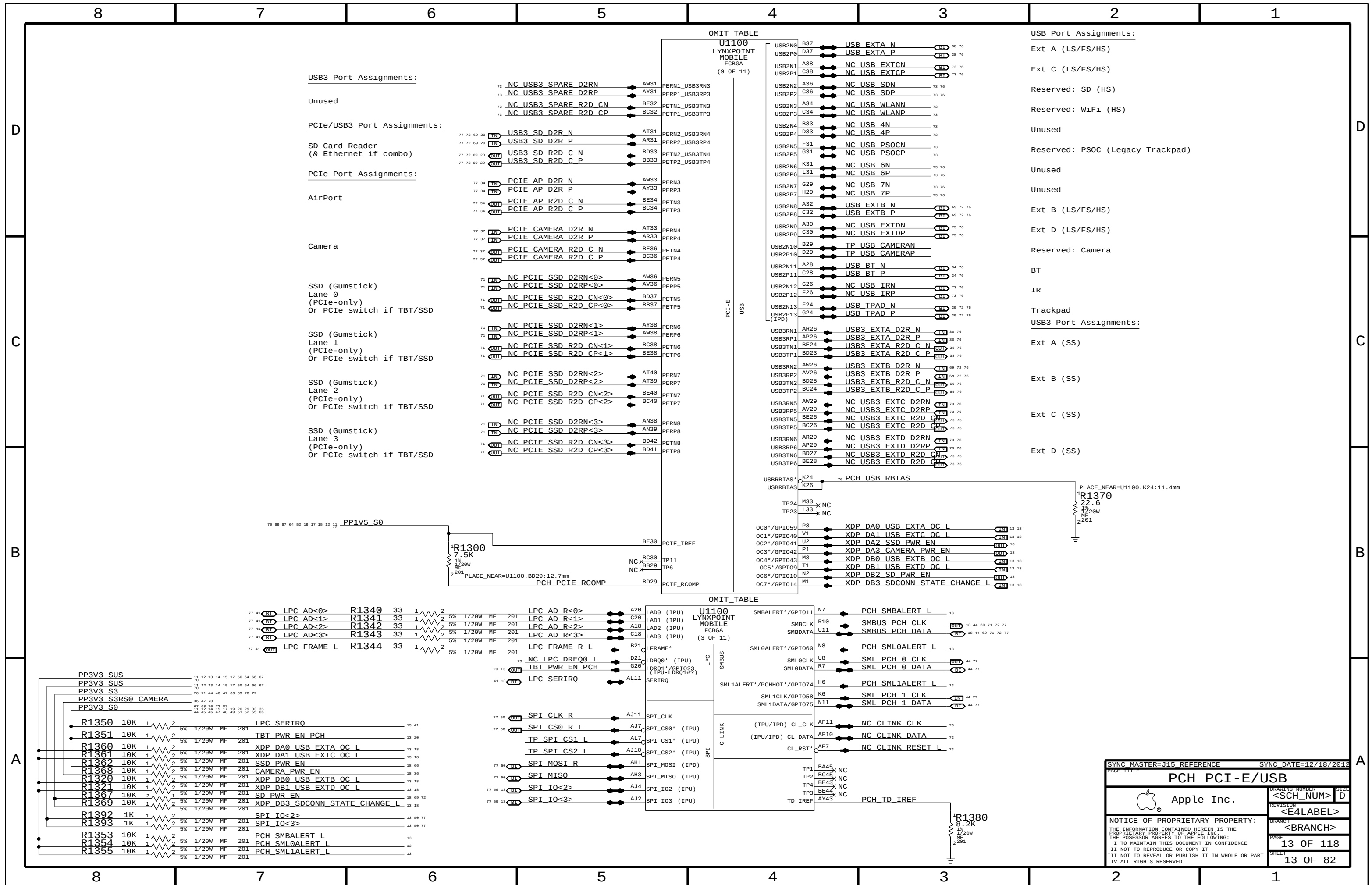
9 OF 118

SHEET

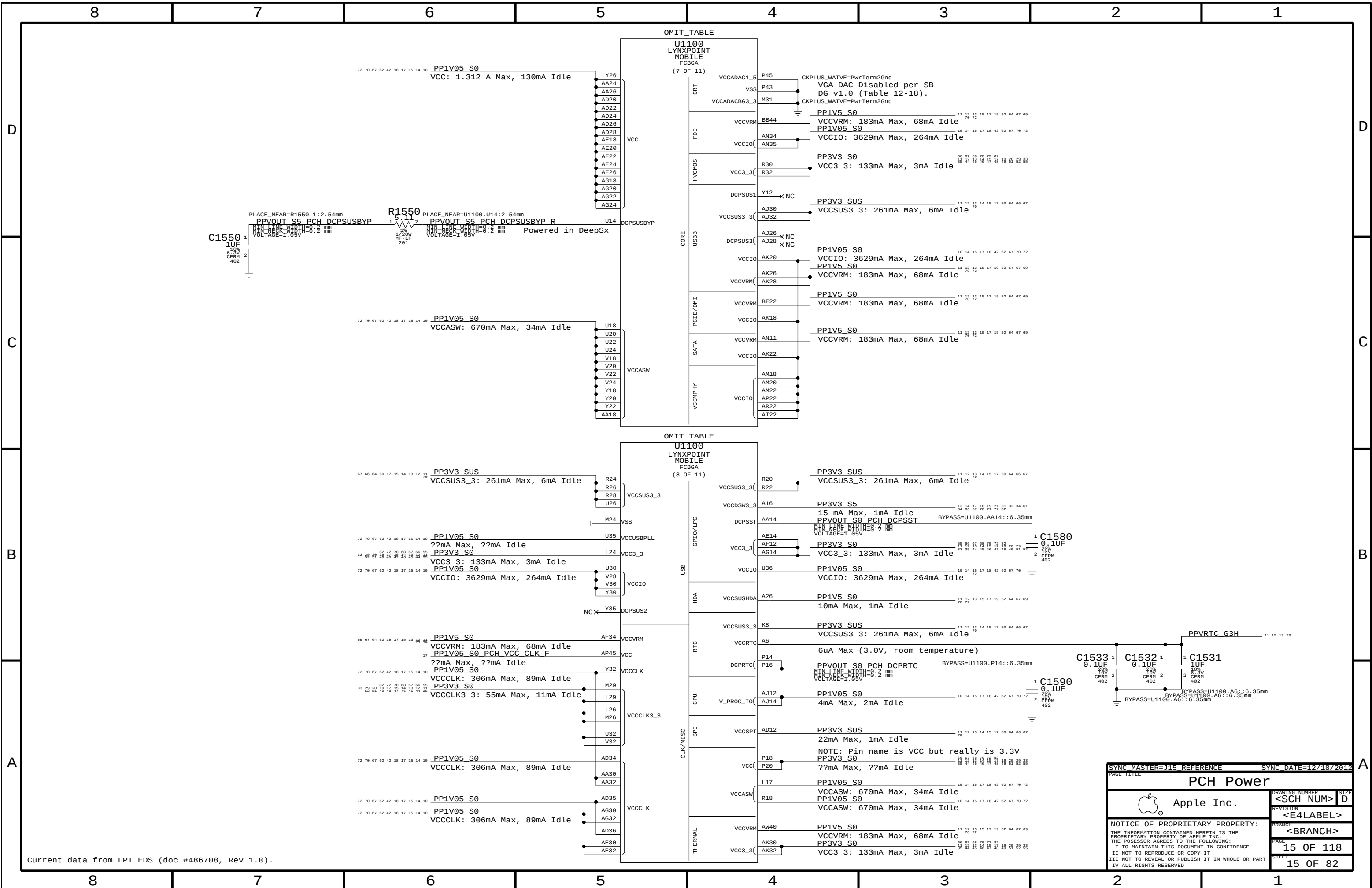
9 OF 82



SYNC MASTER=J15 REFERENCE		SYNC DATE=12/18/2012	
PAGE TITLE			
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		PAGE 12 OF 118	
		SHEET 12 OF 82	







Current data from LPT EDS (doc #486708, Rev 1.0).

SYNC MASTER=J15 REFERENCE

SYNC DATE=12/18/2012

PAGE TITLE

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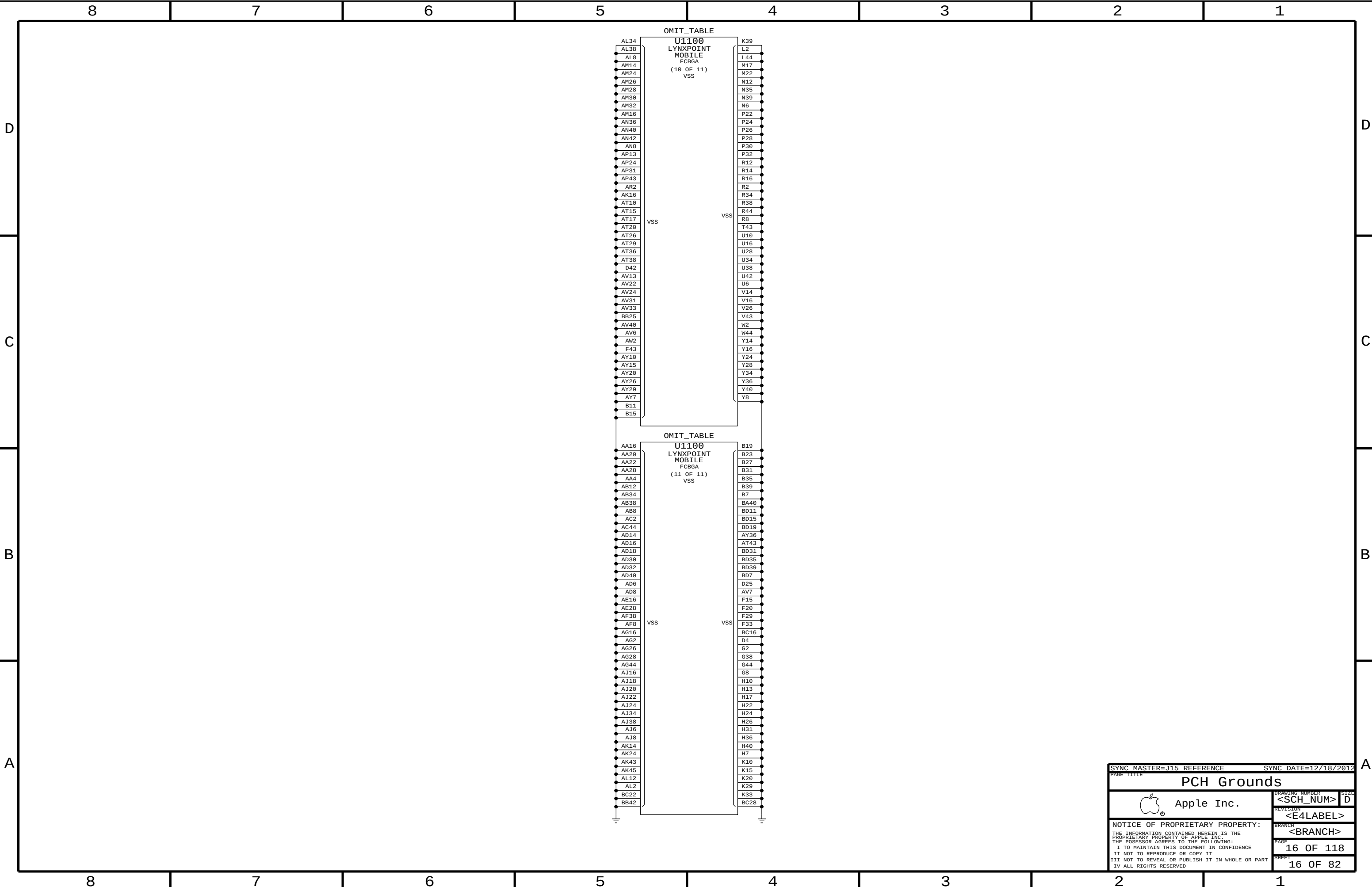
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15 OF 118

SHEET

15 OF 82



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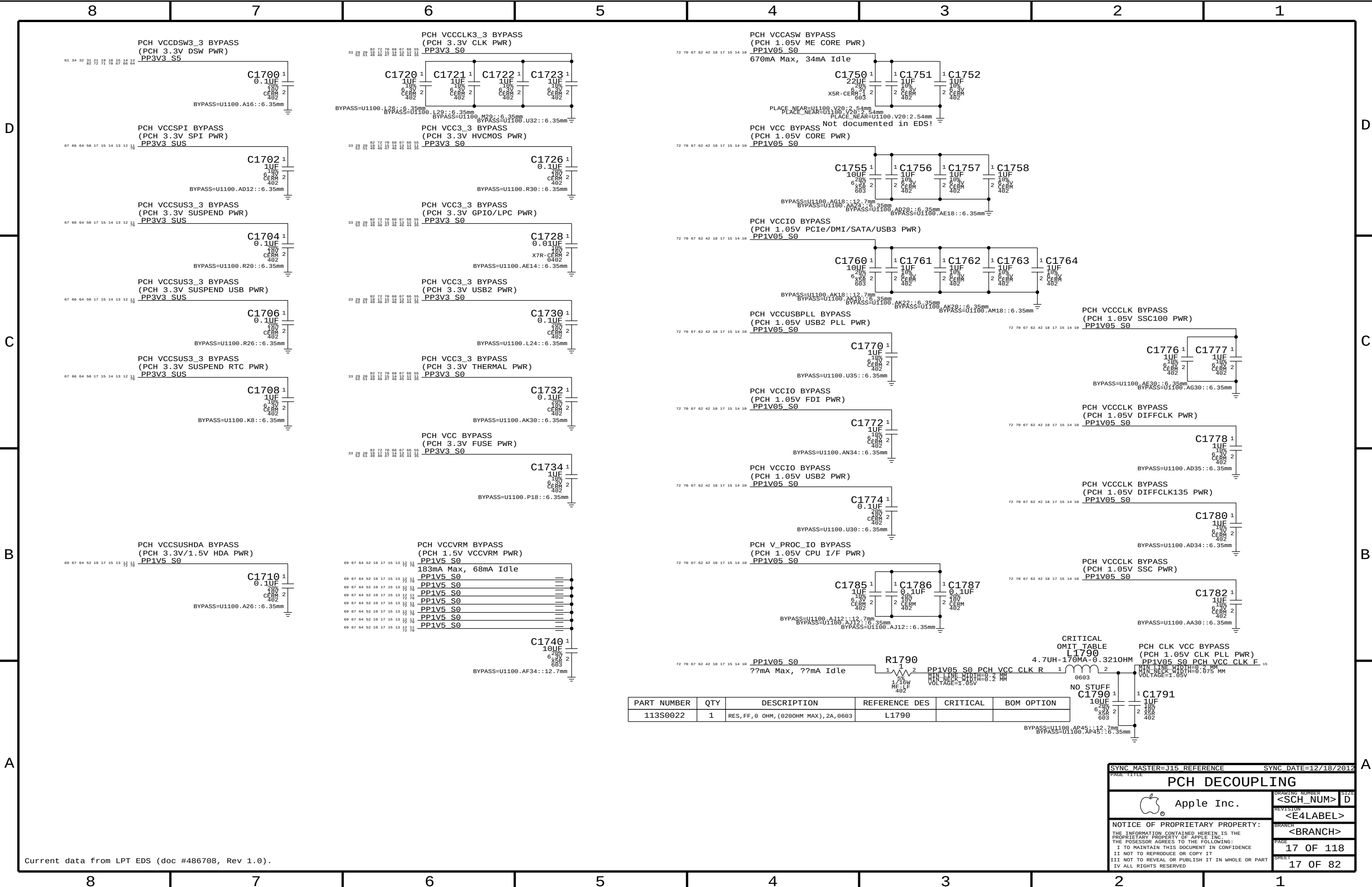
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16 OF 118

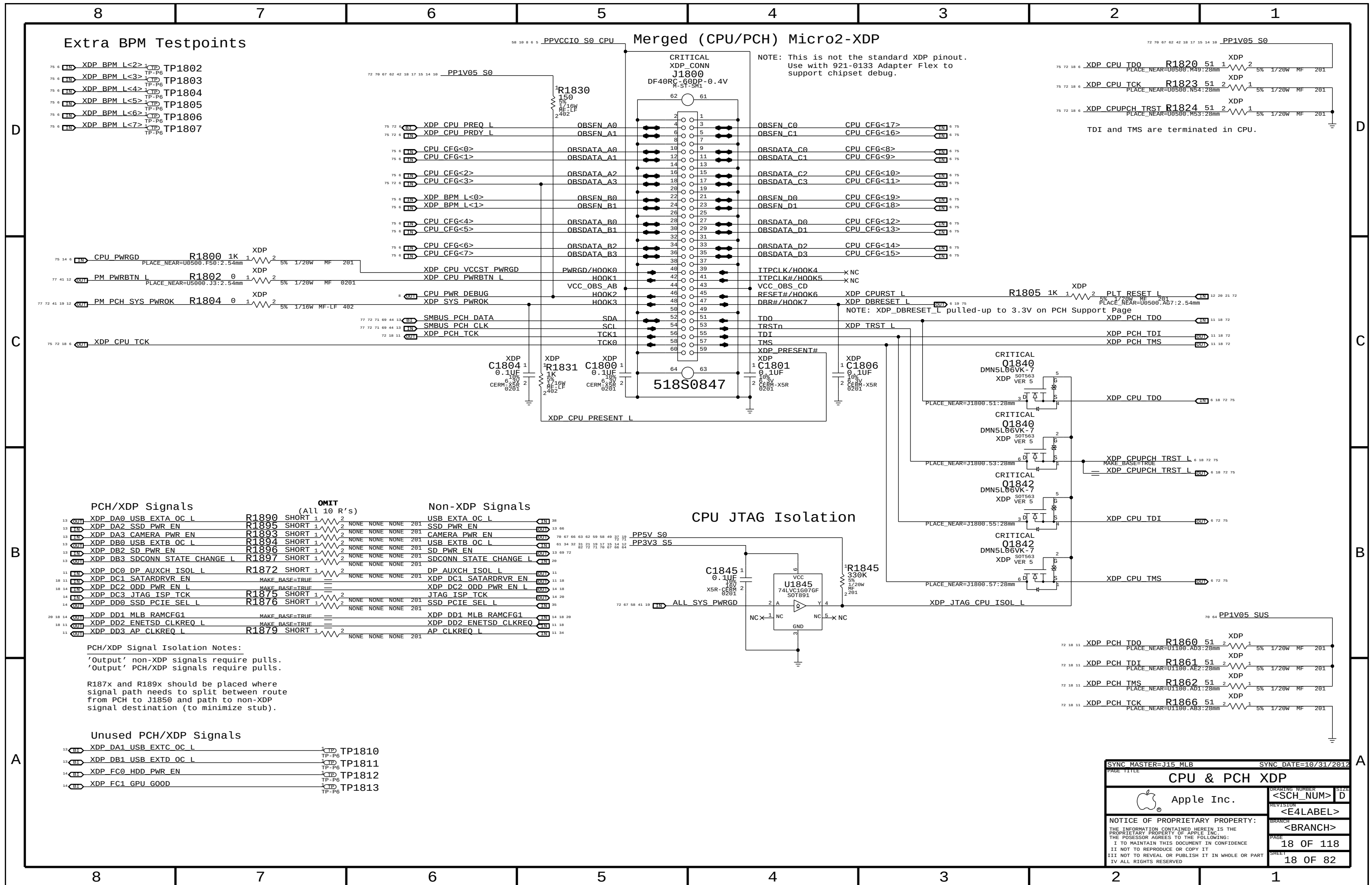
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16 OF 82

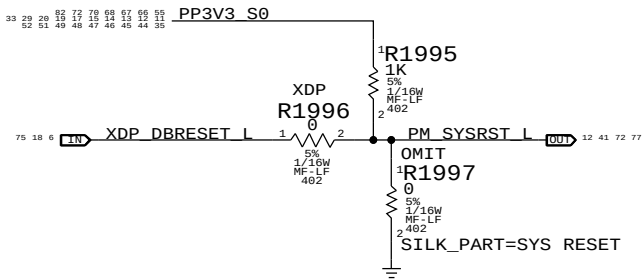


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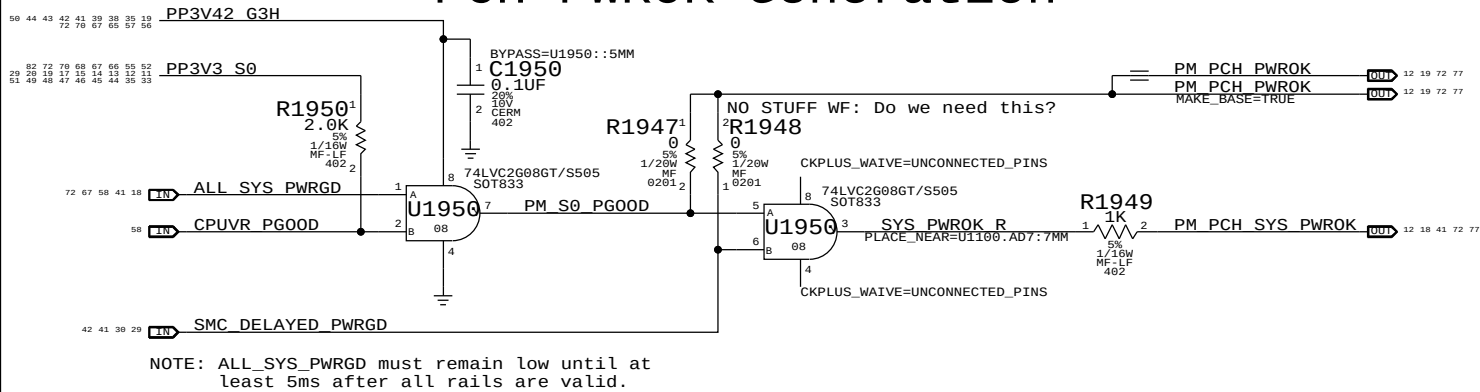
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		SHEET	17 OF 82



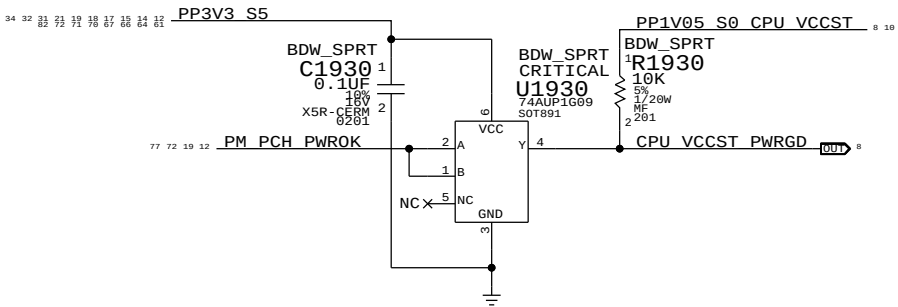
PCH Reset Button



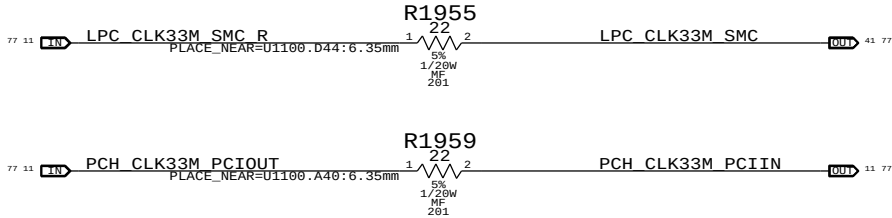
PCH PWROK Generation



VCCST (1.05V S0) PWRGD



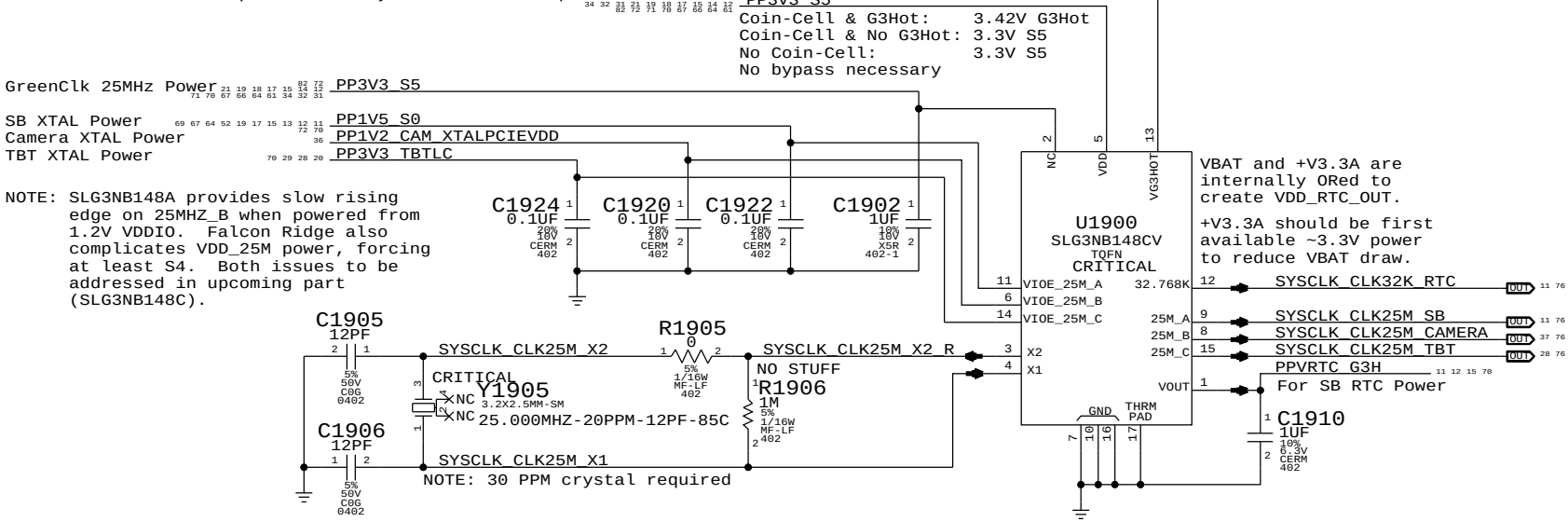
PCH 33MHz Clocks



System RTC Power Source & 32kHz / 25MHz Clock Generator

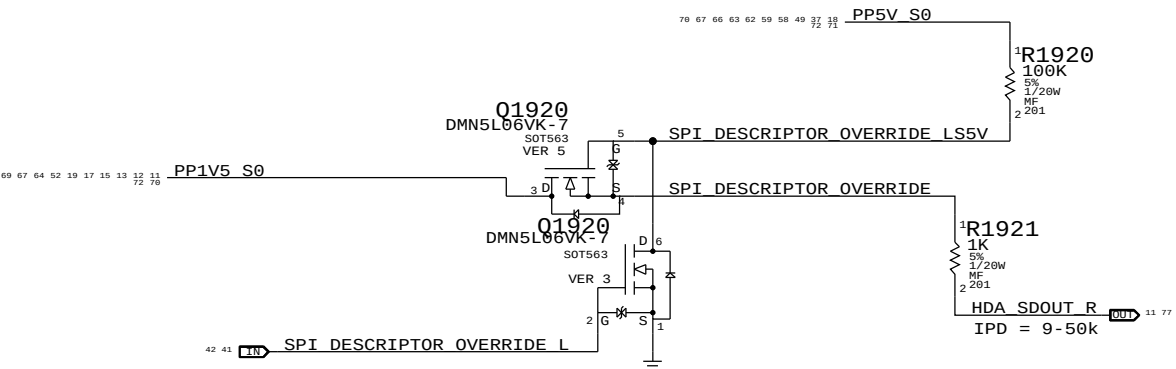
VDDIO_25M_A: SB power rail for XTAL circuit.
VDDIO_25M_B: Camera power rail for XTAL circuit.
VDDIO_25M_C: Thunderbolt power rail for XTAL circuit.

NOTE: VDD_25M must be powered if any VDDIO_25M_x is powered.



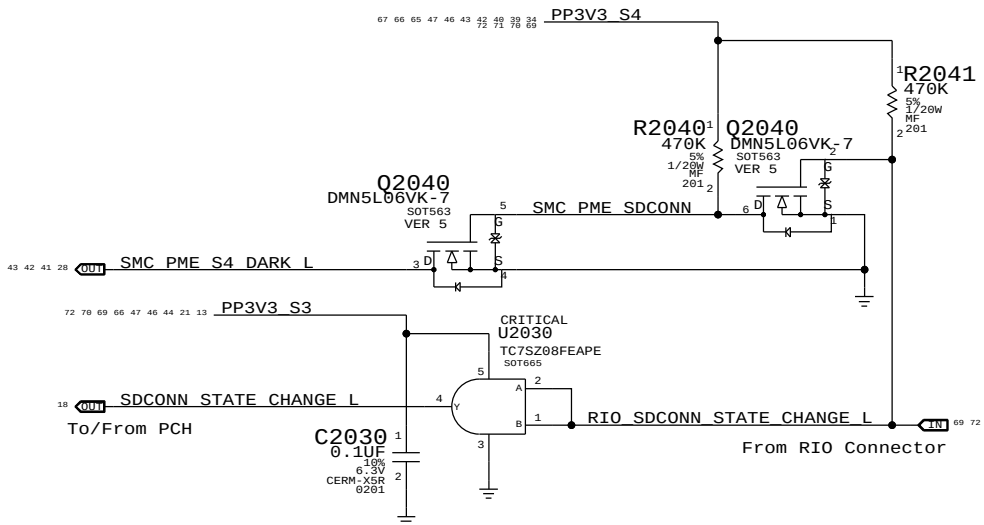
PCH ME Disable Strap

PCH uses HDA_SD0 as a power-up strap. If low, ME functions normally. If high, ME is disabled. This allows for full re-flashing of SPI ROM. SMC controls strap enable to allow in-field control of strap setting. Q1920 & 5V pull-up allows circuit to work regardless of HDA voltage.



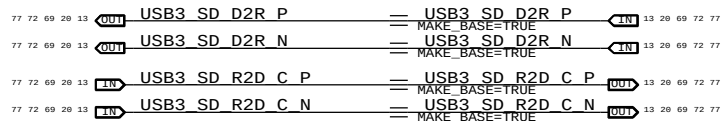
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		SHEET	
		19 OF 82	

RIO SD Card Reader Support



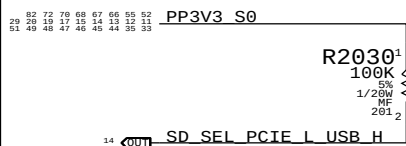
Flexible I/O Aliases

SD Card Reader is always USB3 in this implementaton.

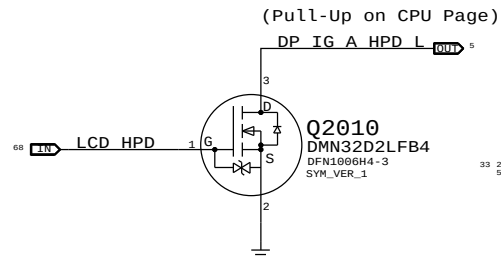


Flexible I/O Configuration Strap

Must pull signal correctly even if always USB or PCIe



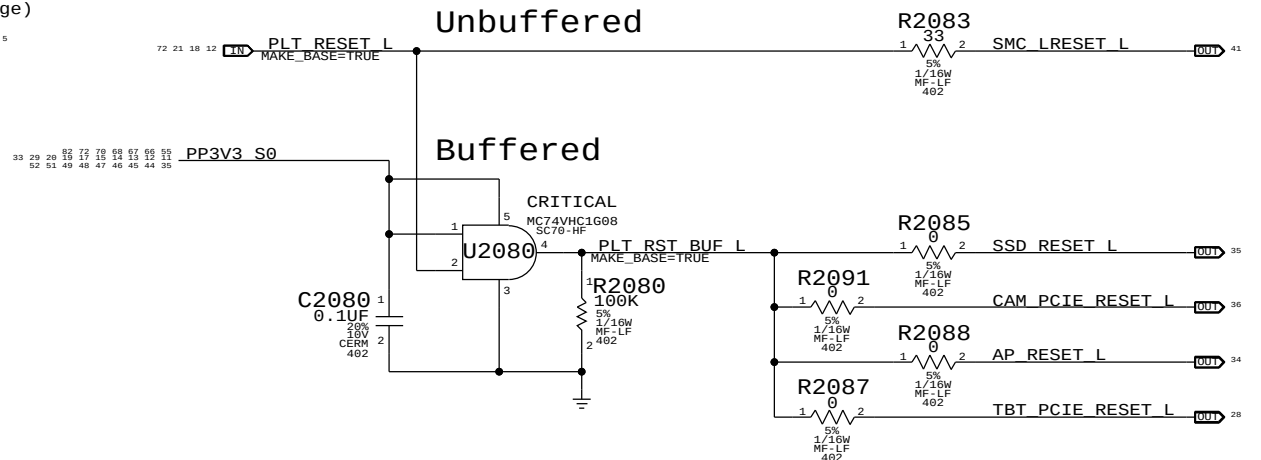
LCD HPD Inverter



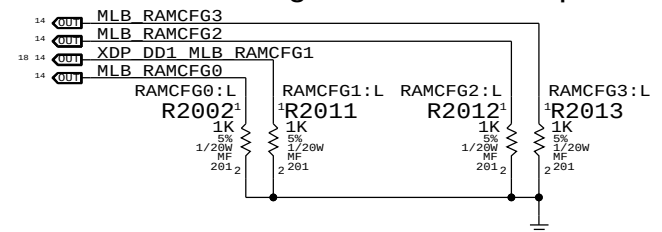
HDMI HPD pull-down



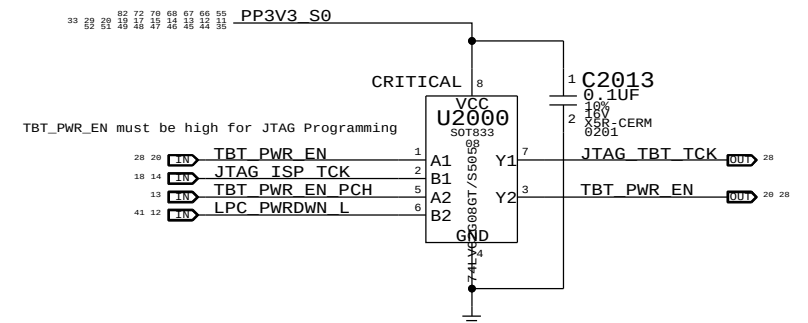
Platform Reset Connections



RAM Configuration Straps

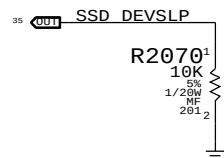


GPIO Glitch Prevention



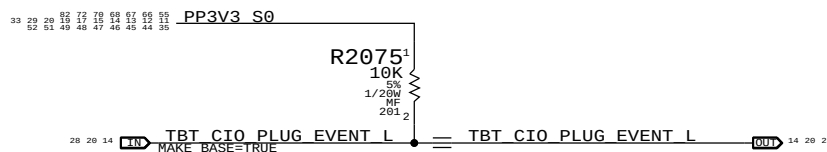
GS3 Connector Support

DEVSLP not supported on LPT-H



Falcon Ridge Support

RR output is open-drain, no isolation necessary

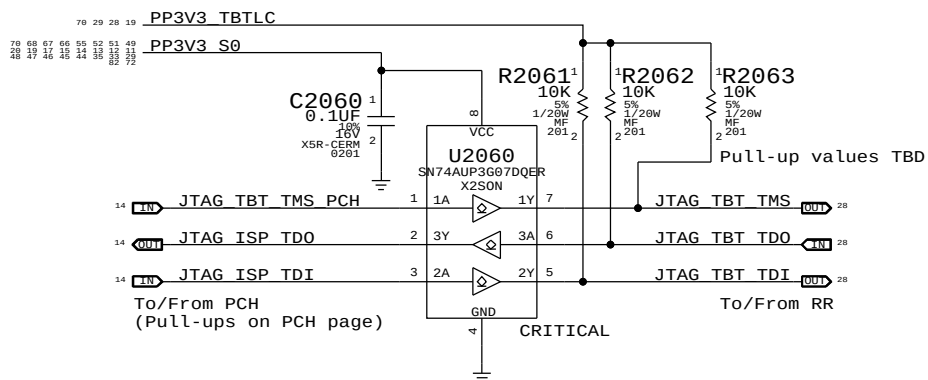


Falcon Ridge JTAG Isolation

TBTLIC can be on when S0 is off, and vice-versa

Isolation ensures no leakage to RR or PCH

U2060 supports I/O's powered when VCC=0V



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PAGE TITLE		Project Chipset Support	
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		PAGE	20 OF 118
		SHEET	20 OF 82

The circuit below handles CPU and VTT power during S0->S3->S0 transitions, as well as isolating the CPU's SM_DRAMRST# output from the S0-DIMMs when necessary.

ISOLATE_CPU_MEM_L GPIO state during S3<->S0 transitions determines behavior of signals.

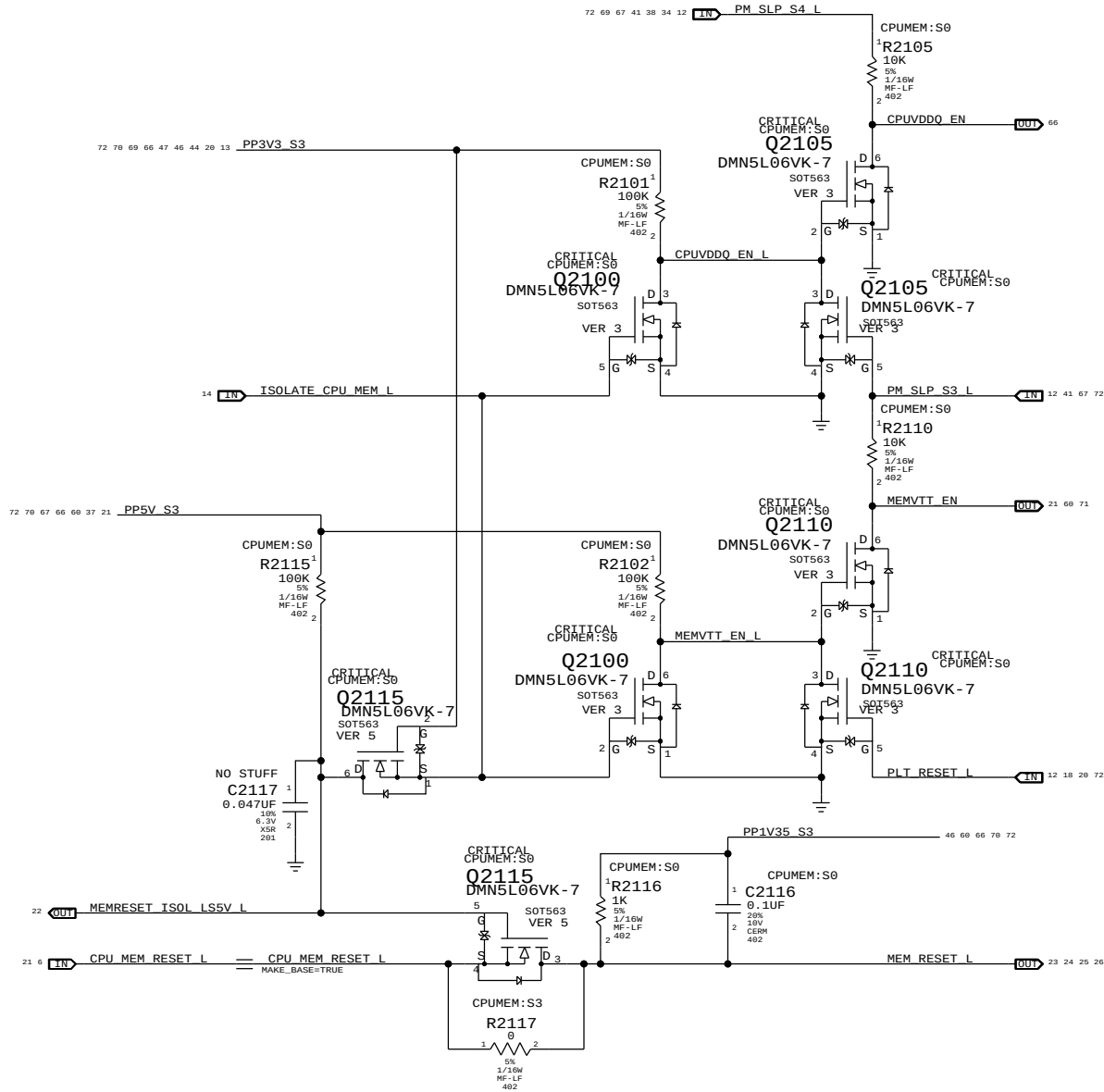
WHEN HIGH: CPU 1.5V remains powered in S3, VTT follows S0 rails, MEM_RESET_L not isolated.

WHEN LOW: CPU 1.5V follows S0 rails, VTT ensures clean CKE transition, MEM_RESET_L isolated.

CPUVDDQ_EN = (ISOLATE_CPU_MEM_L + PM_SLP_S3_L) * PM_SLP_S4_L

MEMVTT_EN = (ISOLATE_CPU_MEM_L + PLT_RST_L) * PM_SLP_S3_L

MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L

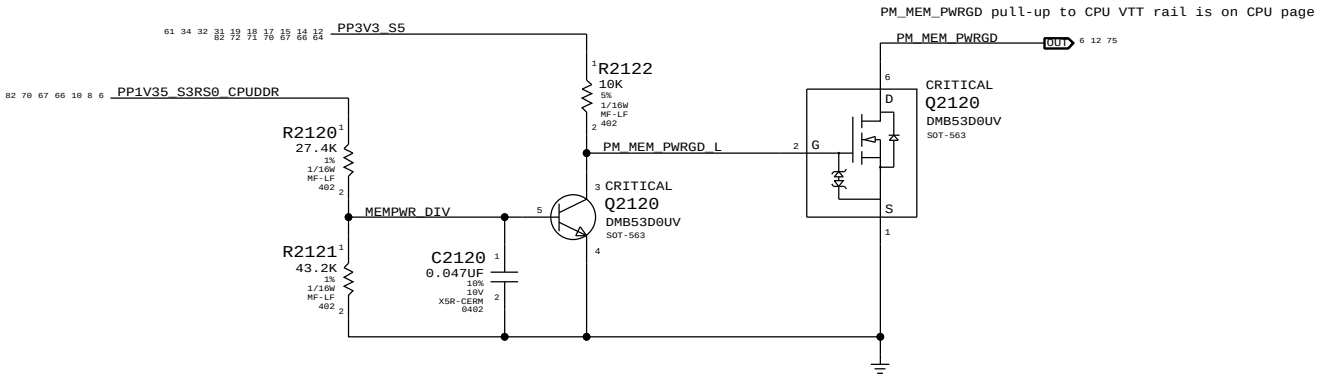


Step	ISOLATE_CPU_MEM_L	PLT_RESET_L	PM_SLP_S3_L	PM_SLP_S4_L	CPU_MEM_RESET_L	MEM_RESET_L	MEMVTT_EN	CPUVDDQ_EN
S0	0	1	1	1	1	CPU_MEM_RESET_L	1	1
1	1	0	1	1	1	1	1	1
2	0	0	1	1	1	1	0	1
3	0	0	0	1	X	1	0	0
4	0	0	1	1	X	1	0	1
5	0	1	1	1	0 (*)	1	1	1
6	0	1	1	1	1	1	1	1
S0	7	1	1	1	1	CPU_MEM_RESET_L	1	1

(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

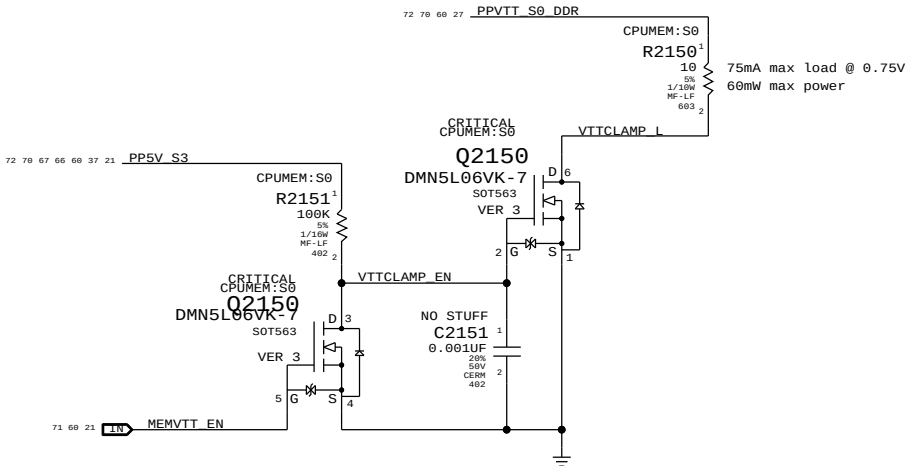
NOTE: In the event of a S3->S5 transition ISOLATE_CPU_MEM_L will still be asserted on next S5->S0 transition. Rails will power-up as if from S3, but MEM_RESET_L will not properly assert. Software must deassert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

MEM S0 "PGOOD" for CPU



MEMVTT Clamp

Ensures CKE signals are held low in S3

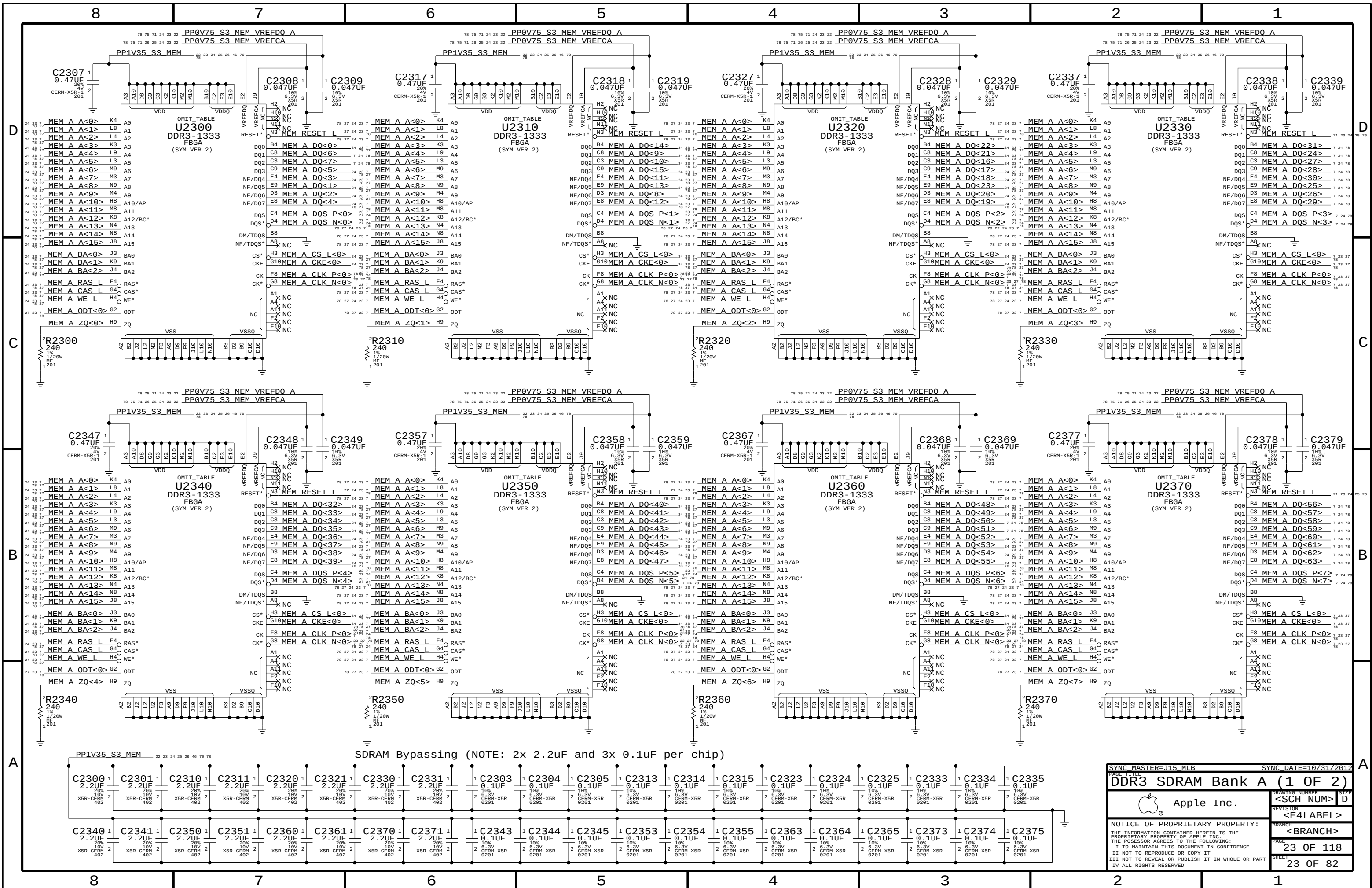


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CPU Memory S3 Support			
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		PAGE	21 OF 118
		SHEET	21 OF 82
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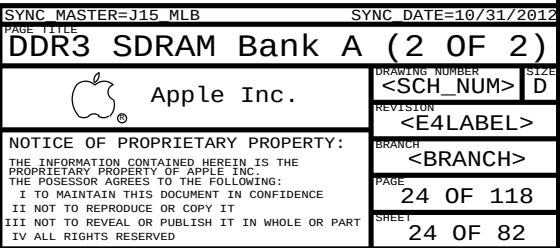
	MEM A VREF DQ	MEM B VREF DQ	MEM A VREF CA	MEM B VREF CA	MEM VREG	
DAC Channel:	A	B	C	C	D	
PCA9557D Pin:	1	2	3	4	5	
	LPDDR3 (1.2V)		DDR3L (1.35V)		LPDDR3 (1.2V)	DDR3L (1.35V) NOTE
Nominal value	0.600V (DAC: 0x2E.5)		0.675V (DAC: 0x34)		1.200V (DAC: 0x5D)	1.343V (DAC: 0x68)
Margined target:	0.300V - 0.900V (+/- 300mV)		0.337V - 1.013V (+/- 337.5mV)		0.800V - 1.600V (+/- 400mV)	0.972V - 1.714V (+/- 371mV)
DAC range:	0.000V - 1.199V (0x00 - 0x5D)		0.000V - 1.354V (0x00 - 0x69)		0.000V - 2.397V (0x00 - 0xBA)	0.000V - 2.694V (0x00 - 0xD1)
Vref current:	+73uA - -73uA (- = sourced)		+82uA - -82uA (- = sourced)		+21uA - -21uA (- = sourced)	+25uA - -25uA (- = sourced)
DAC step size:	6.36mV / step @ output		6.36mV / step @ output		4.28mV / step @ output	3.53mV / step @ output

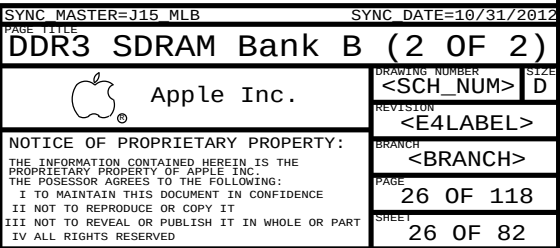
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		SHEET 22 OF 82	

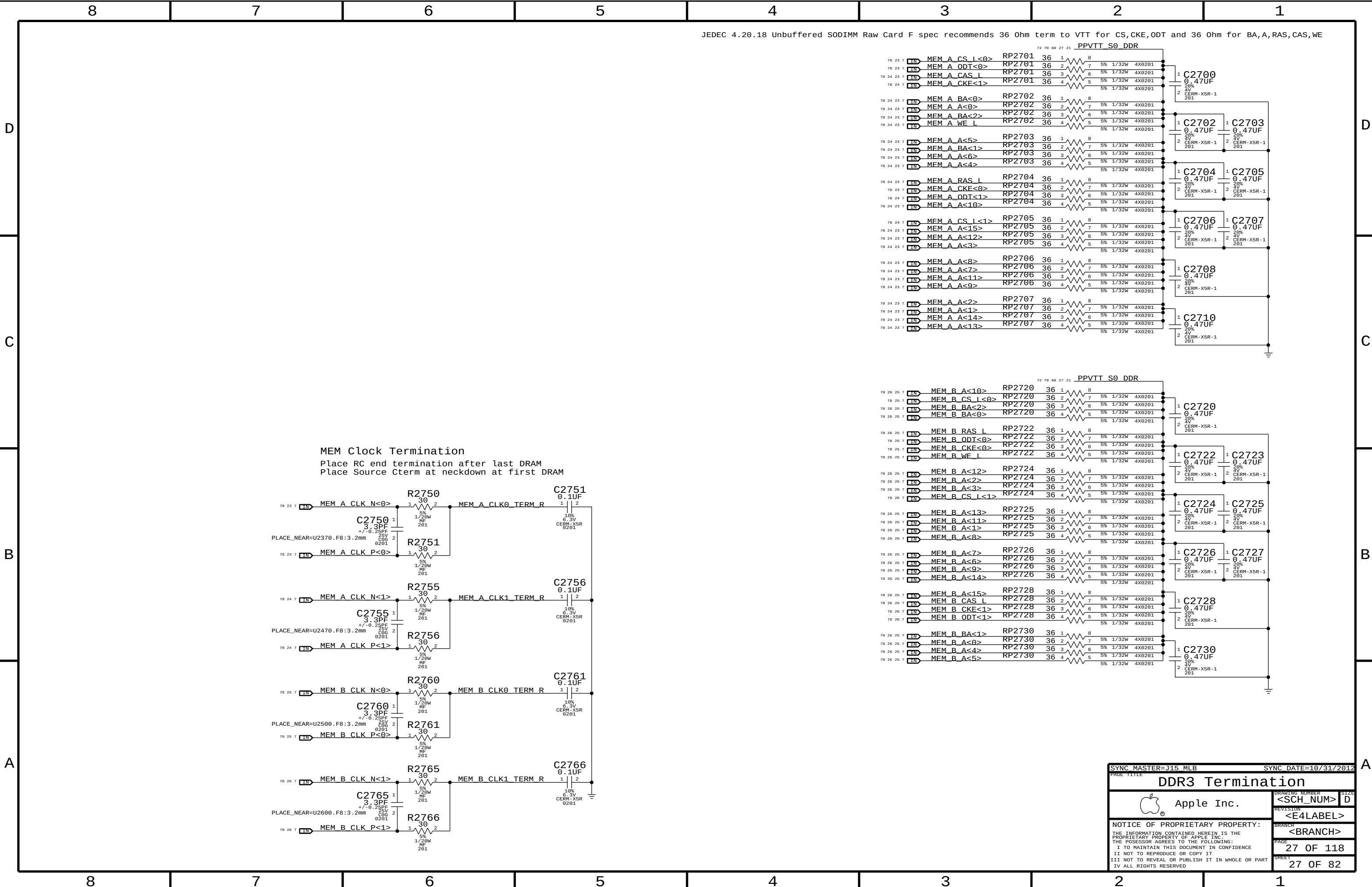


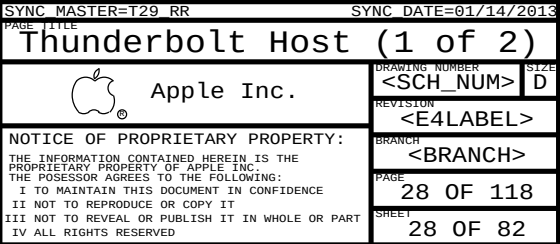
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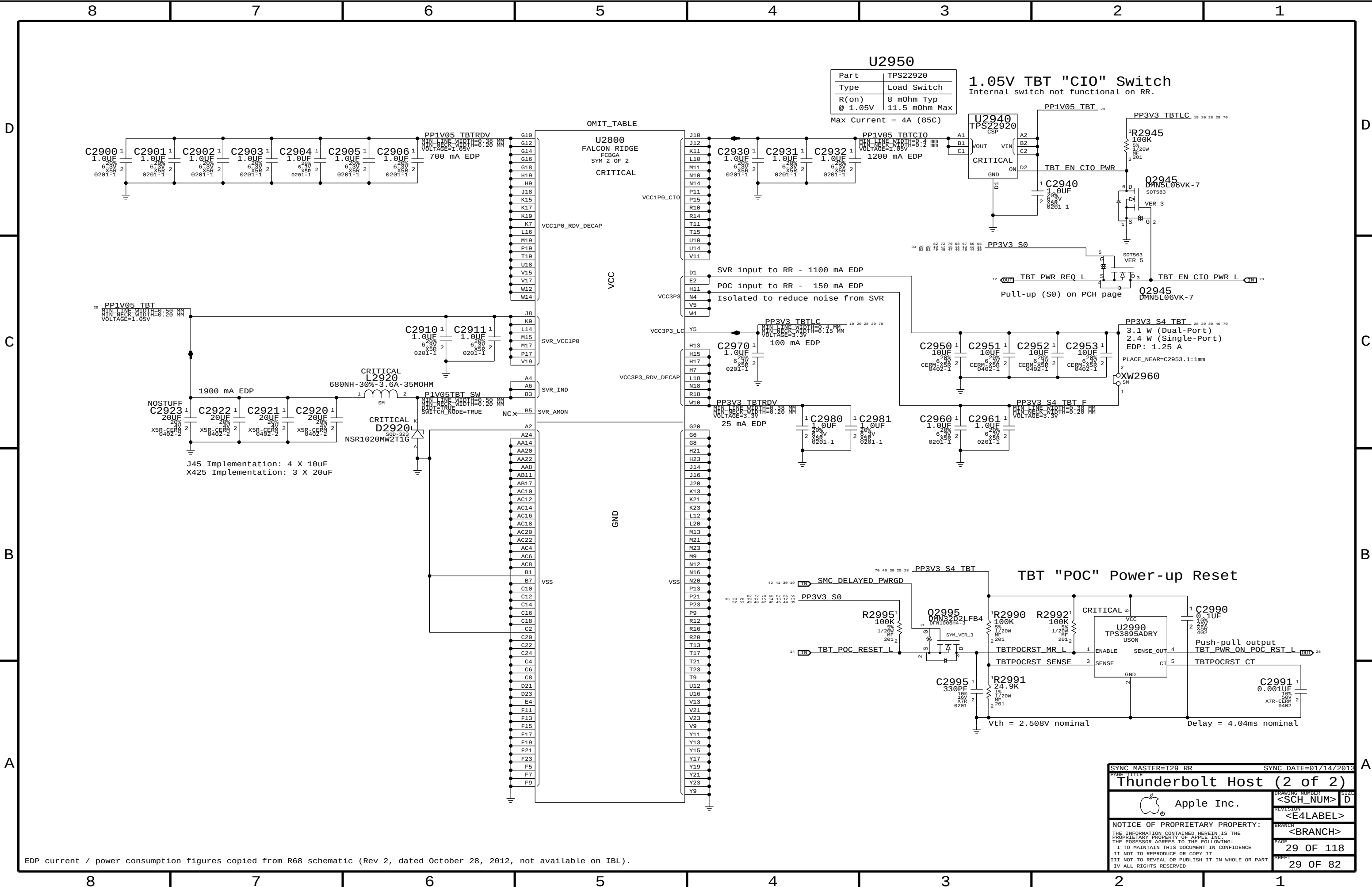
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REVISION
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PAGE
23 OF 118
SHEET
23 OF 82











U2950	
Part	TPS22920
Type	Load Switch
R(on)	8 mOhm Typ
@ 1.05V	11.5 mOhm Max
Max Current = 4A (85C)	

1.05V TBT "CIO" Switch
Internal switch not functional on RR.

SYNC MASTER=T29 RR		SYNC DATE=01/14/2013	
PAGE TITLE		Thunderbolt Host (2 of 2)	
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		SHEET	29 OF 82

EDP current / power consumption figures copied from R68 schematic (Rev 2, dated October 28, 2012, not available on IBL).

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8

Thunderbolt 15V Boost Regulator

7

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4

3

2

1

Power aliases required by this page:

- PPVIN_SW_TBTBST (8-13V Boost Input)
- PP15V_TBT_REG (15V Boost Output)

Signal aliases required by this page:

(NONE)

BOM options provided by this page:

(NONE)

The schematic shows a boost converter circuit. Key components include:

- MOSFET Q3080 (SI8409DB) as the main switching device.
- Inductor L3095 (PIMB063T-SM) for energy storage.
- Diode D3095 (PDS540XF) for output rectification.
- Sense amplifier U3090 (LT3957-QFN) for feedback.
- Various resistors (R3080-R3099) and capacitors (C3085-C3099) for timing, compensation, and filtering.
- A second FET stage at the bottom right for PM BATLOW isolation.

UVLO(falling) = 1.22 * (R1 + R2) / R2

UVLO(rising) = UVLO(falling) + (2uA * R1)

UVLO = 4.55V (falling), 4.95 (rising)

Vout = 15.47V

Max Current = 2A?

Freq = 480KHz

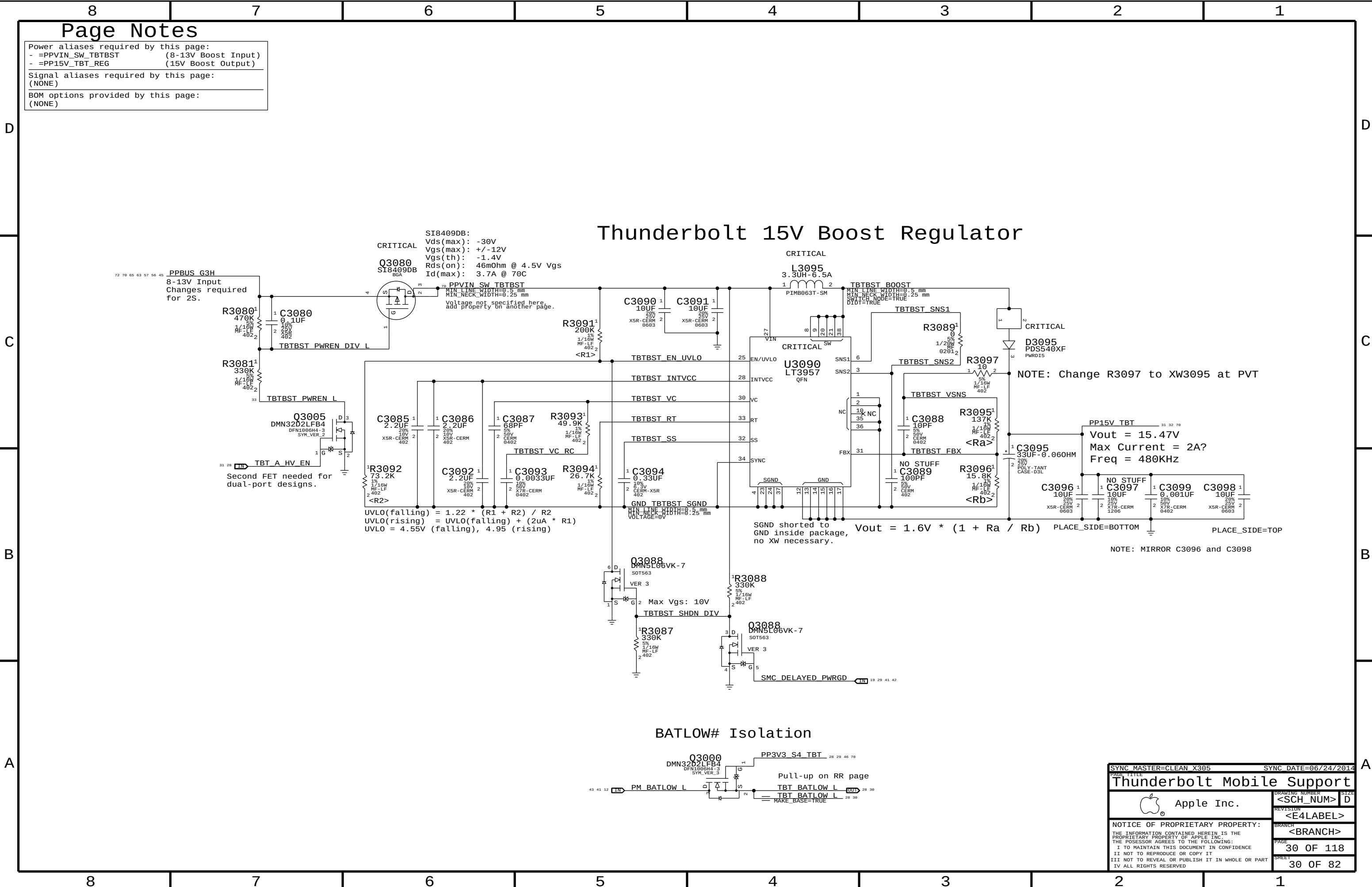
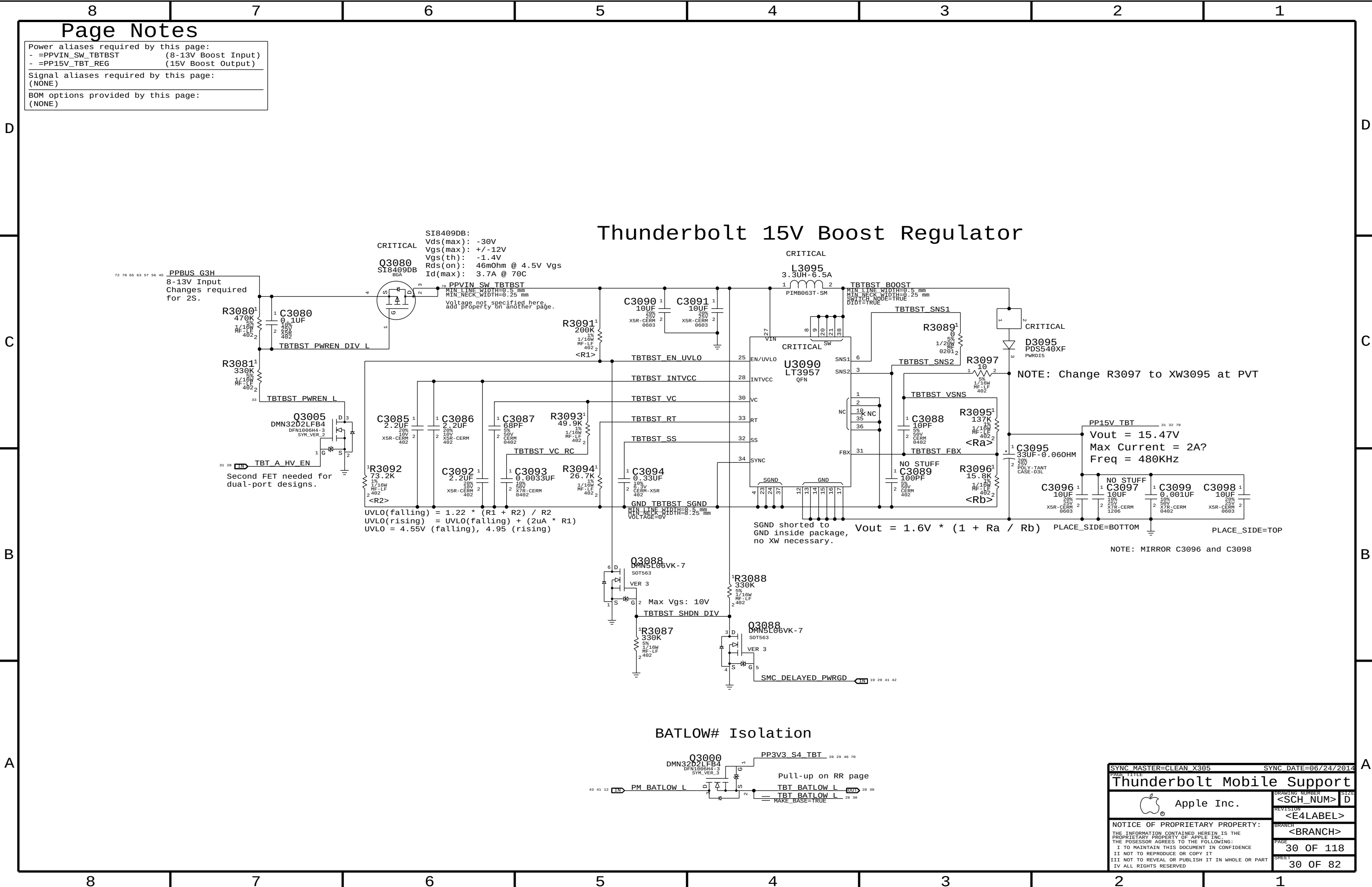
PM BATLOW# Isolation

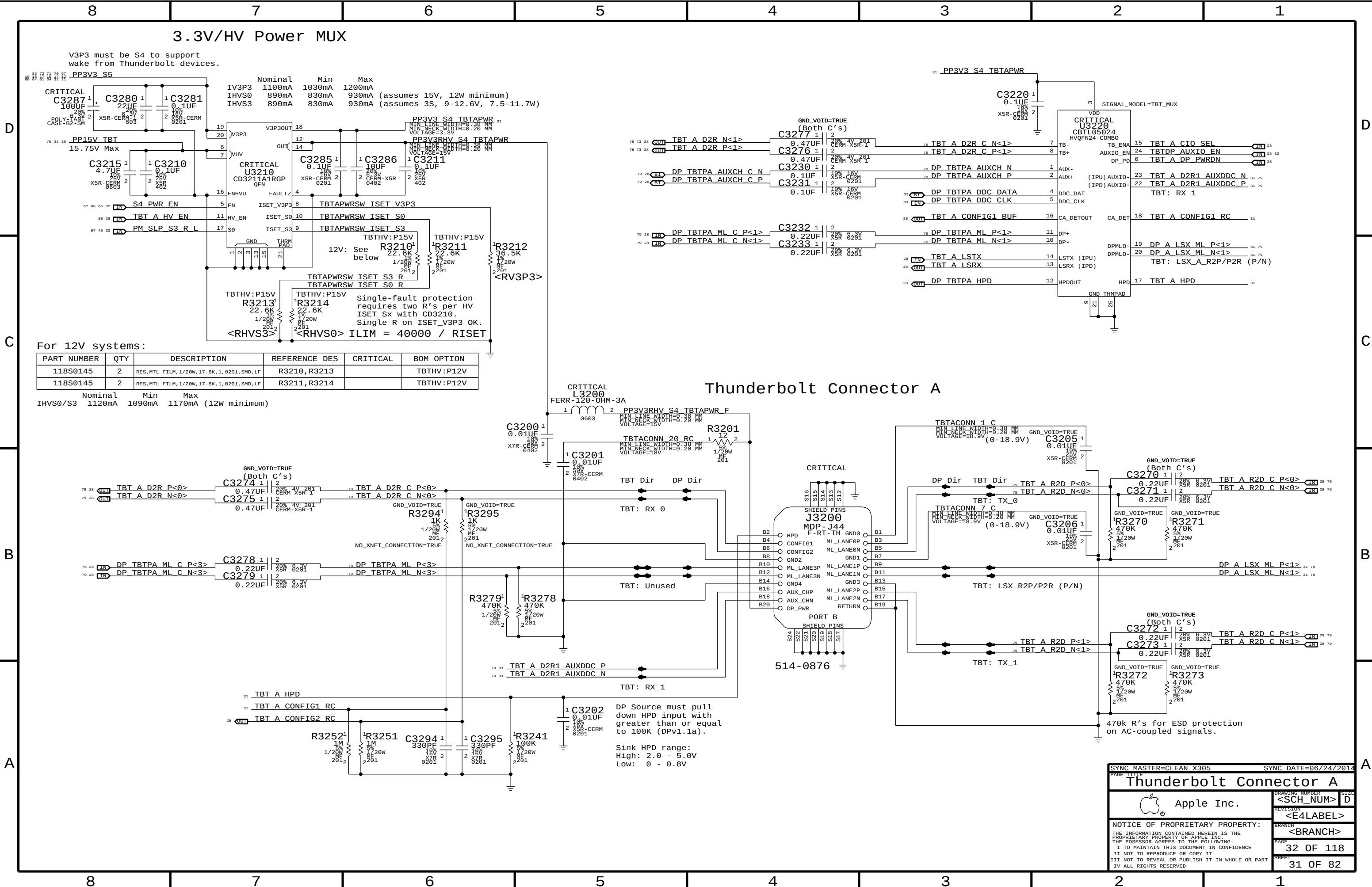
This section shows a pull-up resistor connected to the PM BATLOW signal through a MOSFET (Q3000).

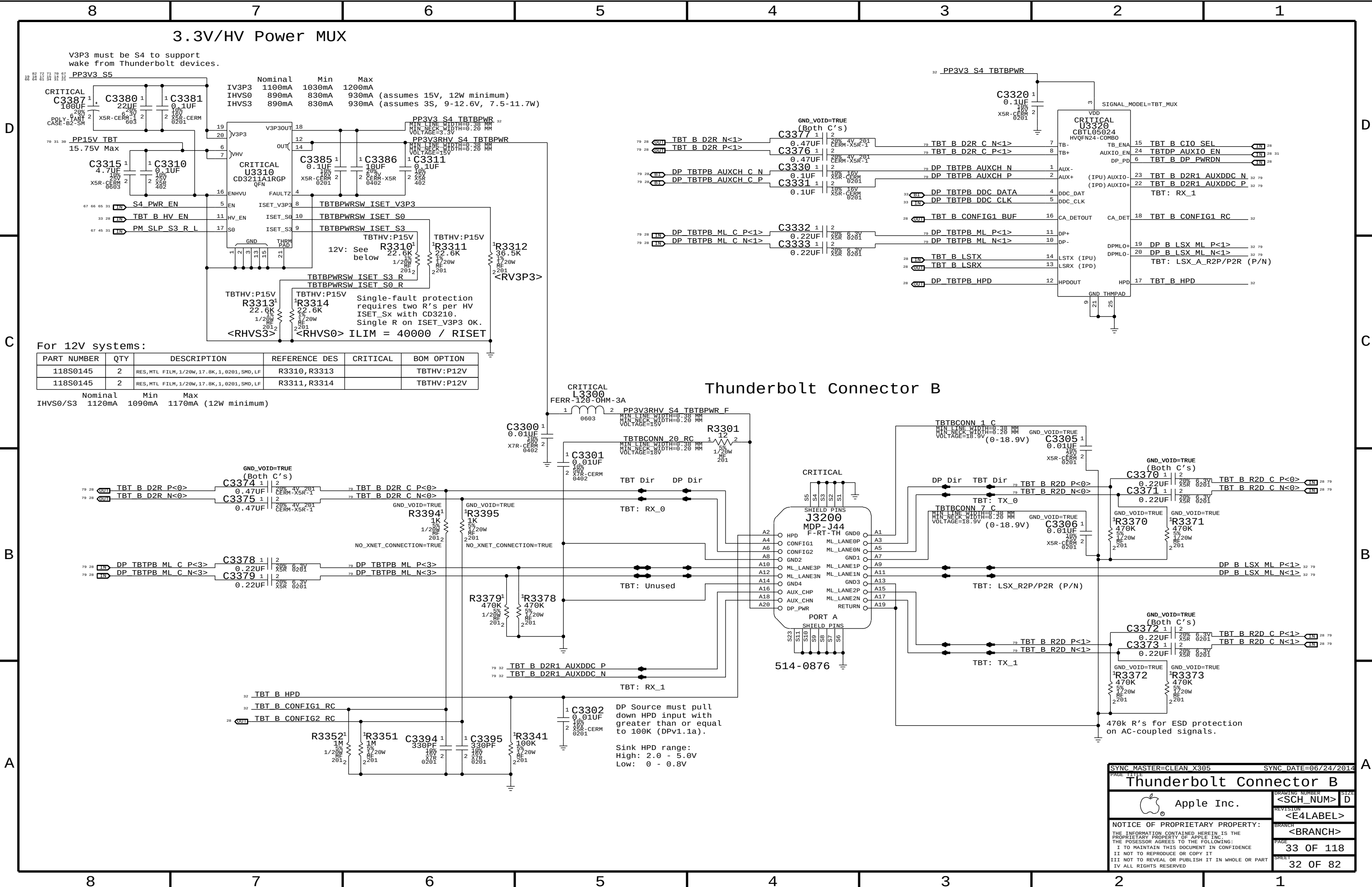
Pull-up on RR page

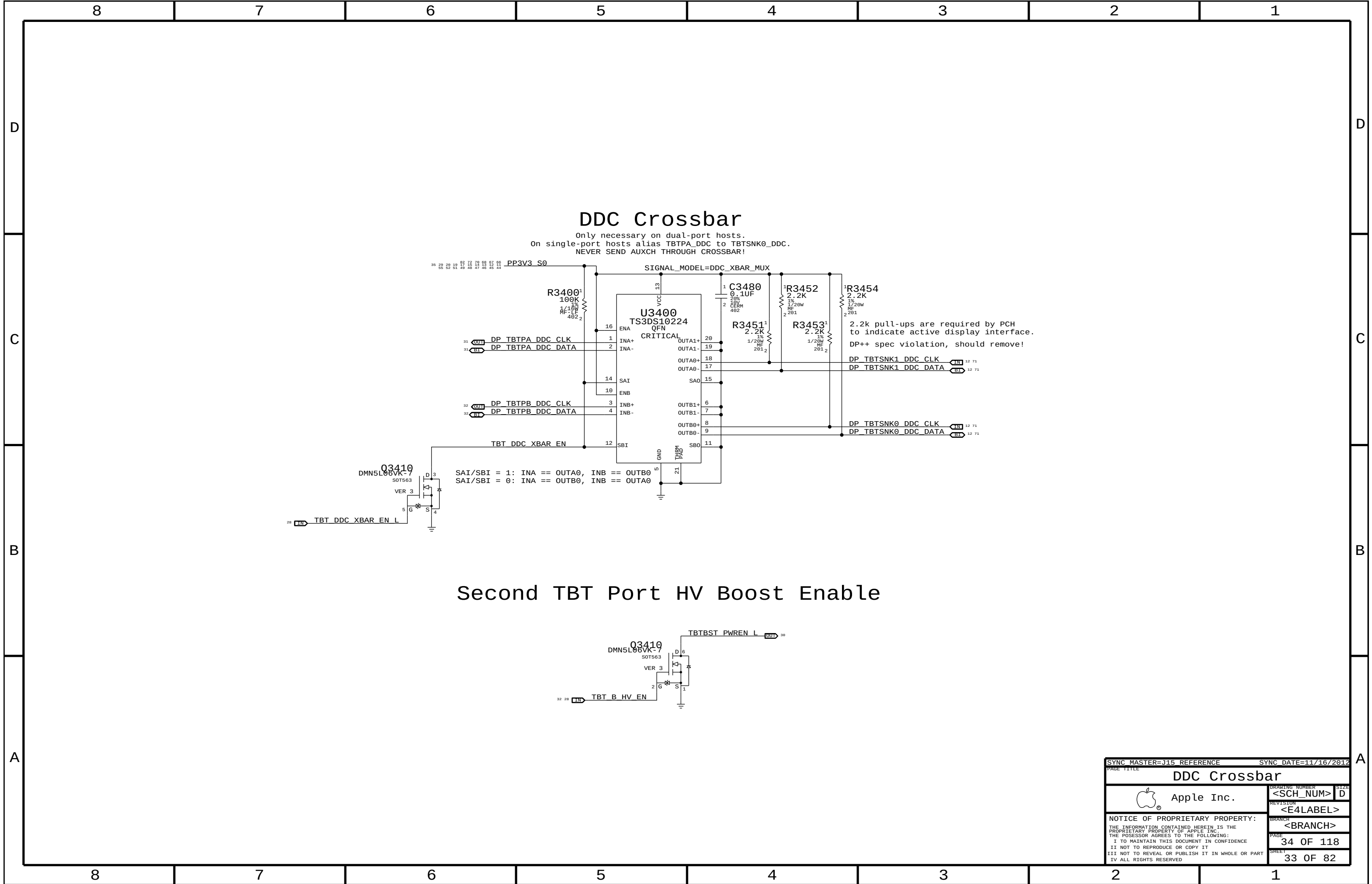
Shows a pull-up resistor connected to the TBT_BATLOW_L signal.

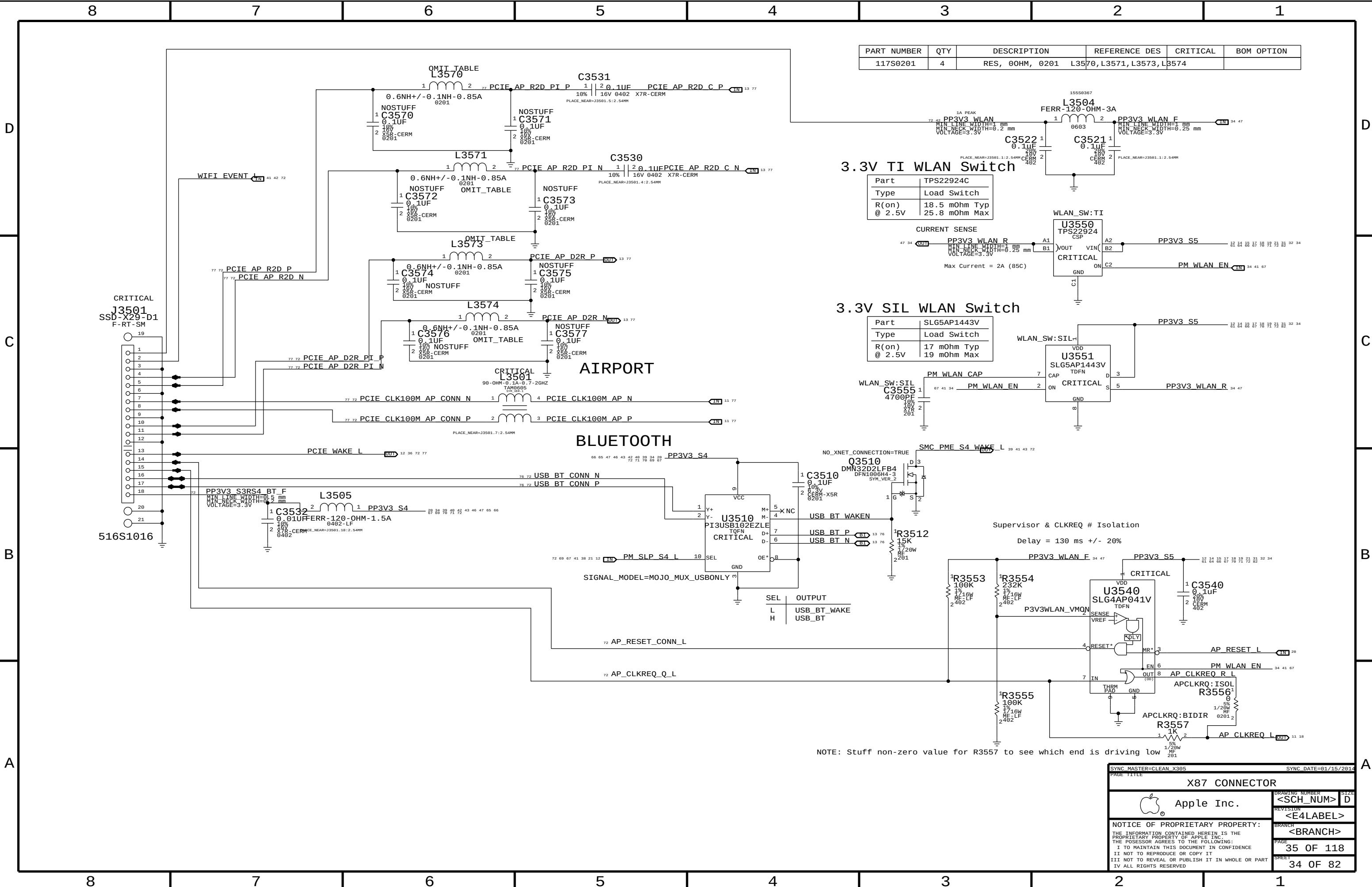
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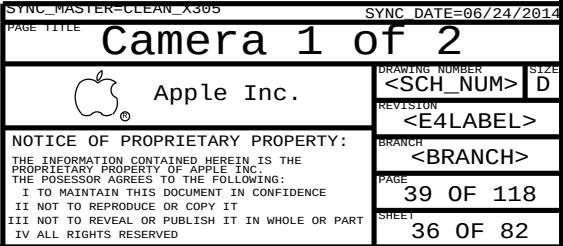


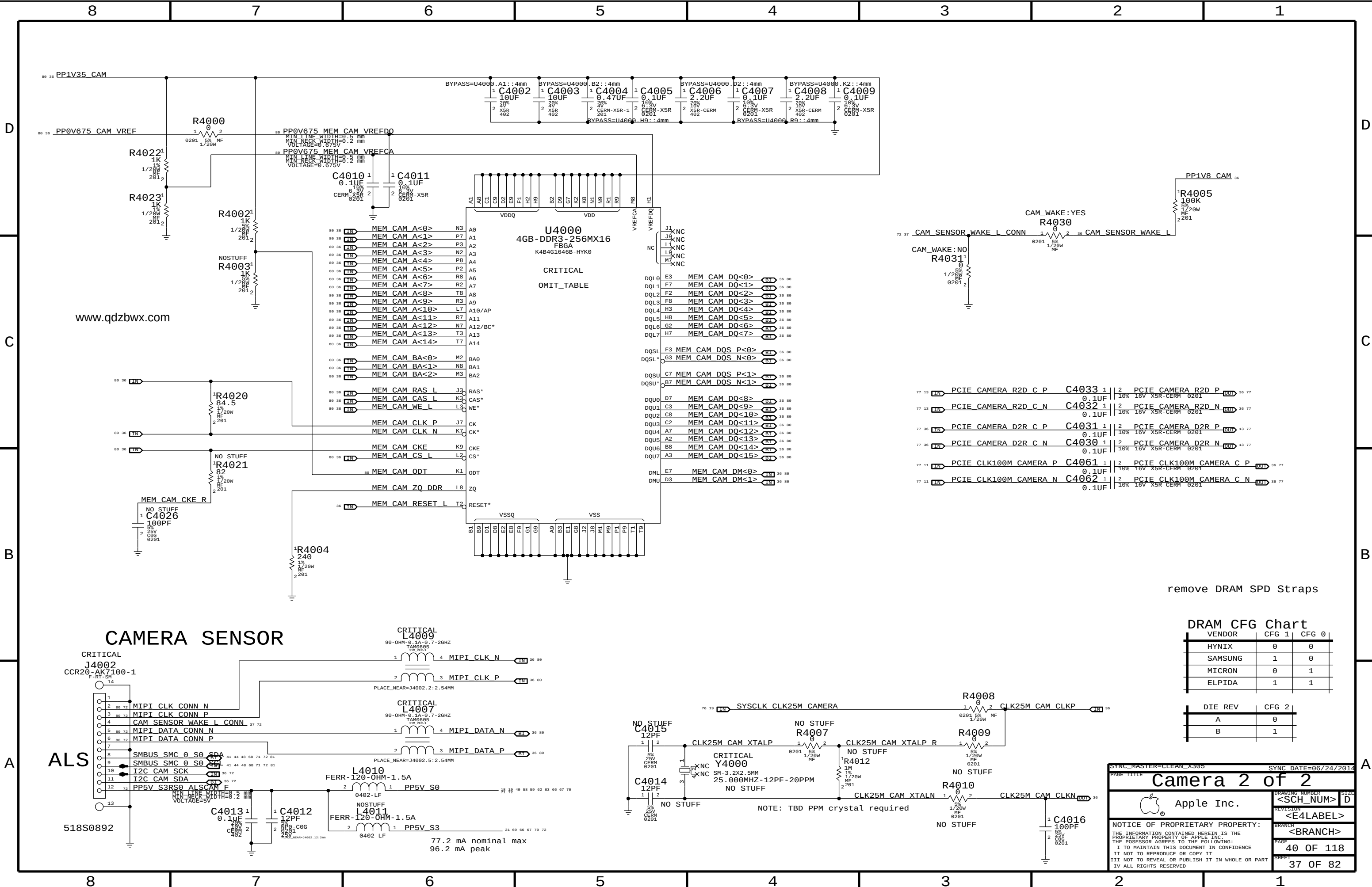
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0201	4	RES, 00HM, 0201	L3570,L3571,L3573,L3574		

Part	TPS22924C
Type	Load Switch
R(on)	18.5 mOhm Typ
@ 2.5V	25.8 mOhm Max

Part	SLG5AP1443V
Type	Load Switch
R(on)	17 mOhm Typ
@ 2.5V	19 mOhm Max

SYNC MASTER=CLEAN X305		SYNC DATE=01/15/2014	
PAGE TITLE		X87 CONNECTOR	
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		PAGE	35 OF 118
		SHEET	34 OF 82





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ALS

remove DRAM SPD Straps

DRAM CFG Chart

VENDOR	CFG 1	CFG 0
HYNIX	0	0
SAMSUNG	1	0
MICRON	0	1
ELPIDA	1	1

DIE REV	CFG 2
A	0
B	1

SYNC_MASTER=CLEAN_X305

SYNC_DATE=06/24/2014

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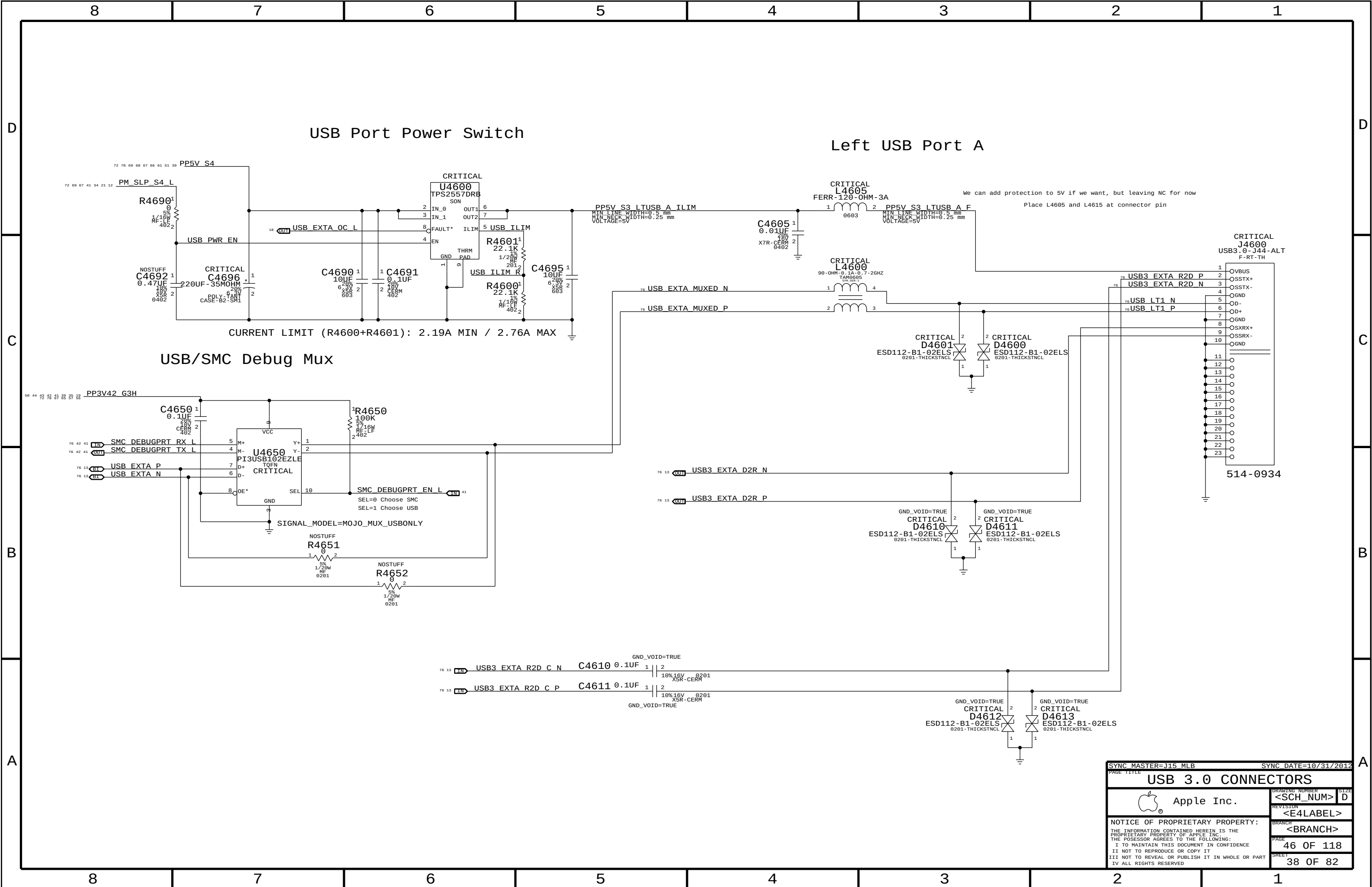
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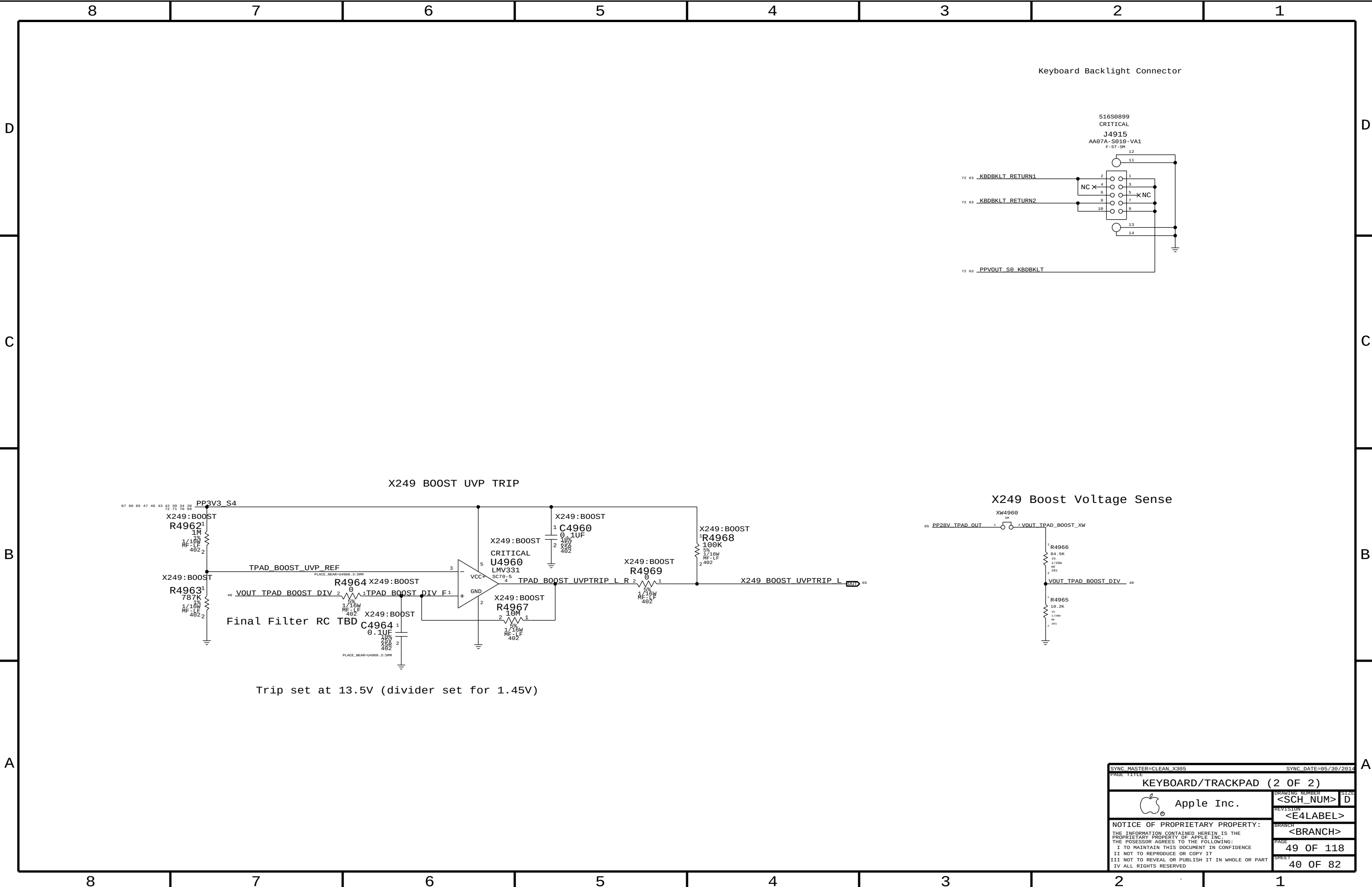
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40 OF 118

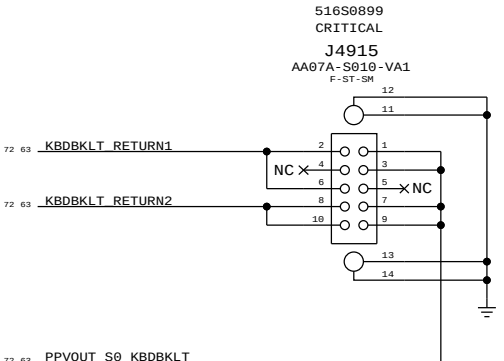
SHEET

37 OF 82

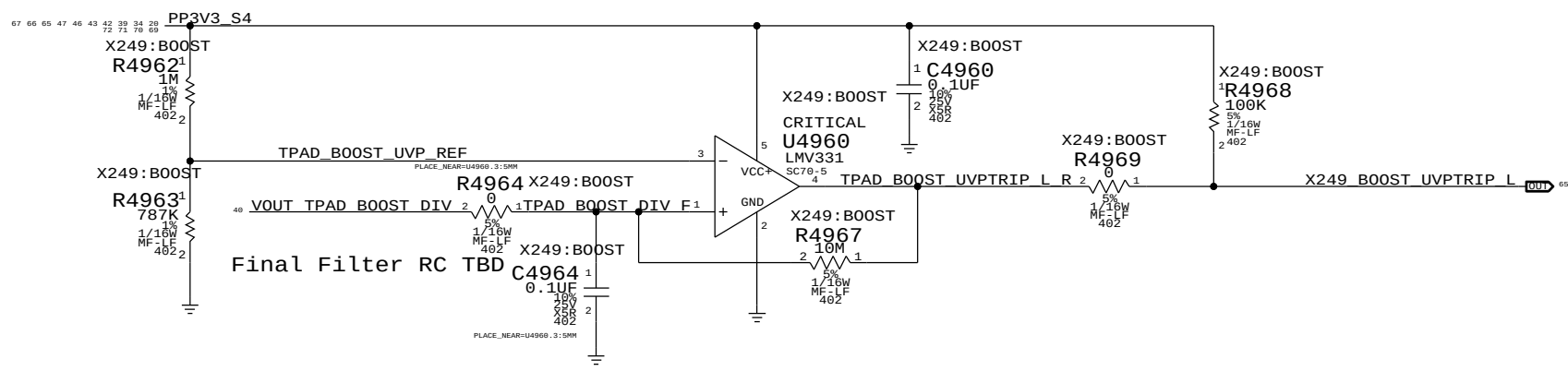




Keyboard Backlight Connector

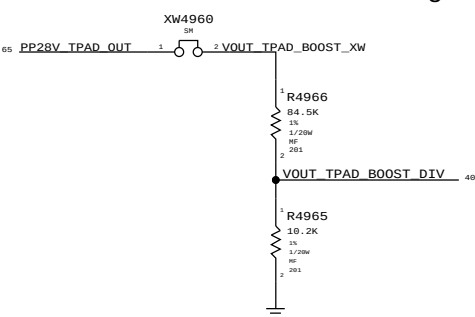


X249 B00ST UVP TRIP

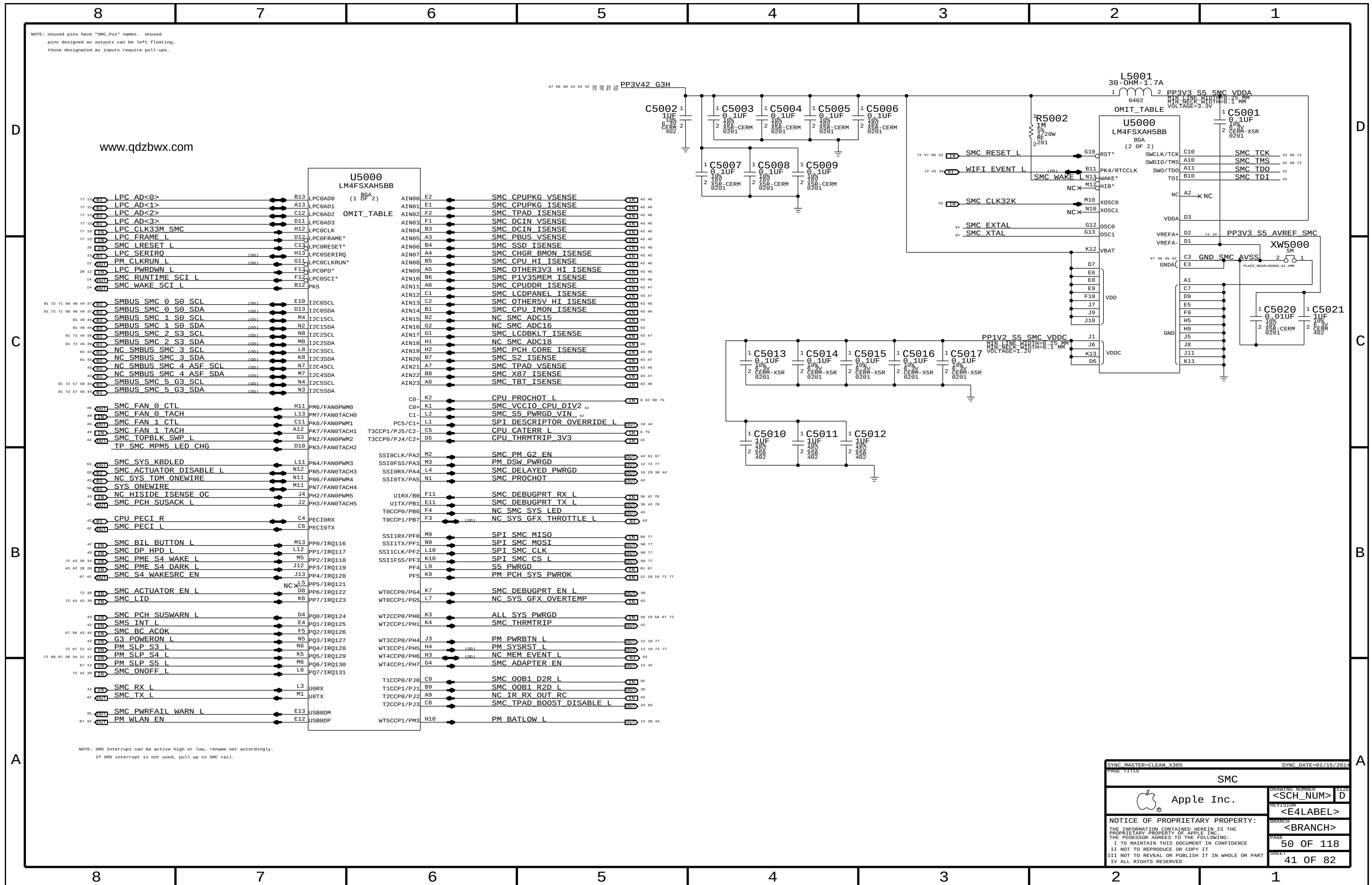


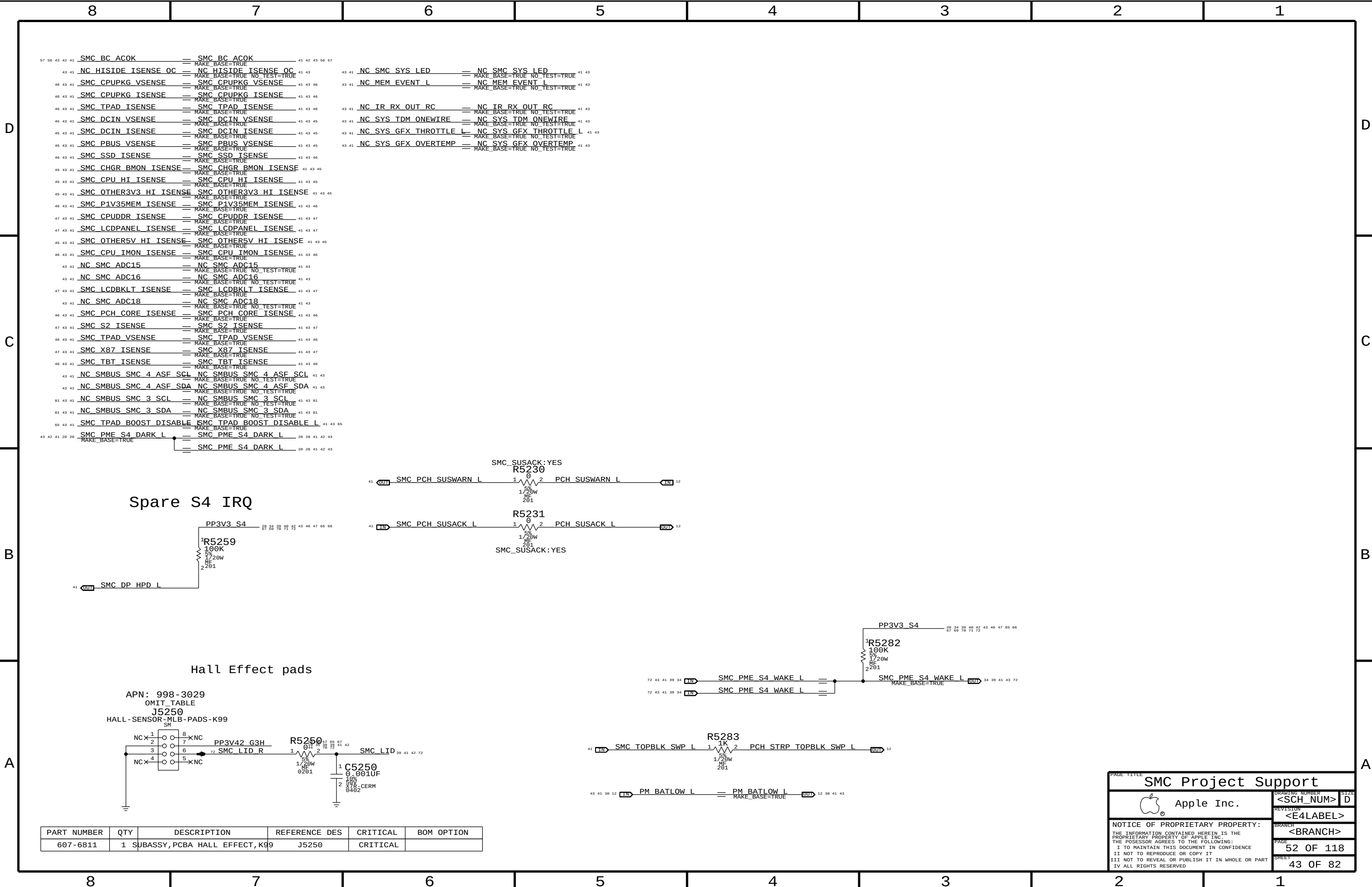
Trip set at 13.5V (divider set for 1.45V)

X249 Boost Voltage Sense

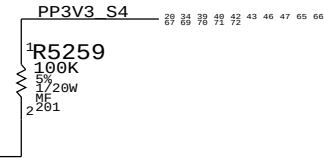


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PAGE TITLE			
KEYBOARD/TRACKPAD (2 OF 2)			
 Apple Inc.		DRAWING NUMBER	SIZE
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		40 OF 82	



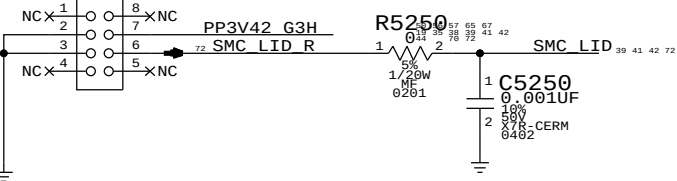


Spare S4 IRQ



Hall Effect pads

APN: 998-3029
OMIT_TABLE
J5250
HALL-SENSOR-MLB-PADS-K99



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
607-6811	1	SUBASSY,PCBA HALL EFFECT,K99	J5250	CRITICAL	

SMC Project Support

Apple Inc.

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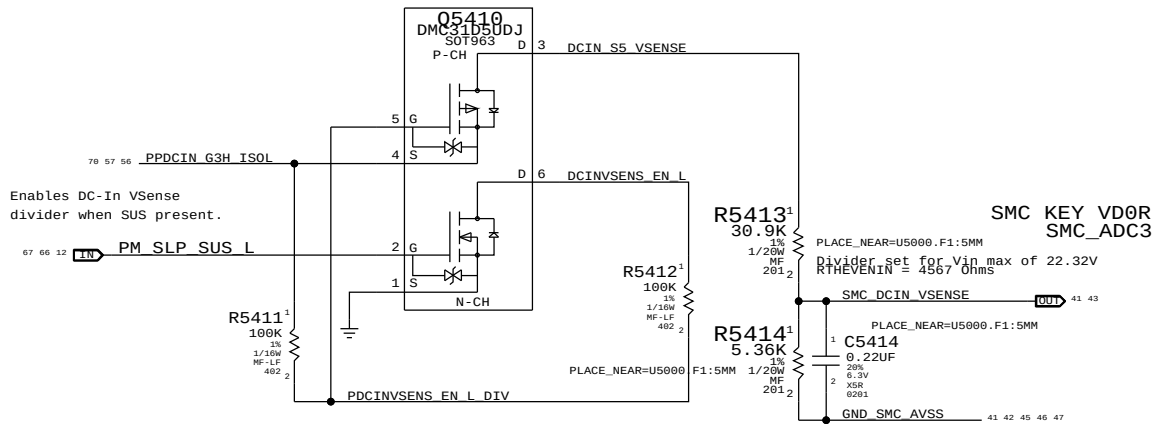
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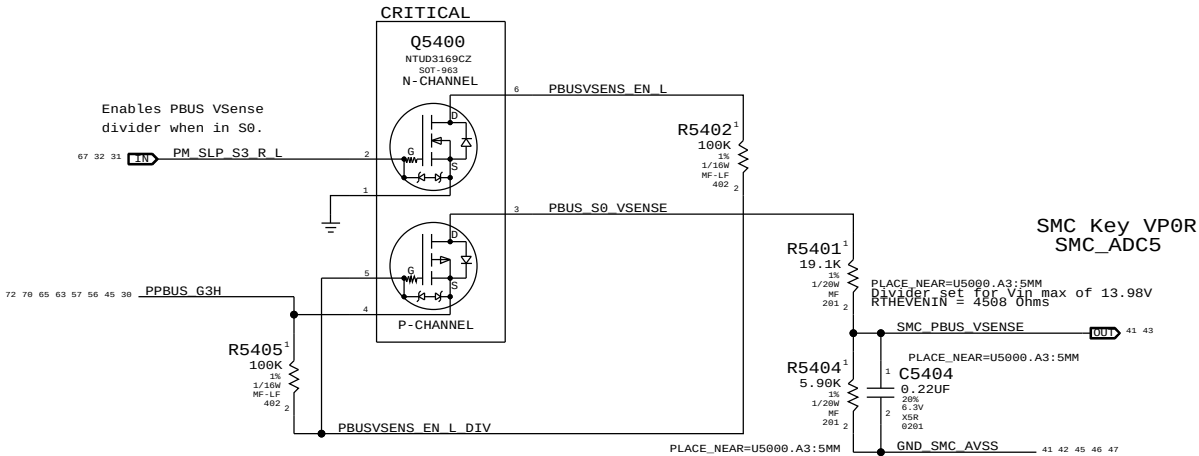
PAGE
52 OF 118

SHEET
43 OF 82

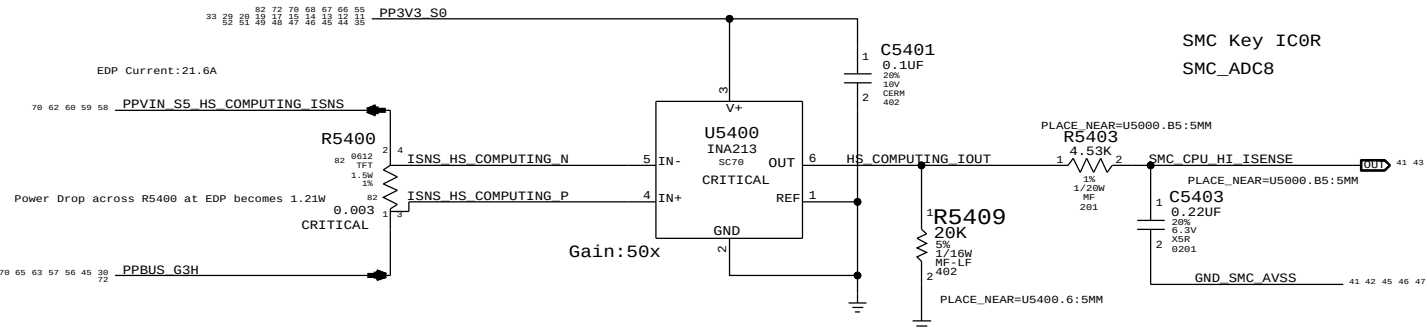
DC-In Voltage Sense Enable & Filter



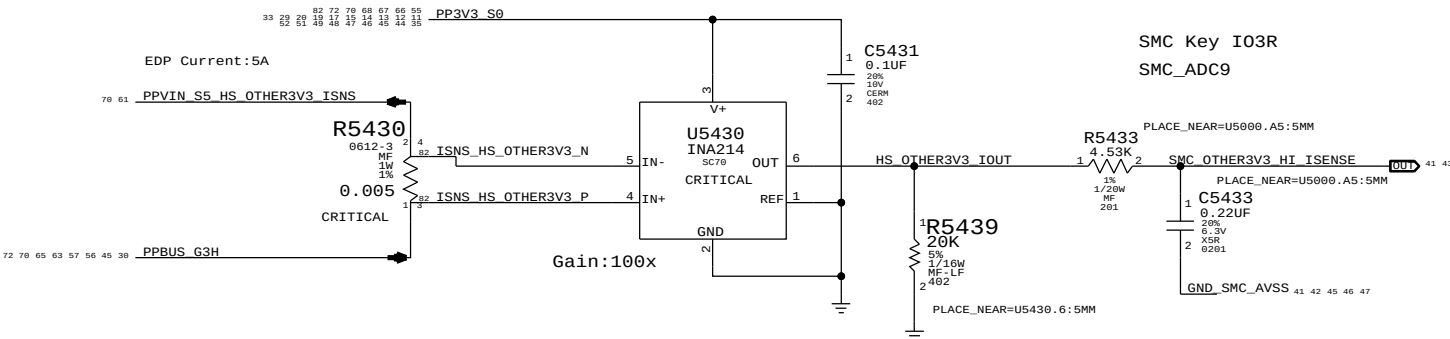
PBUS Voltage Sense Enable & Filter



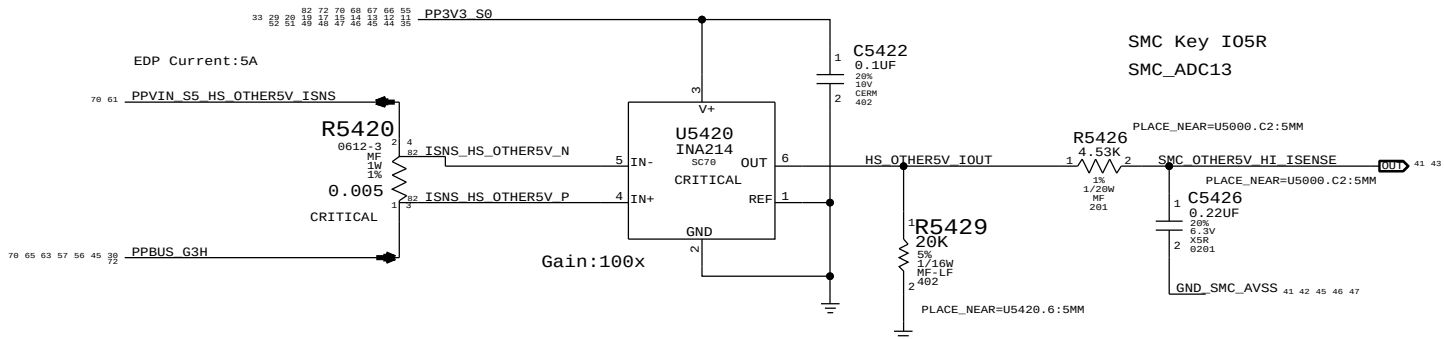
COMPUTING High Side Current Sense / Filter



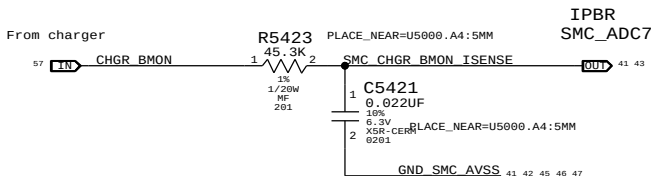
OTHERS (3.3V) High Side Current Sense / Filter



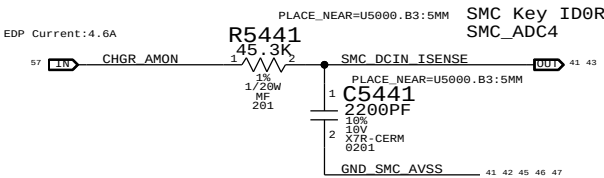
OTHERS (5V) High Side Current Sense / Filter



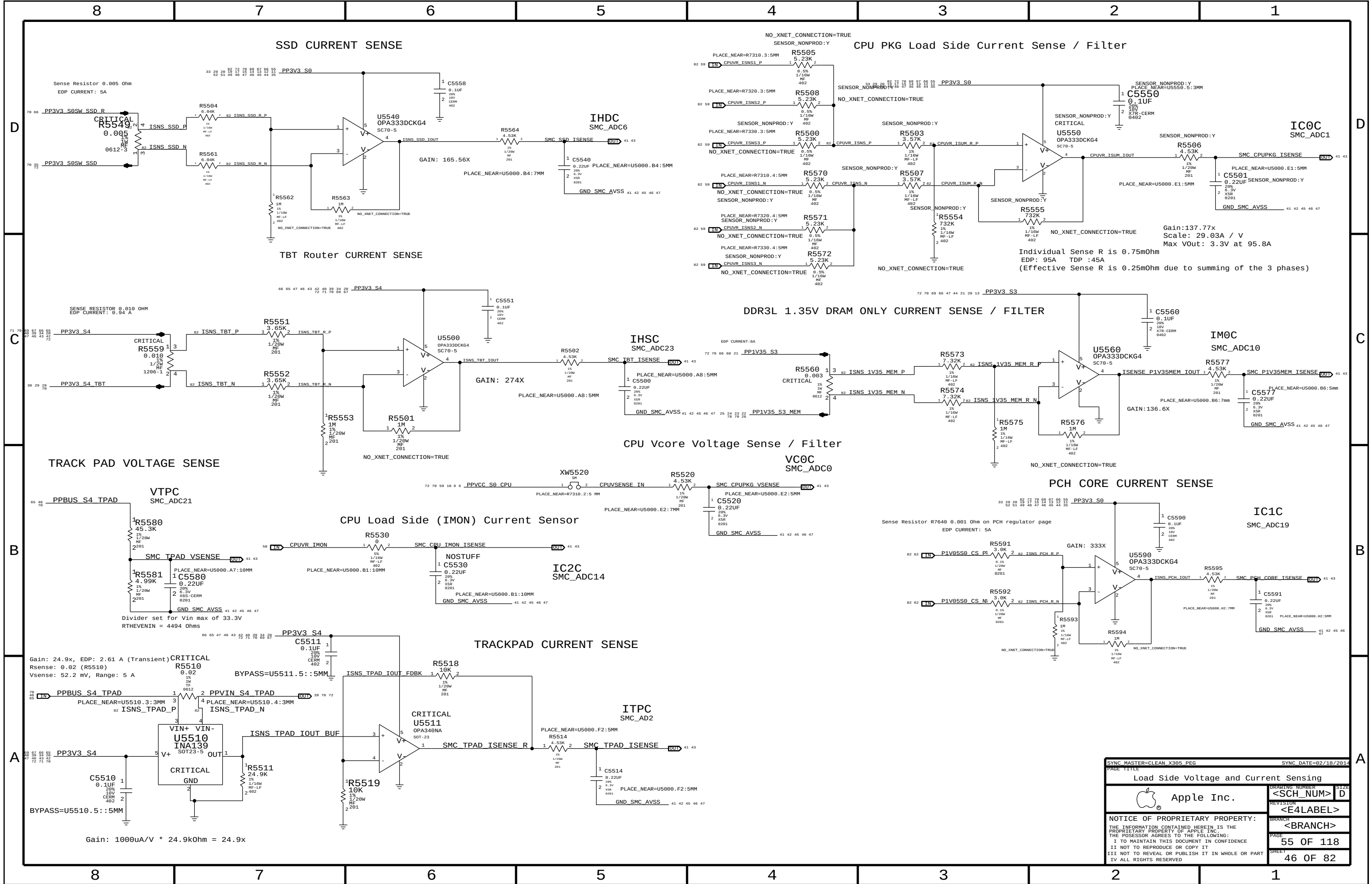
CHARGER BMON HIGH SIDE (BATTERY DISCHARGE) CURRENT SENSE & FILTER

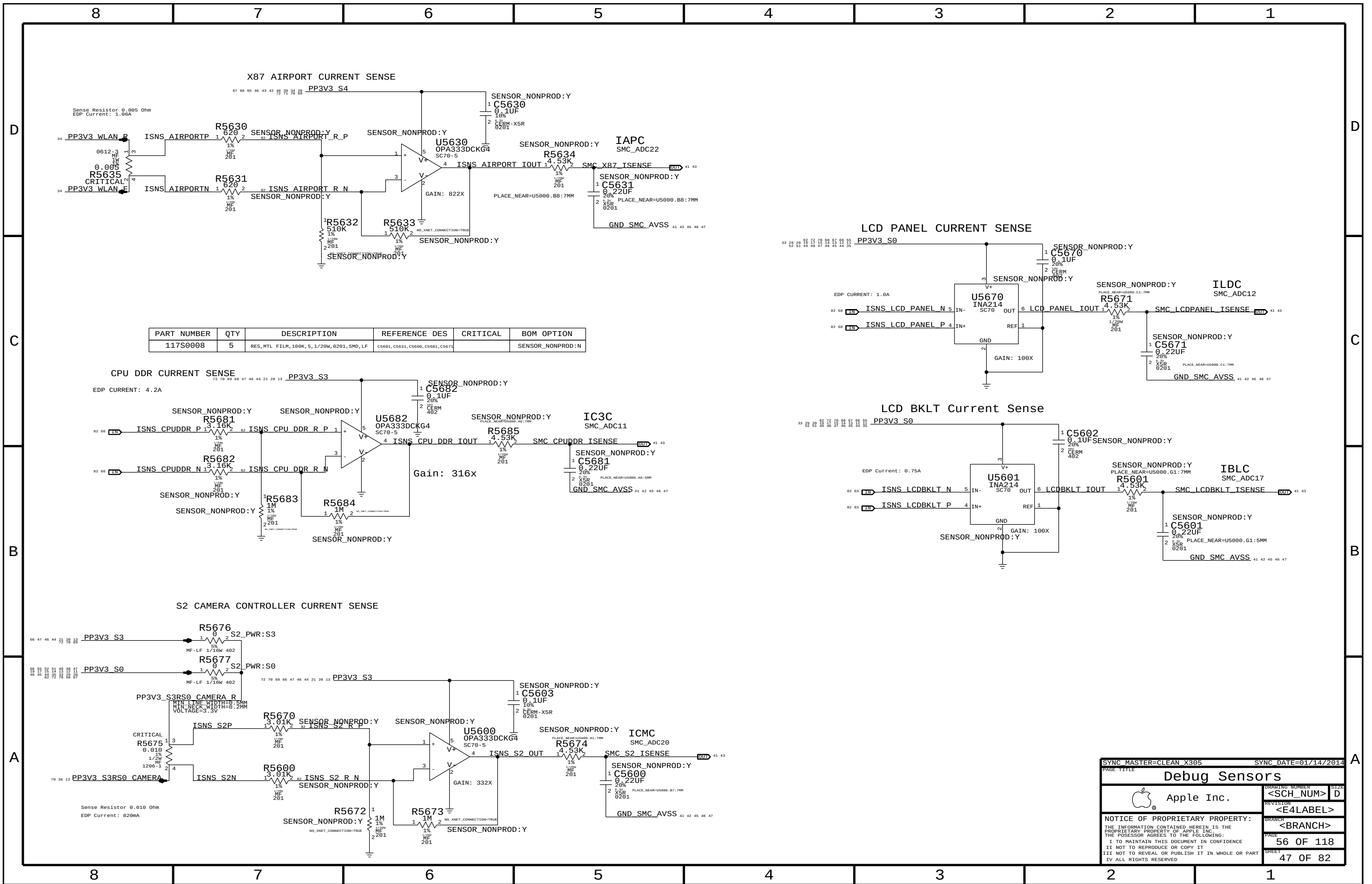


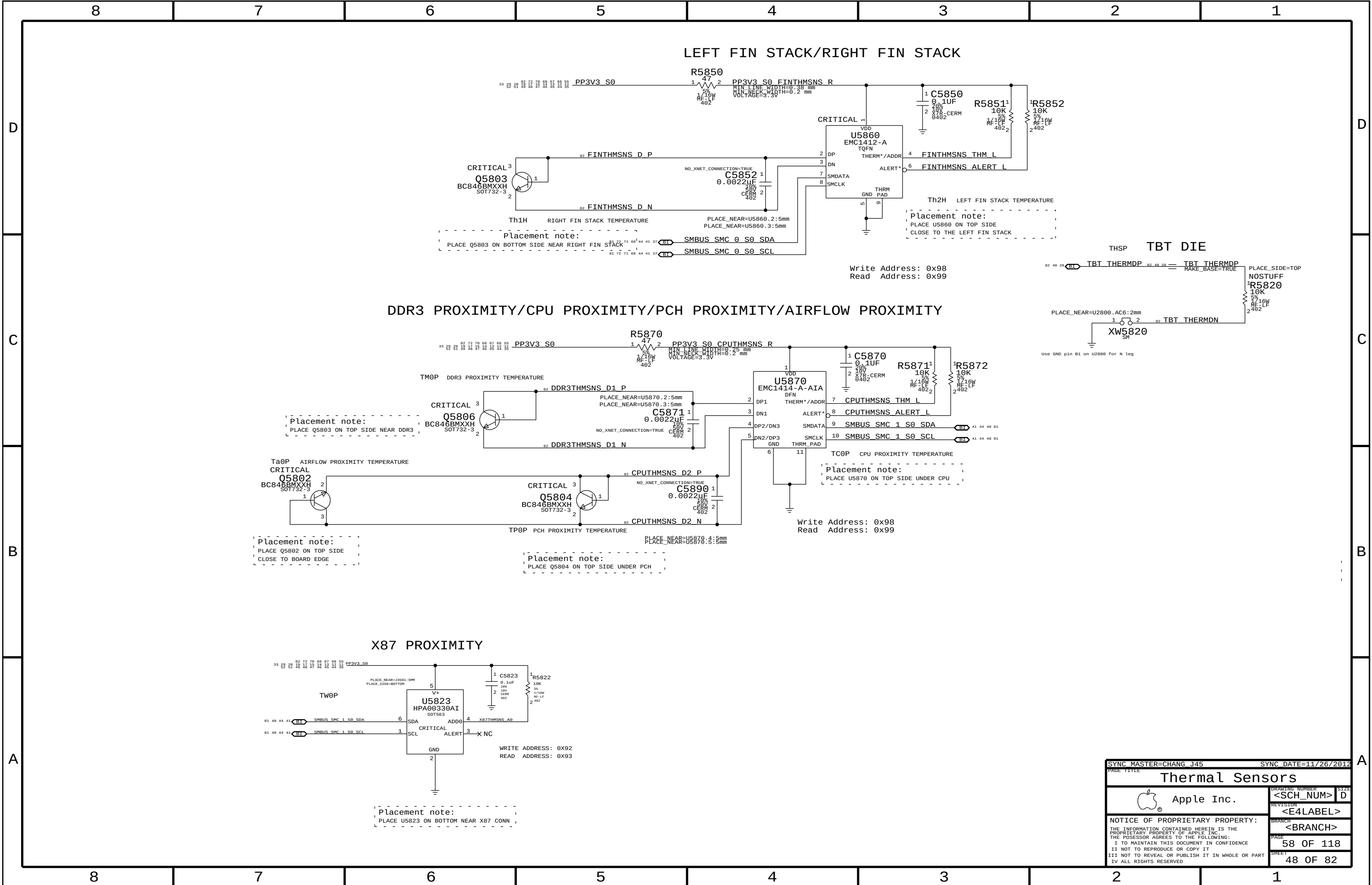
DC-IN (AMON) Current Sense Filter



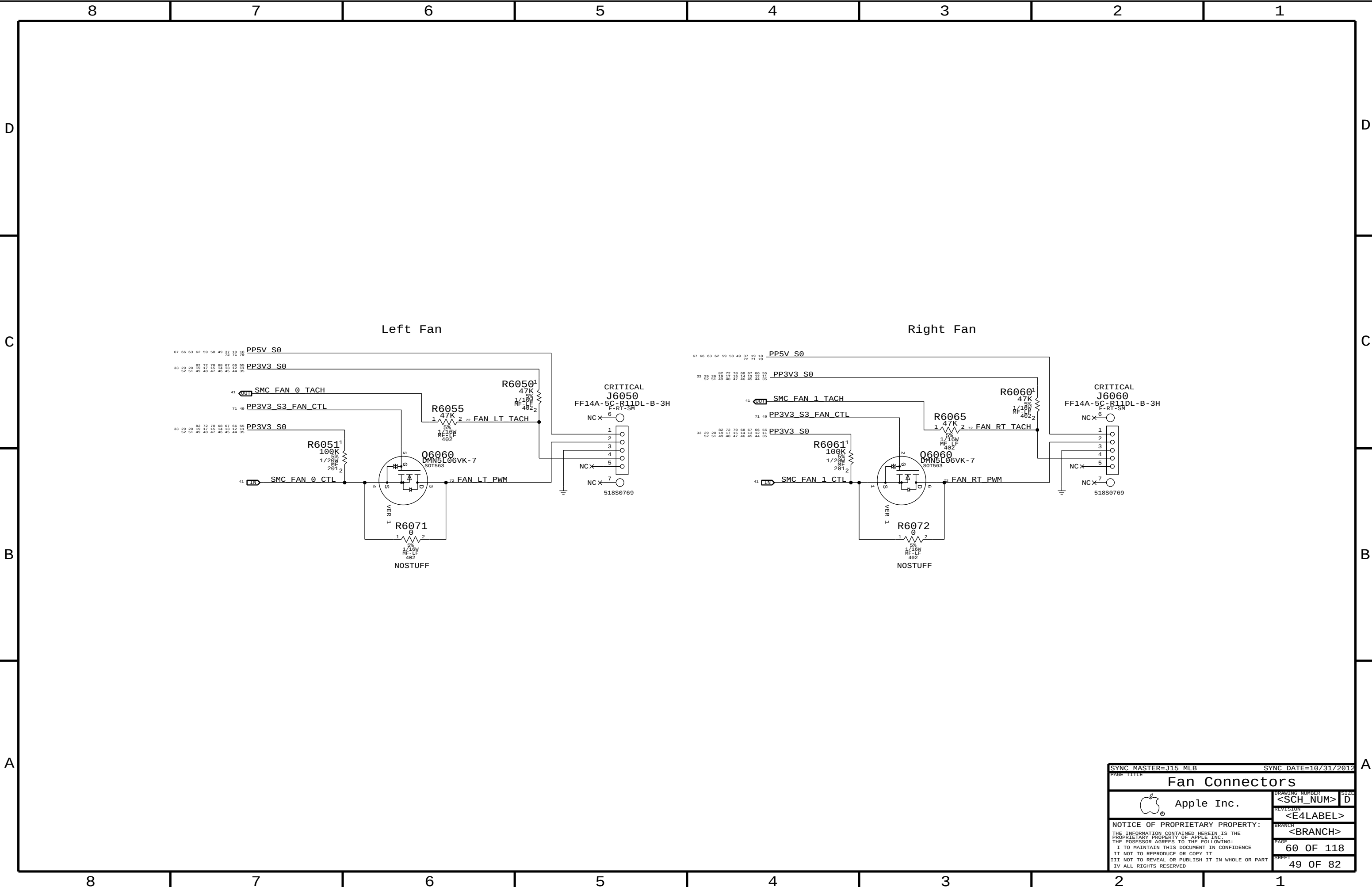
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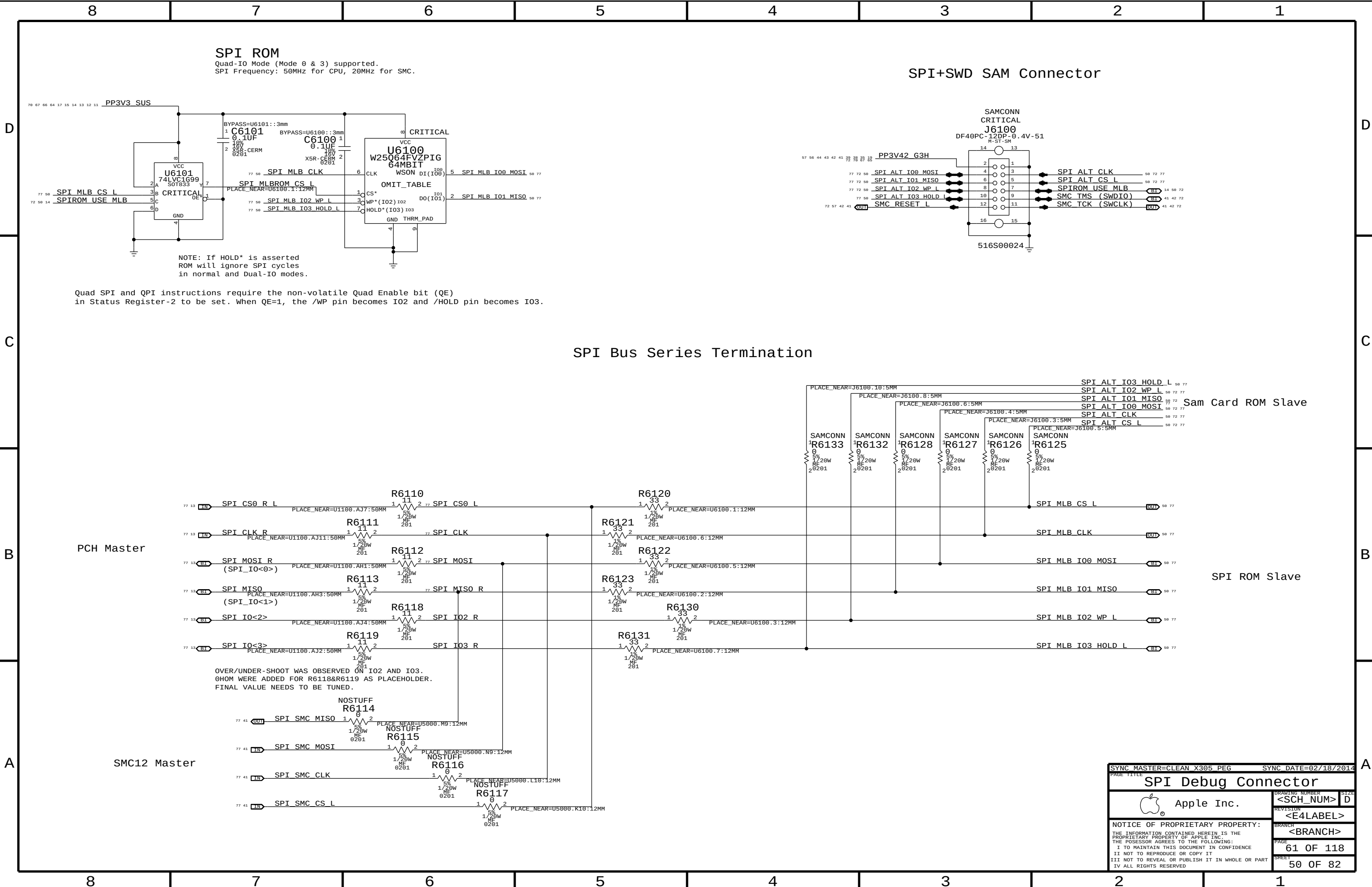




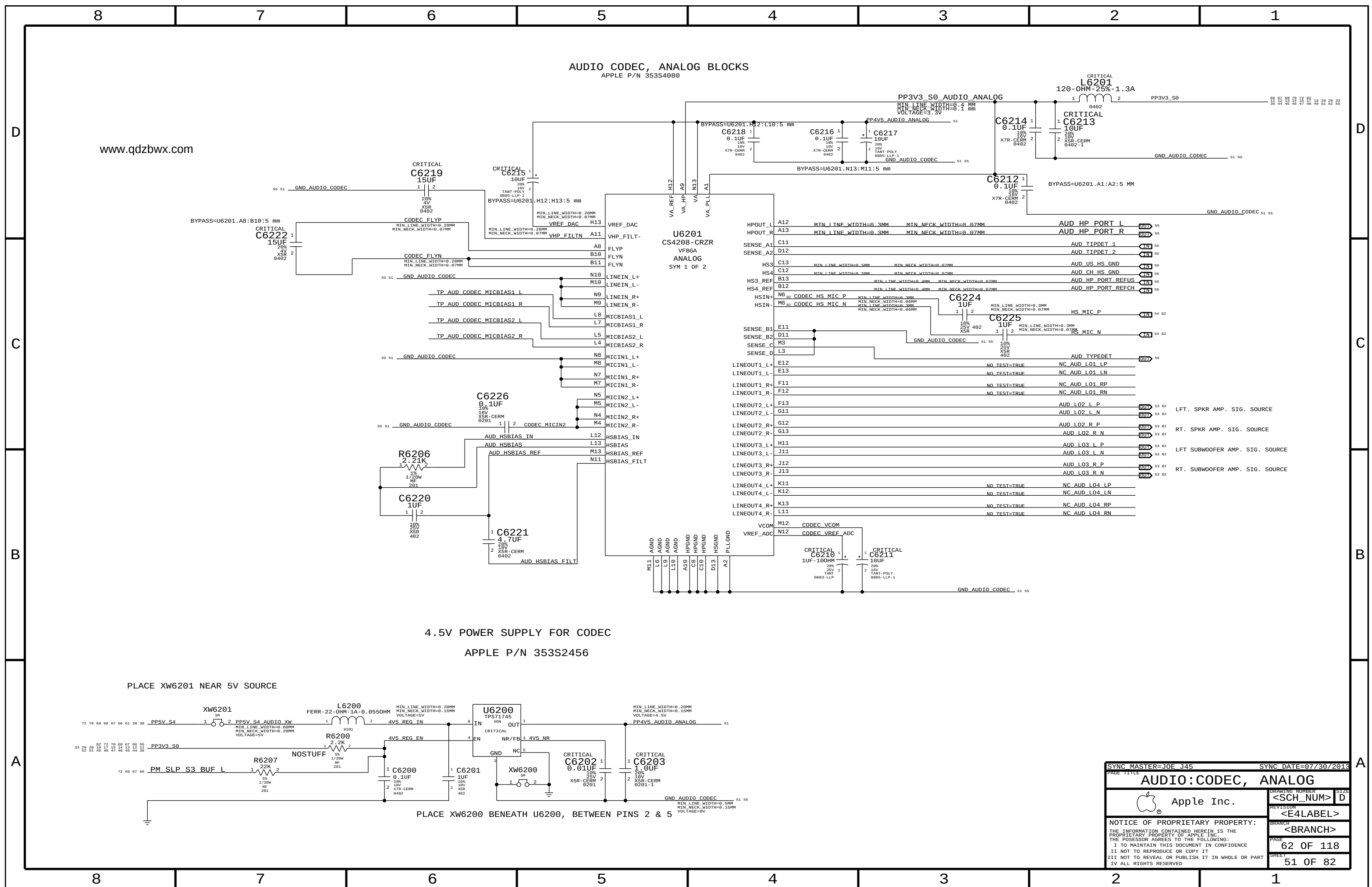


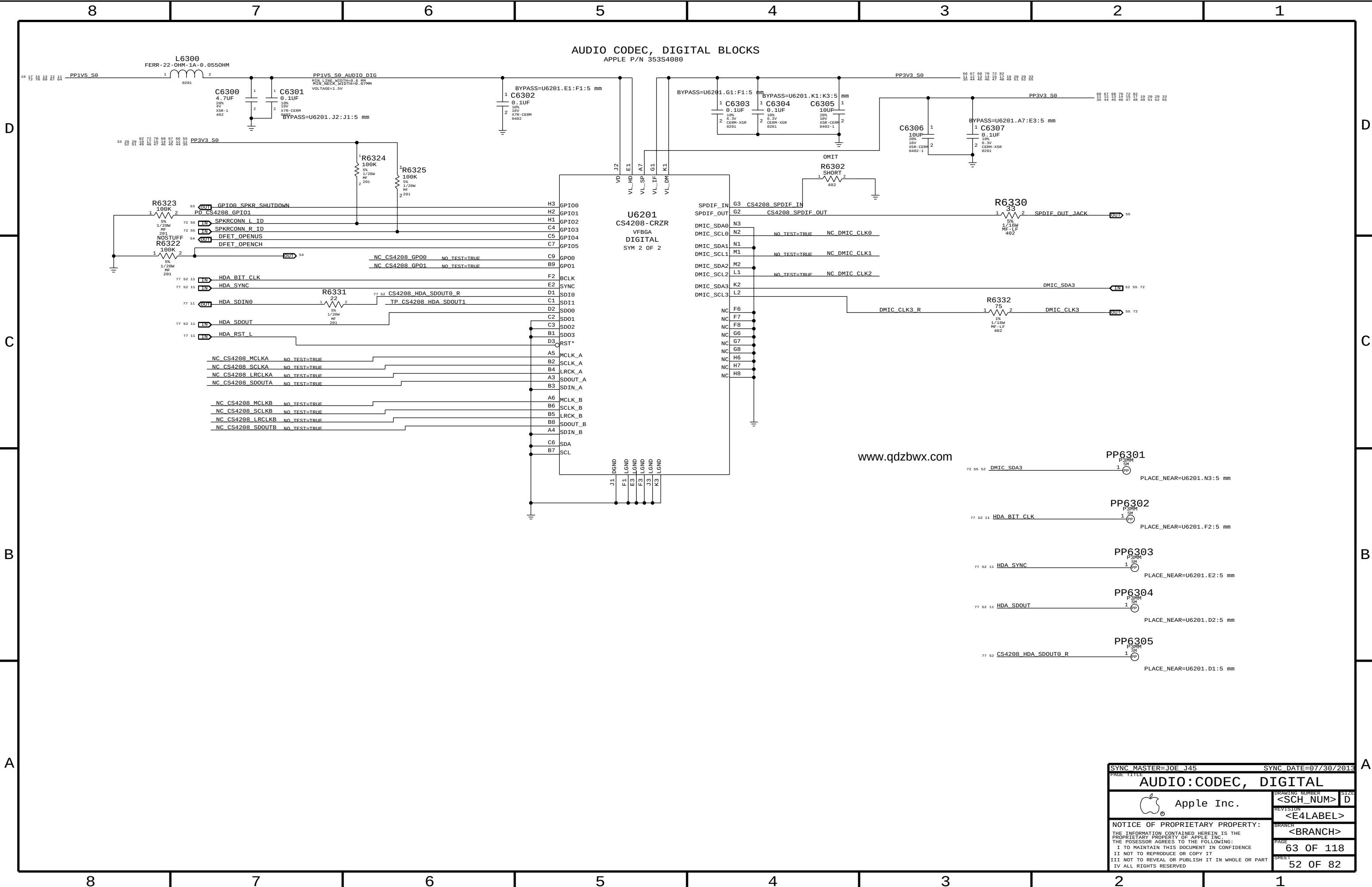
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Thermal Sensors			
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	58 OF 118		
	SHEET		
	48 OF 82		



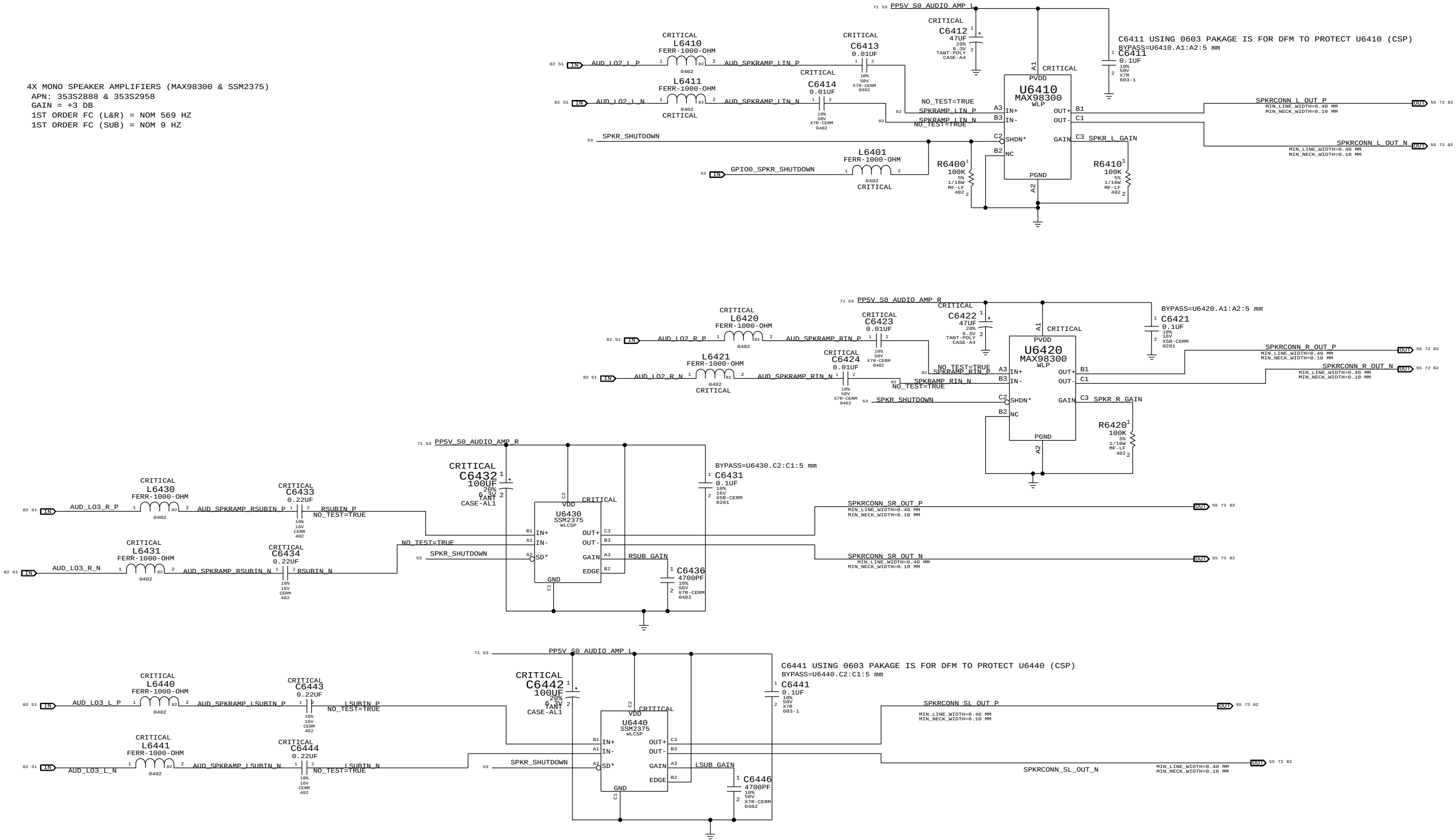


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SPI Debug Connector			
 Apple Inc.		DRAWING NUMBER	
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		SHEET	
		50 OF 82	

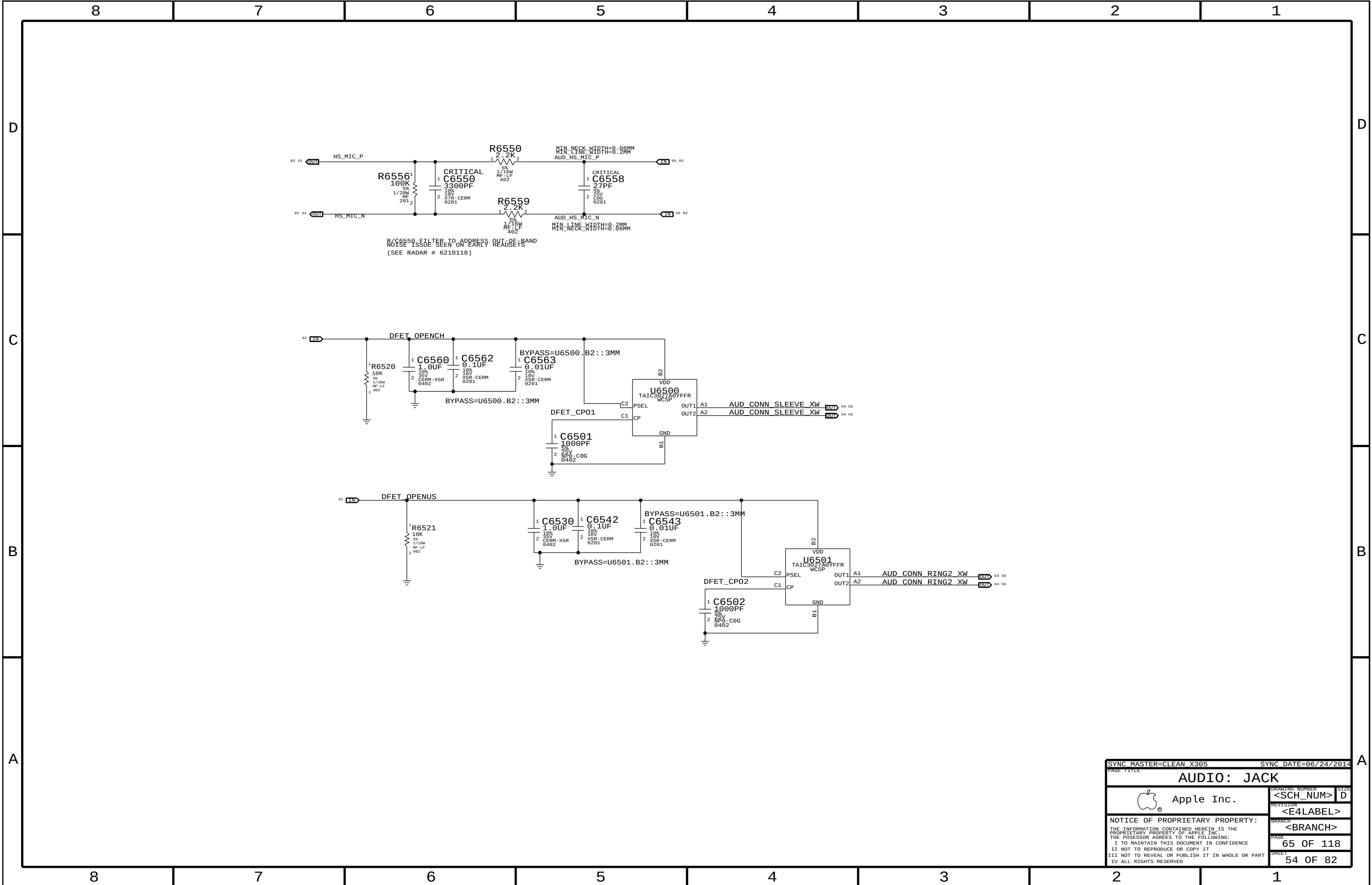




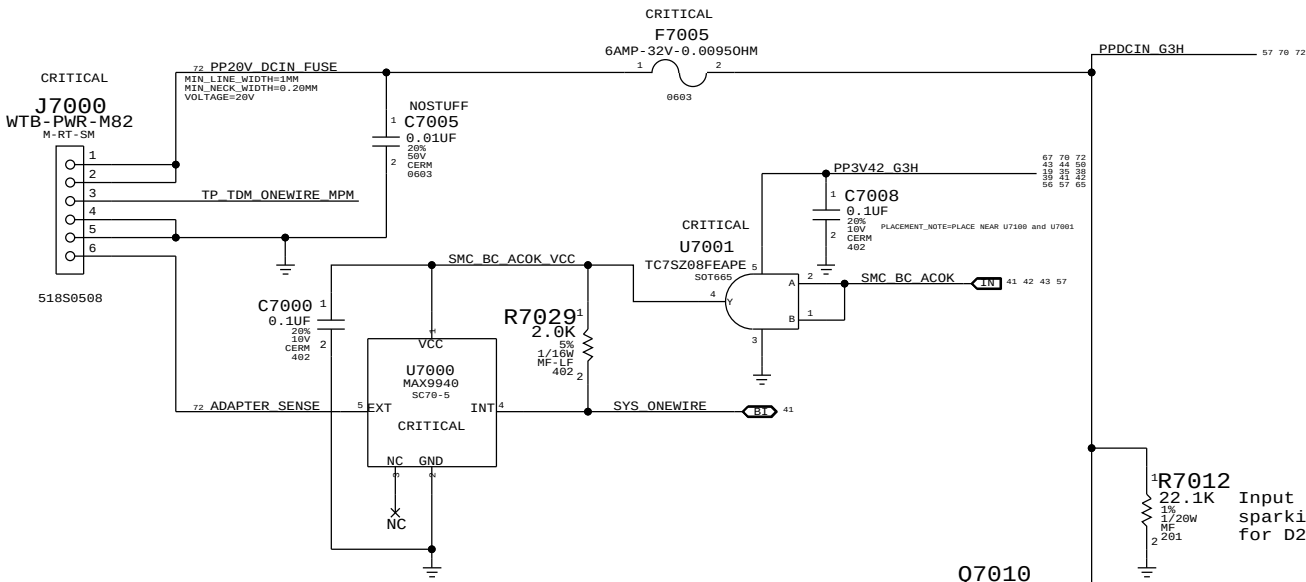
4X MONO SPEAKER AMPLIFIERS (MAX98300 & SSM2375)
APN: 353S2888 & 353S2958
GAIN = +3 DB
1ST ORDER FC (L&R) = NOM 569 HZ
1ST ORDER FC (SUB) = NOM 9 HZ



SYNC MASTER=JOE J45		SYNC DATE=07/30/2013	
PAGE TITLE			
AUDIO: SPEAKER AMP			
 Apple Inc.		DRAWING NUMBER	SIZE
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NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
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		SHEET	53 OF 82

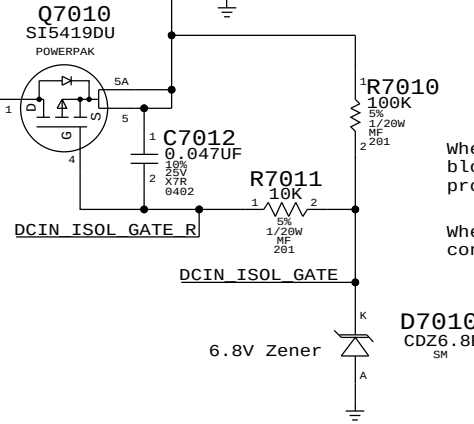


MagSafe DC Power Jack



1-Wire OverVoltage Protection

The chassis ground will otherwise float and can send transients onto ADAPTER_SENSE when AC is connected.



Input impedance of 22.1K meets sparkiterture requirements for D2 design only

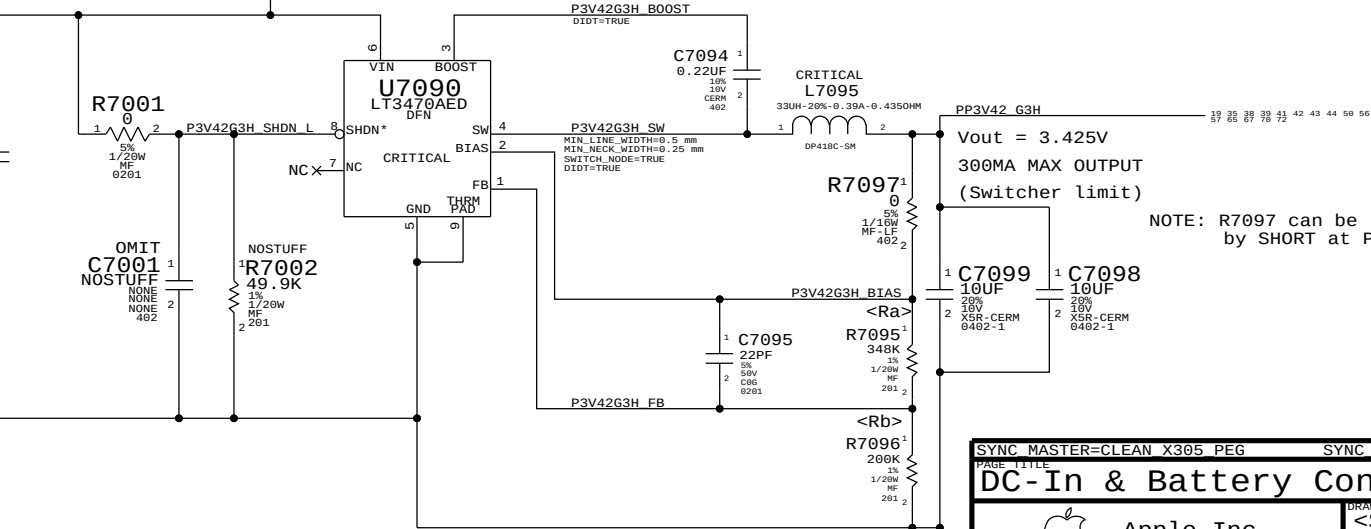
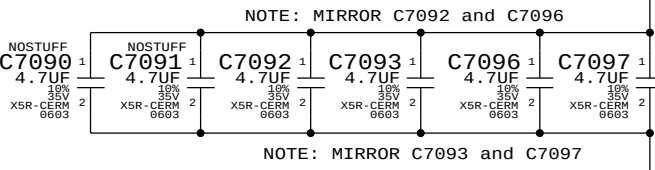
When input voltage is 2V the FET will be off blocking the leakage path and 22.1K can be properly detected.

When input voltage is at 16V+, FET will conduct and power charger and 3.42V reg

www.qdzbwx.com

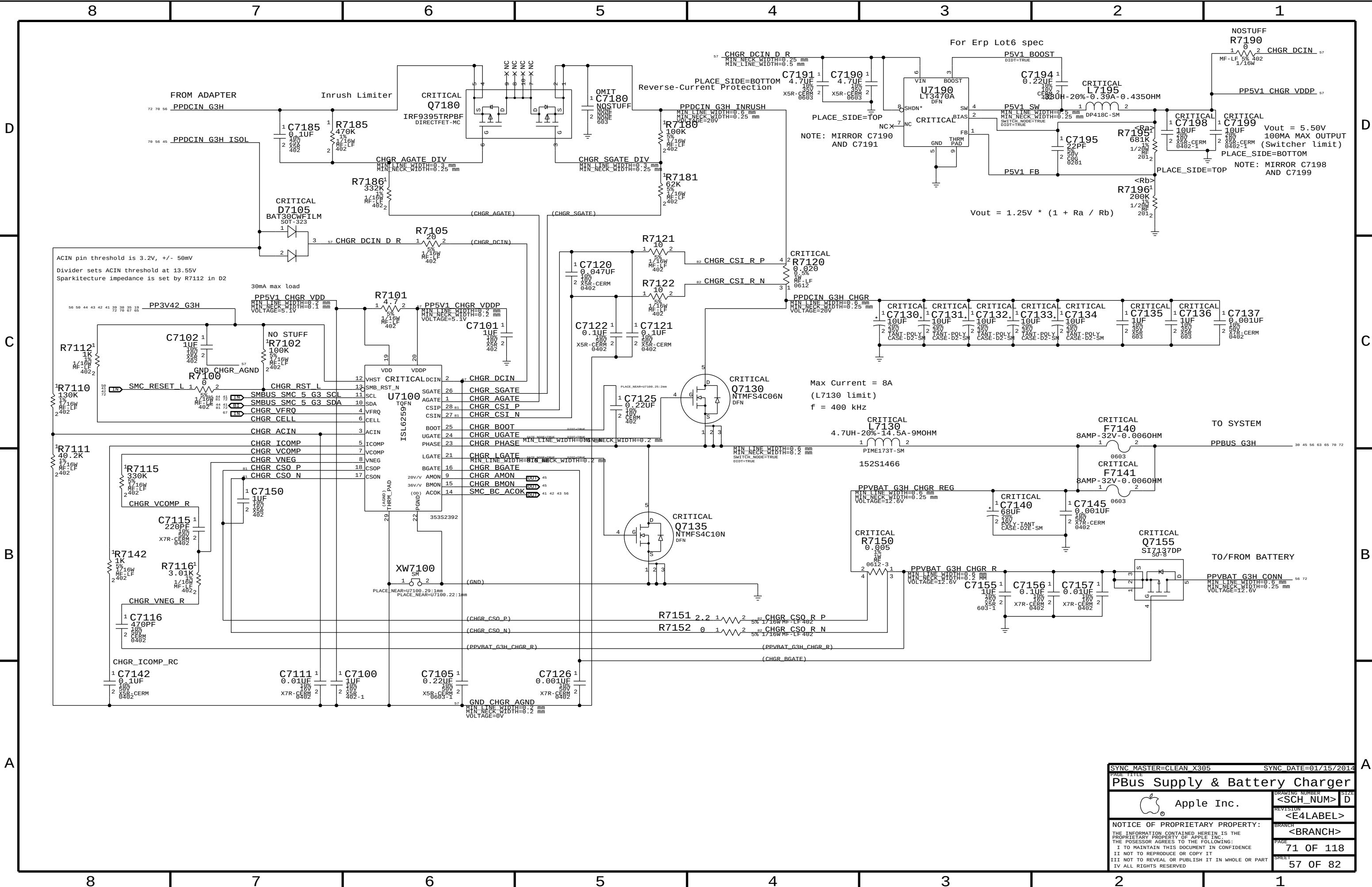
3.425V "G3Hot" Supply

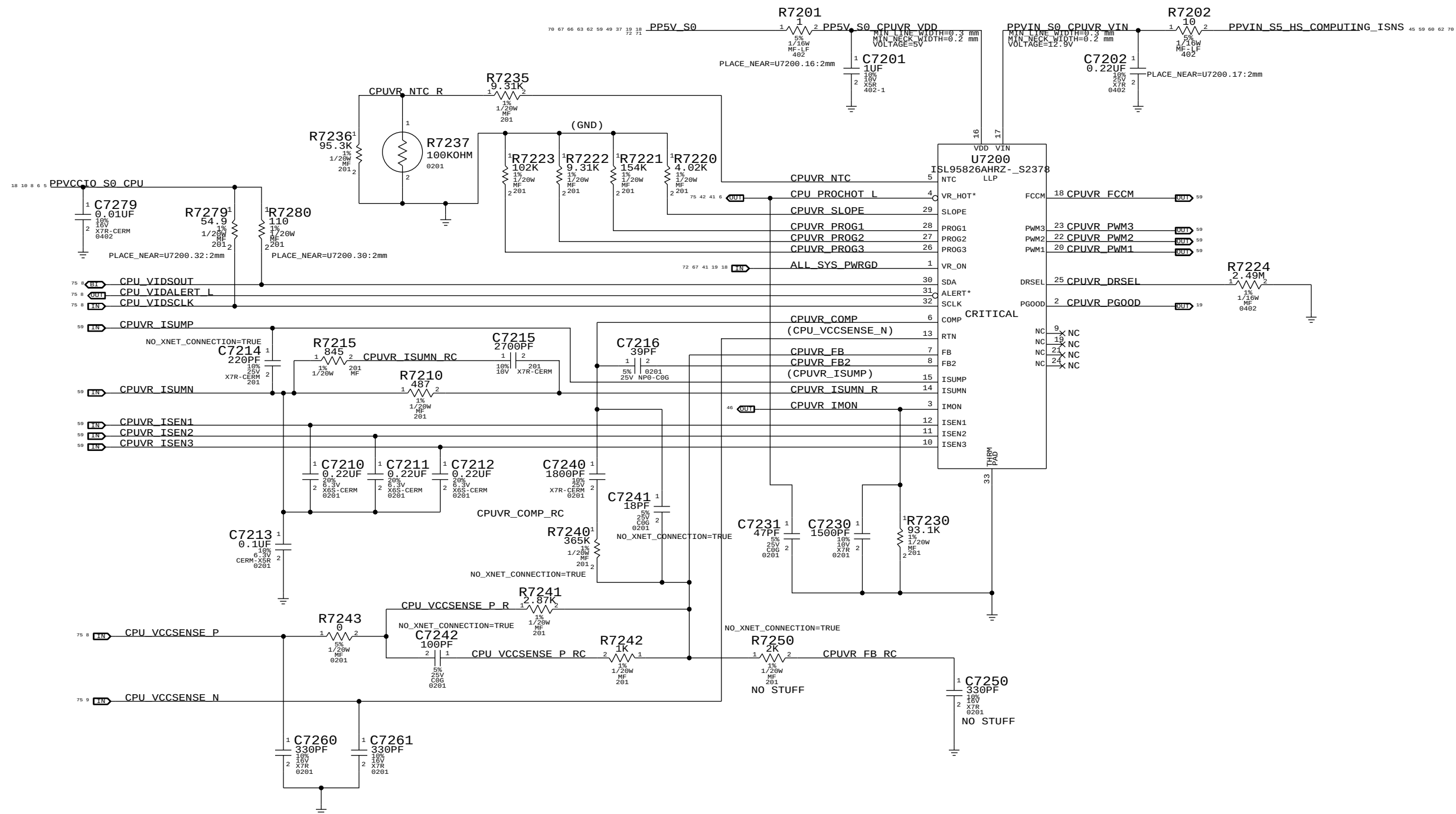
Supply needs to guarantee 3.31V delivered to SMC VRef generator



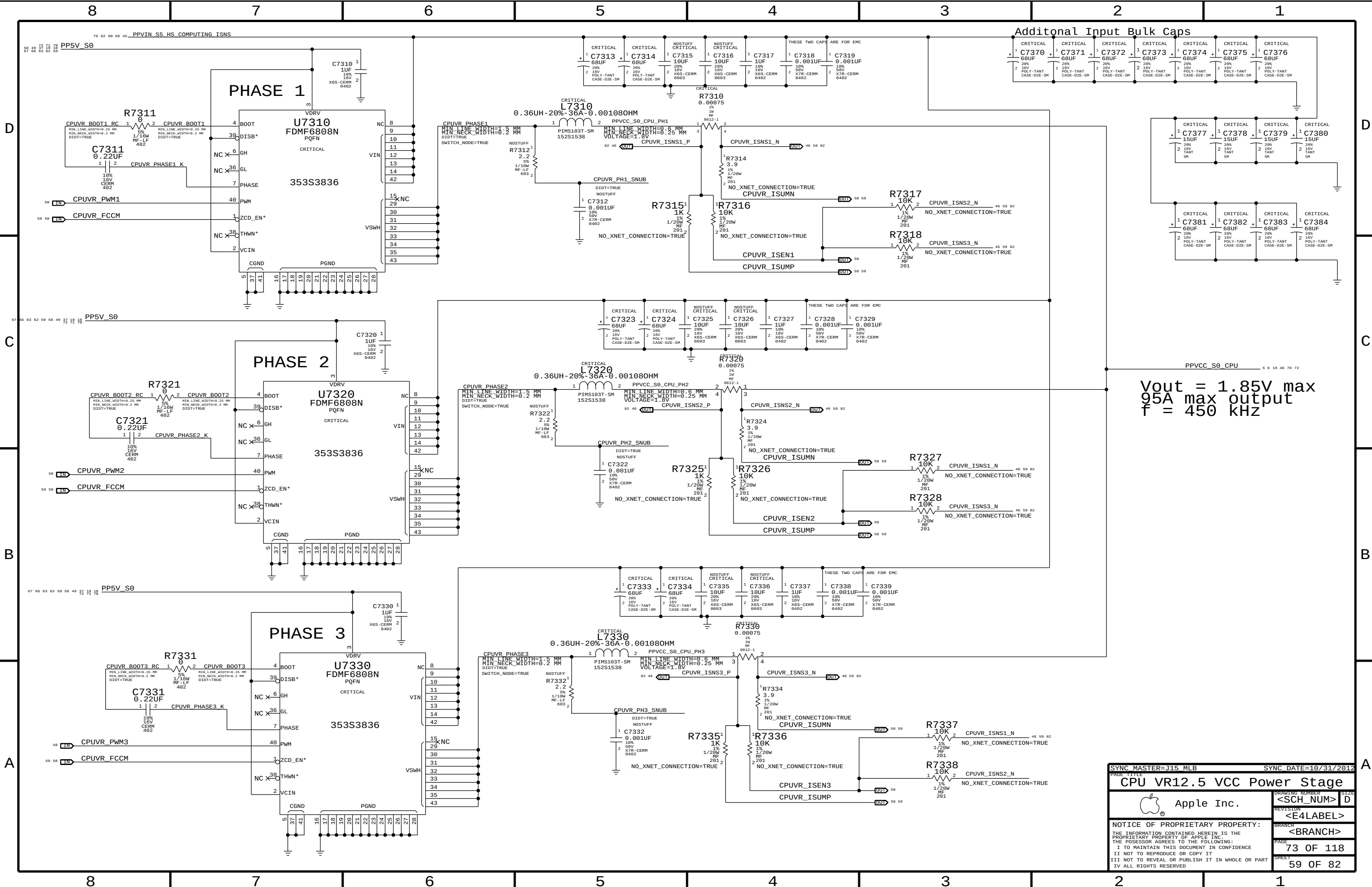
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DC-In & Battery Connectors		DRAWING NUMBER	
Apple Inc.		<SCH_NUM> D	
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		SHEET	
		56 OF 82	

$V_{out} = 1.25V * (1 + R_a / R_b)$





SYNC MASTER=CLEAN X305 PEG		SYNC DATE=02/24/2014	
PAGE TITLE			
CPU VR12.5 VCC Regulator IC			
 Apple Inc.		DRAWING NUMBER	SIZE
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		SHEET	58 OF 82



Vout = 1.85V max
95A max output
f = 450 kHz

SYNC MASTER=J15 MLB		SYNC DATE=10/31/2012	
PAGE TITLE		CPU VR12.5 VCC Power Stage	
		DRAWING NUMBER	SIZE
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		PAGE	73 OF 118
		SHEET	59 OF 82

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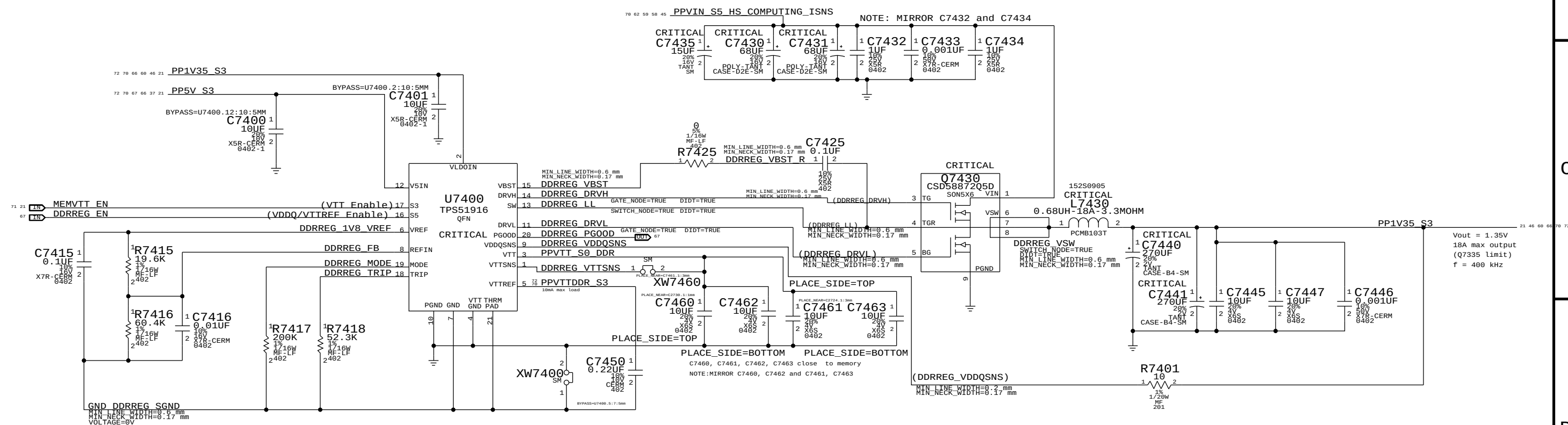
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DDR3L (1V35 S3) REGULATOR



SYNC MASTER=CLEAN X305		SYNC DATE=01/15/2014	
PAGE TITLE			
1.35V		DDR3L SUPPLY	
 Apple Inc.		DRAWING NUMBER	SIZE
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		<BRANCH>	
		PAGE	74 OF 118
		SHEET	60 OF 82

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NOTE: MIRROR C7540 and C7541

NOTE: MIRROR C7581 and C7586

NOTE: Change R7522 to XW7522 at PVT

NOTE: Change R7562 to XW7562 at PVT

SYNC MASTER=CLEAN X305 PEG		SYNC DATE=02/18/2014	
PAGE TITLE			
5V / 3.3V Power Supply		Supply	
 Apple Inc.		DRAWING NUMBER	SIZE
		<SCH_NUM>	D
		REVISION	
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[illegible]

1V05 S0 REGULATOR

PP1V05 S0

PP5V S0

PP1V05 S0 REG R2

PP1V05 S0

Vout = 1.05V
12A MAX OUTPUT
f = 300 kHz

Vout = 0.5V * (1 + Ra / Rb)

OCP = R7641 x 8.5uA / R7640
OCP = 14.4A

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1V05V POWER SUPPLY

76 OF 118

62 OF 82

1V05 S0 REGULATOR

PP1V05 S0

PP5V S0

PP1V05 S0 REG R2

PP1V05 S0

Vout = 1.05V
12A MAX OUTPUT
f = 300 kHz

Vout = 0.5V * (1 + Ra / Rb)

OCP = R7641 x 8.5uA / R7640
OCP = 14.4A

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1V05V POWER SUPPLY

76 OF 118

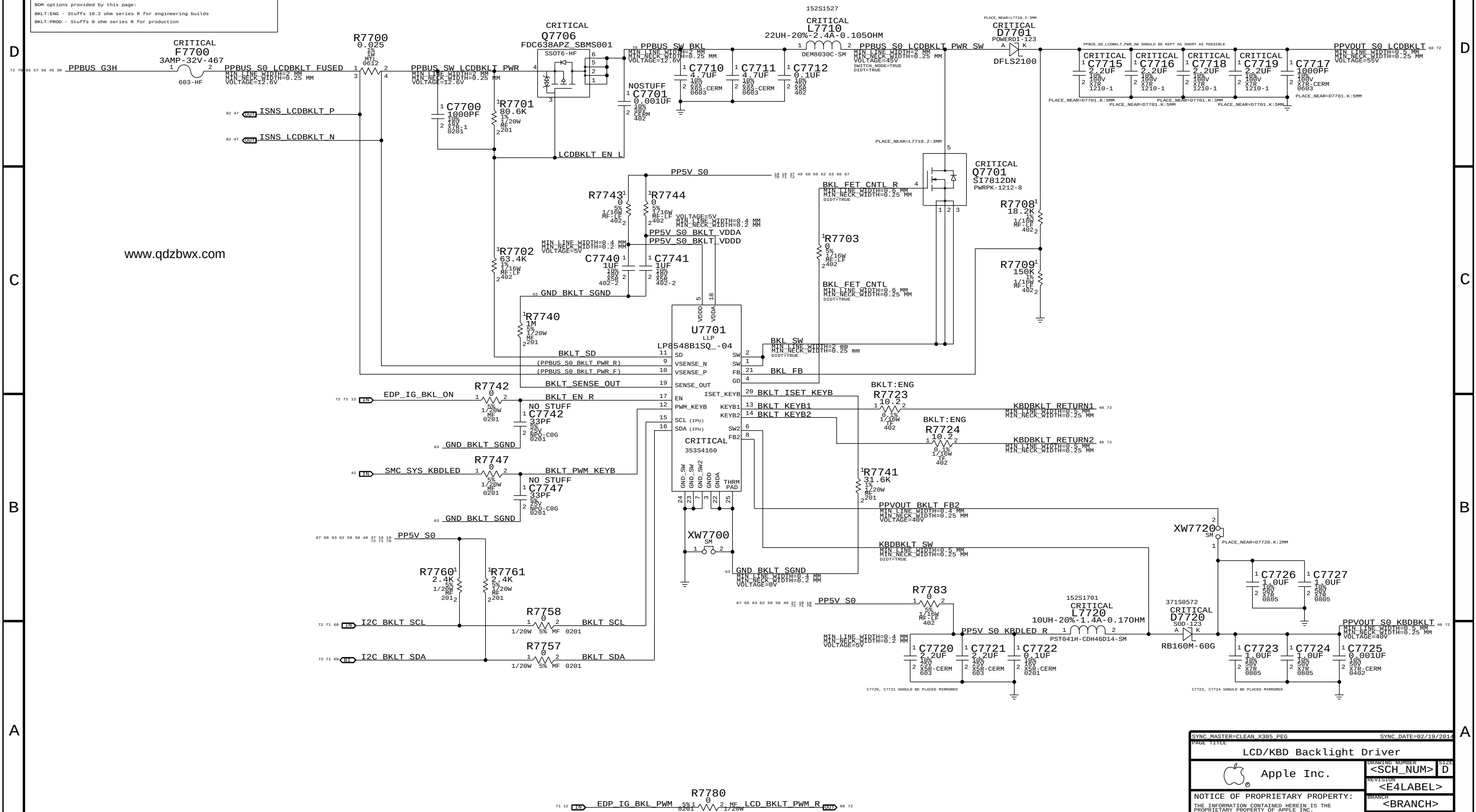
62 OF 82

Page Notes

Power aliases required by this page:
- =PPVIN_S0_LCDBKLT (9-12.6V LCD Backlight Input)
- =PPSV_S0_BKLTCTRL (5V Backlight Driver Input)
- =PPSV_S0_KBDLED (5V Keyboard Backlight Input)

BOM options provided by this page:
BKLT:ENG - Stuffs 10.2 ohm series R for engineering builds
BKLT:PROD - Stuffs 0 ohm series R for production

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
116S0004	2	RES,MTL FILM,0 OHM,1A MAX,0402,SMD	R7723,R7724		BKLT:PROD



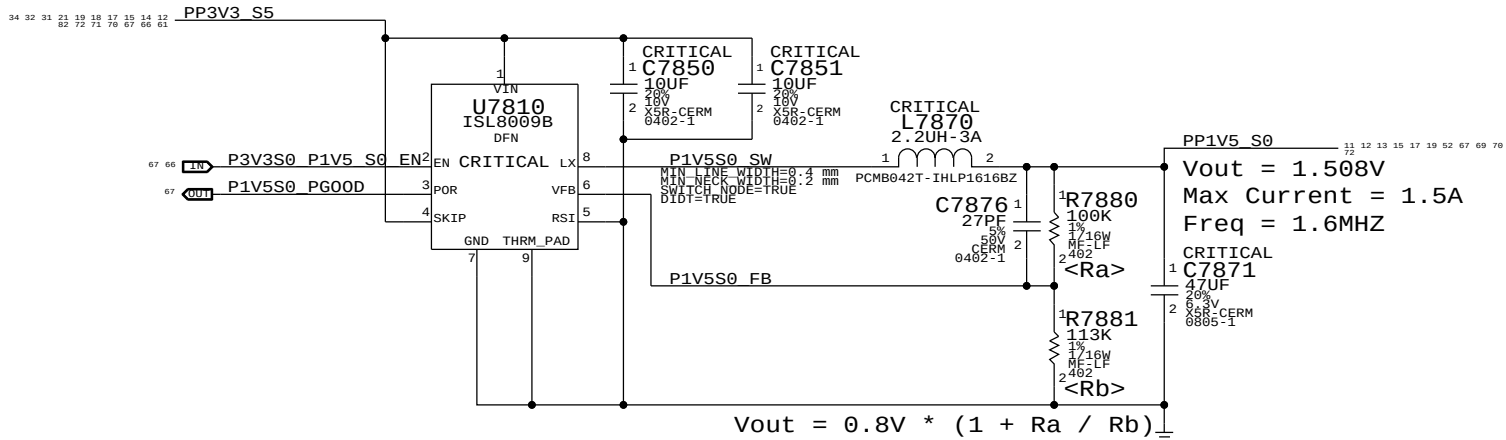
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LCD/KBD Backlight Driver			
 Apple Inc.		DRAWING NUMBER	SIZE
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		REVISION	
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		PAGE	77 OF 118
		SHEET	63 OF 82
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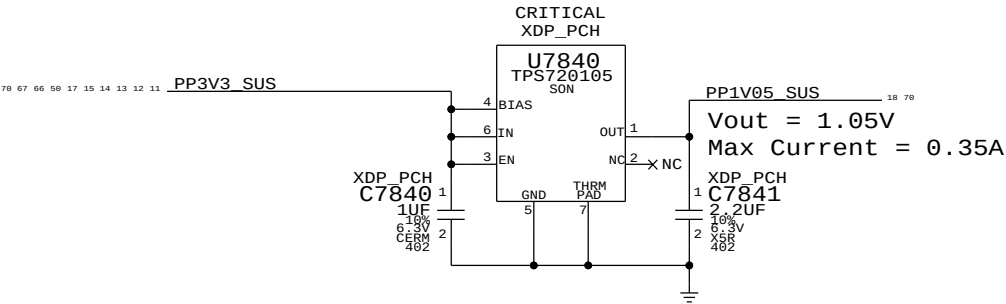
www.qdzbx.com

1.5V S0 Regulator



1.05V SUS LD0

Lynx Point-H requires JTAG pull-ups to be powered at 1.05V in SUS.
Pull-ups (3) must be 51 ohms to support XDP (not required in production).
70mA is required to support pull-ups. Alternative is strong voltage
dividers (200/100) to 3.3V SUS, which burns 100mW in all S-states.

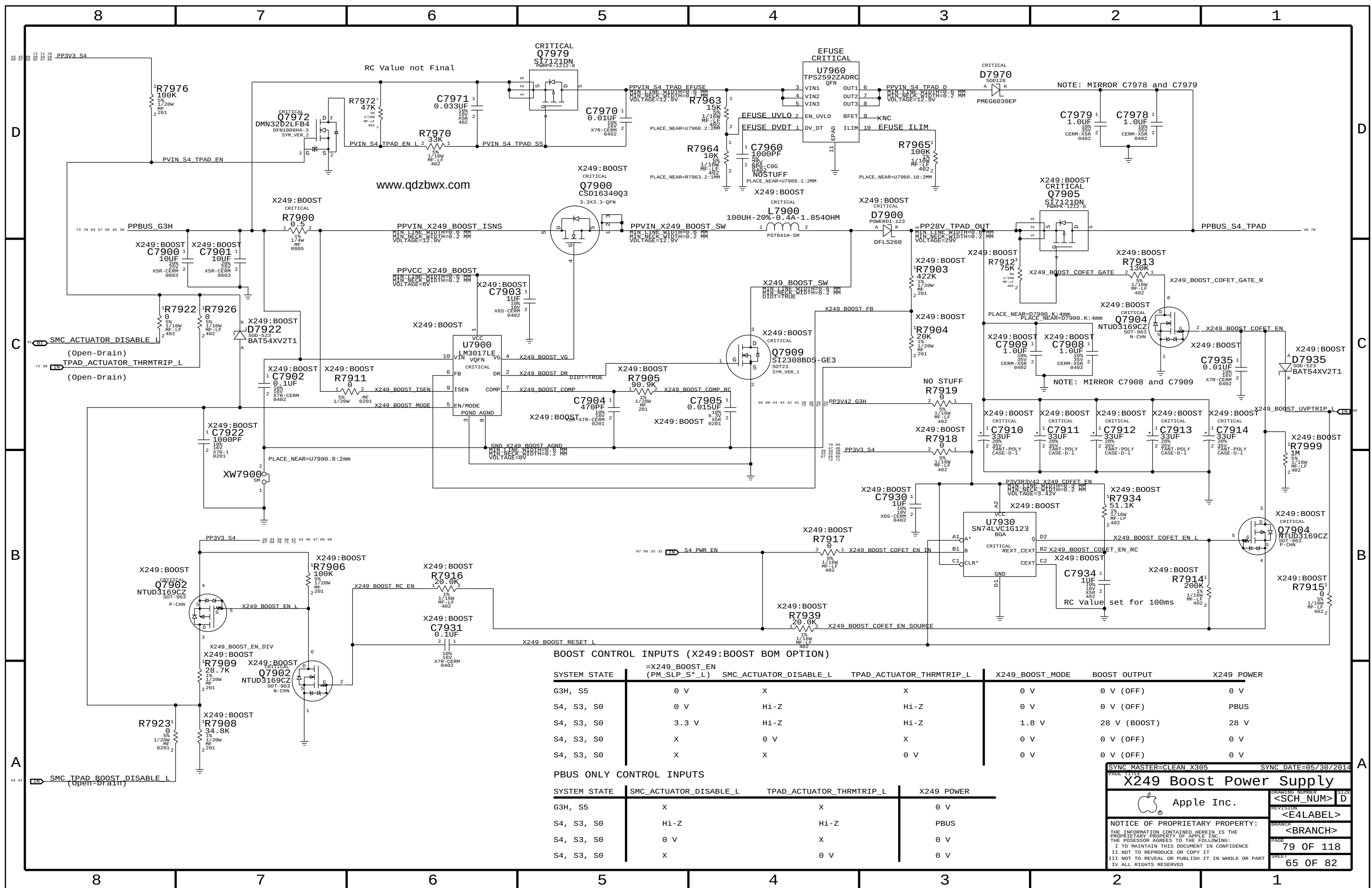


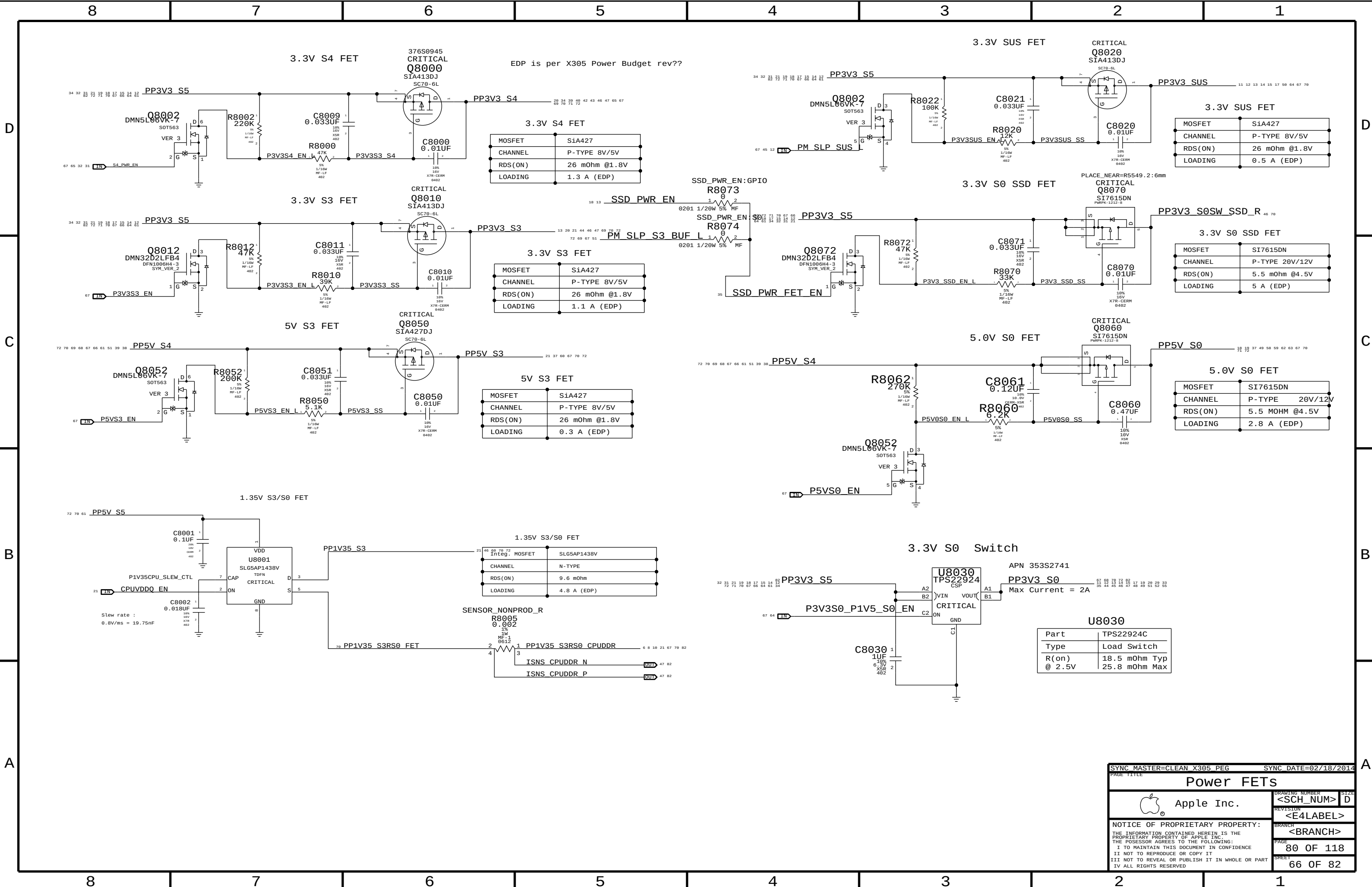
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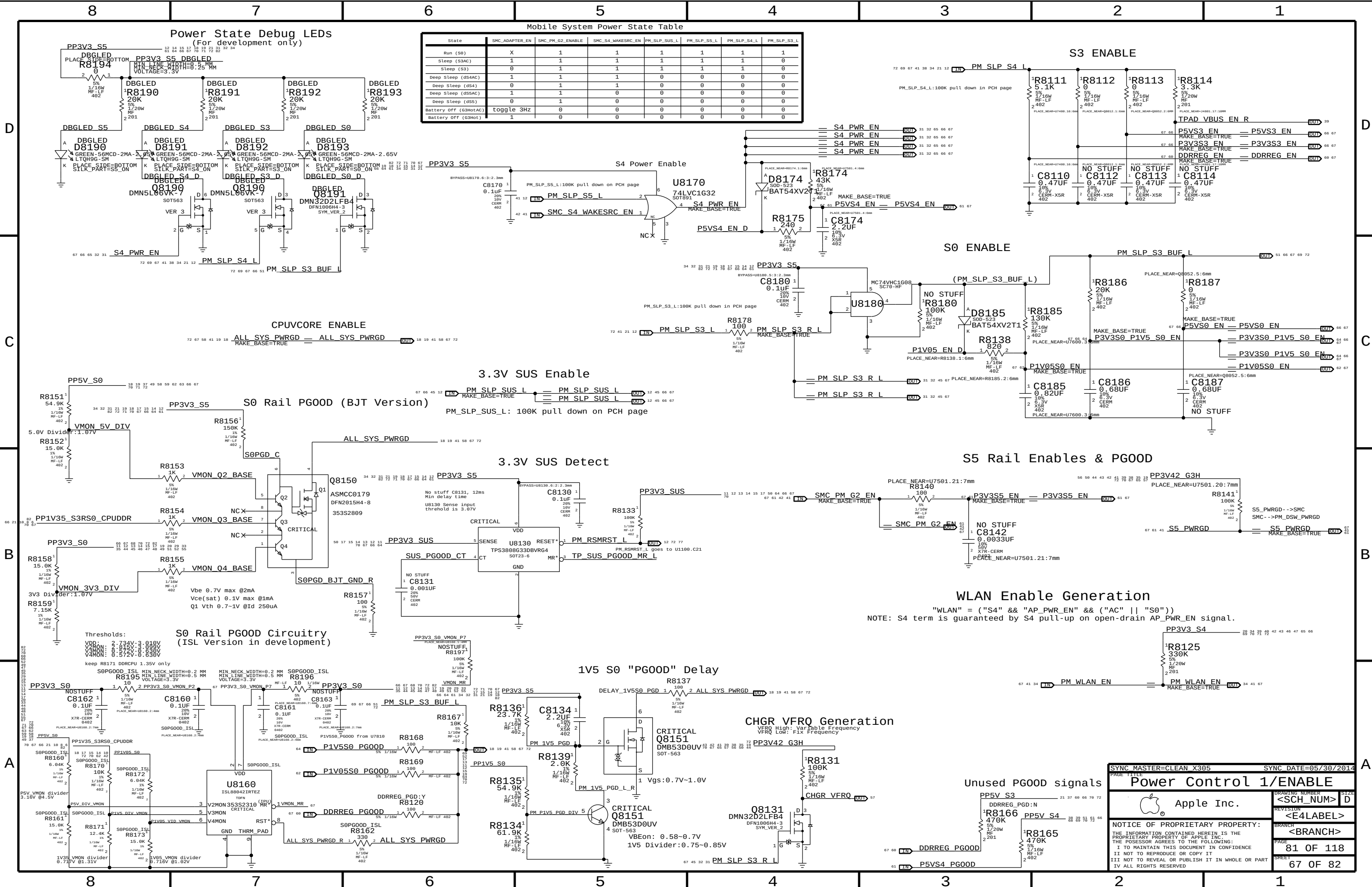
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A







Mobile System Power State Table							
State	SMC_ADAPTER_EN	SMC_PM_G2_ENABLE	SMC_S4_WAKESRC_EN	PM_SLP_SUS_L	PM_SLP_S5_L	PM_SLP_S4_L	PM_SLP_S3_L
Run (S0)	X	1	1	1	1	1	1
Sleep (S3AC)	1	1	1	1	1	1	0
Sleep (S3)	0	1	1	1	1	1	0
Deep Sleep (ds4AC)	1	1	1	0	0	0	0
Deep Sleep (ds4C)	0	1	1	0	0	0	0
Deep Sleep (ds5AC)	1	1	0	0	0	0	0
Deep Sleep (ds5C)	0	1	0	0	0	0	0
Battery Off (G3HotC)	toggle 3Hz	0	0	0	0	0	0
Battery Off (G3Hot)	1	0	0	0	0	0	0

SYNC MASTER=CLEAN X305

SYNC DATE=05/30/2014

Power Control 1/ENABLE

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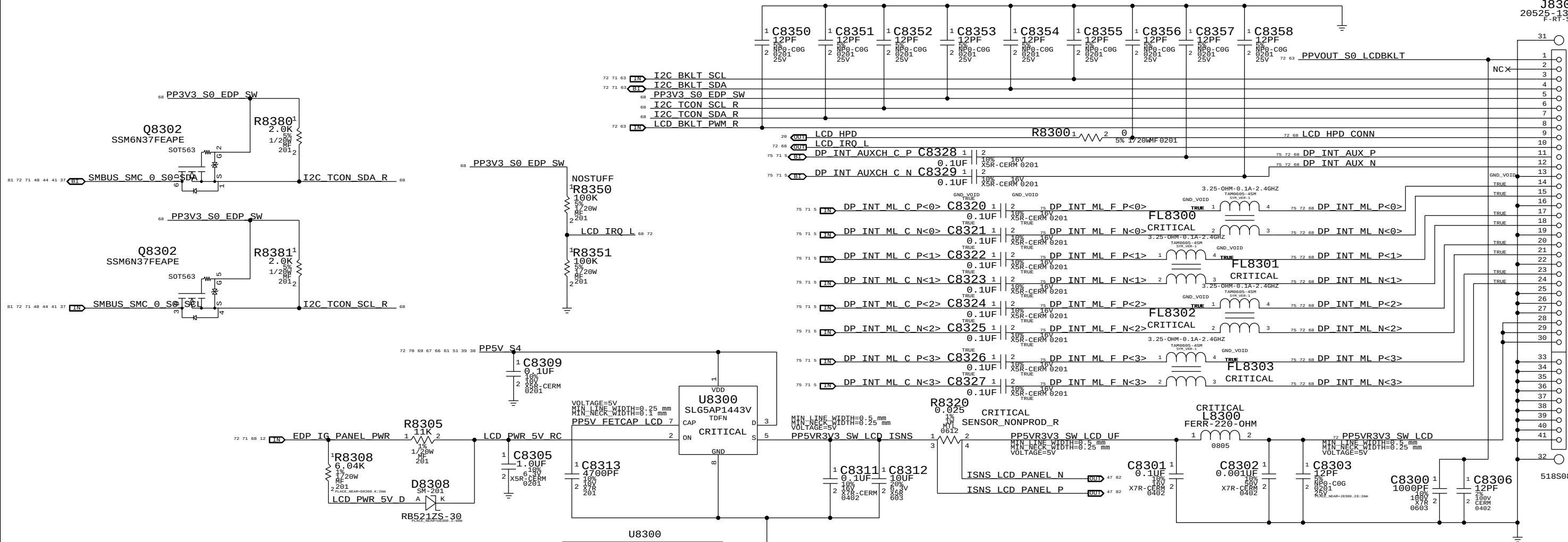
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PAGE
81 OF 118

SHEET
67 OF 82

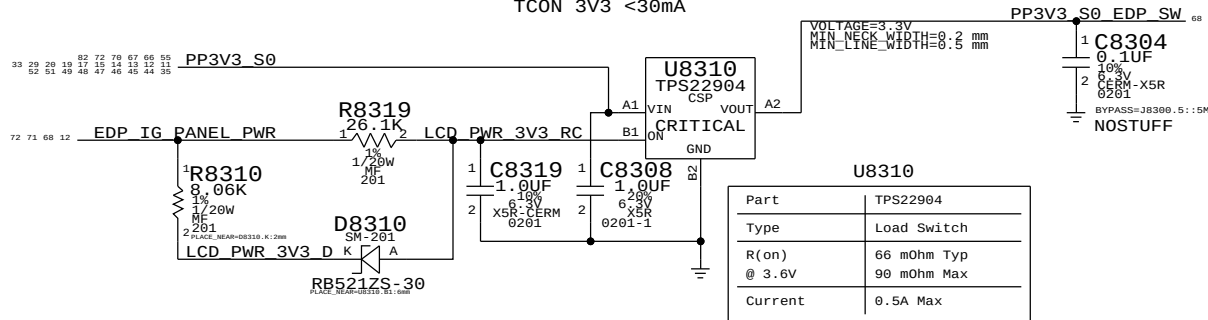
LCD PANEL INTERFACE (eDP)

CRITICAL
J8300
20525-130E-01
F-RT-SM

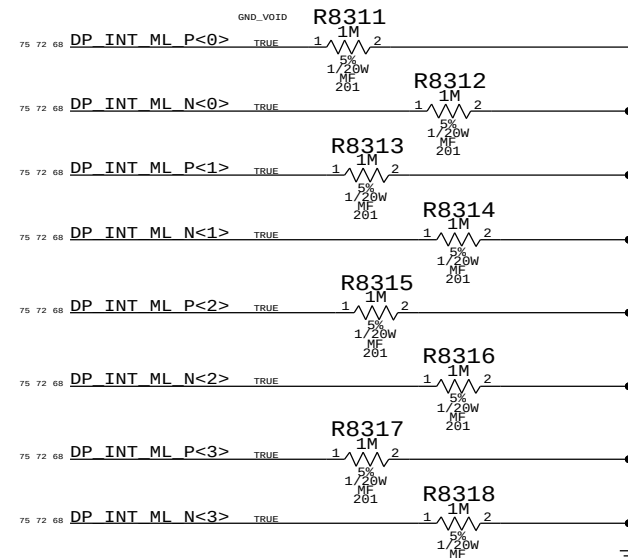


Part	SLG5AP1443V
Type	Load Switch
R(on) @100mA	17 mOhm Typ 19 mOhm Max
Current	2.5 A Max

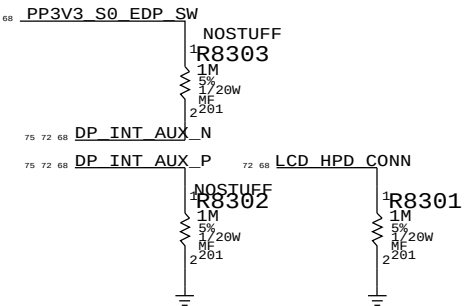
3.3V TCON Switch
TCON 3V3 <30mA



Part	TPS22904
Type	Load Switch
R(on) @ 3.6V	66 mOhm Typ 90 mOhm Max
Current	0.5A Max

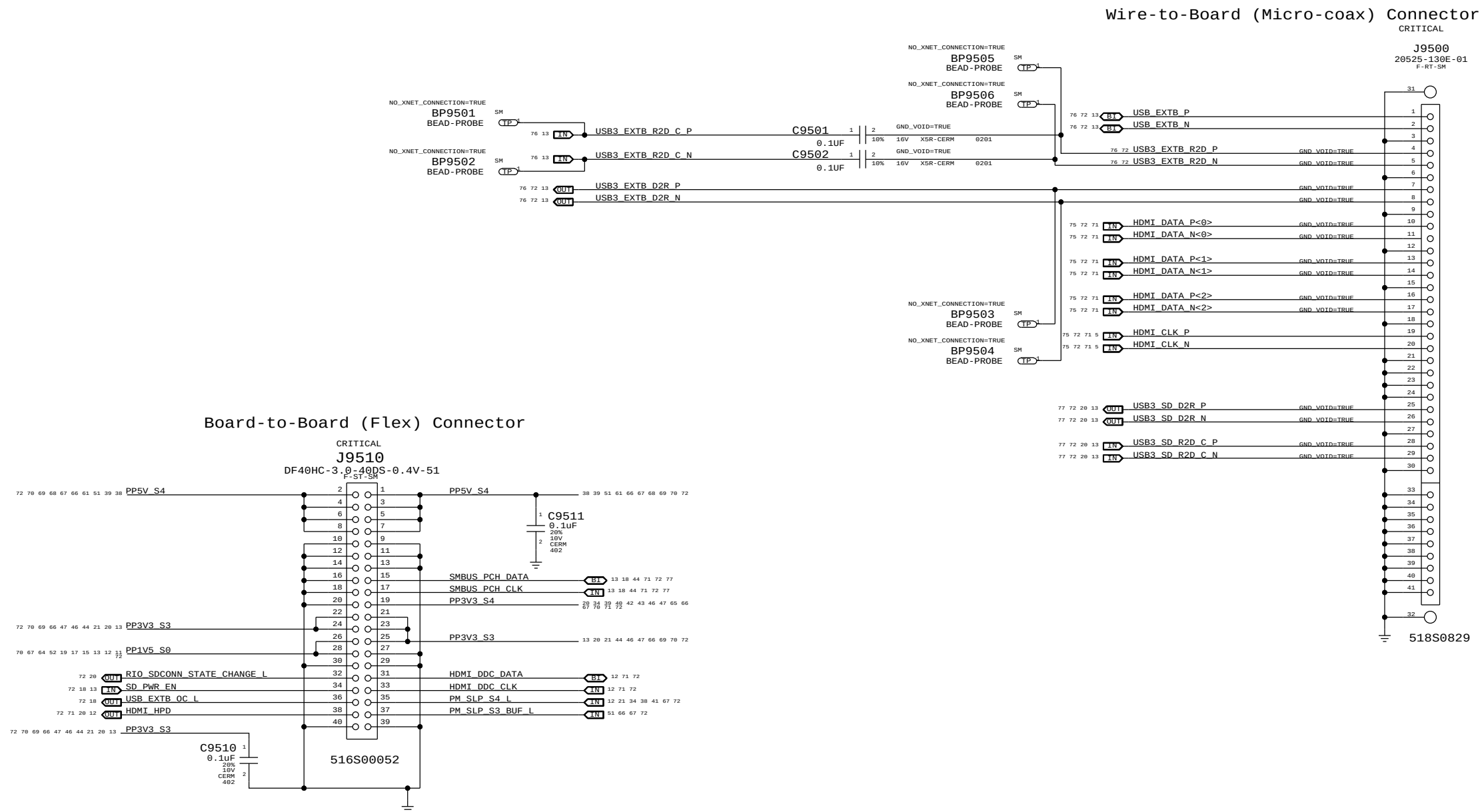


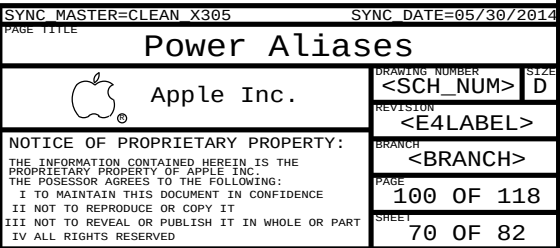
LCD Panel HPD & AUX strapping



SYNC MASTER=CLEAN X305.PEG		SYNC DATE=02/18/2014	
PAGE TITLE		DRAWING NUMBER	
eDP Display Connector		<SCH_NUM>	
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		<E4LABEL>	
		PAGE	
		83 OF 118	
		SHEET	
		68 OF 82	

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Display Aliases

72 71 68 12	EDP IG PANEL PWR	==	EDP IG PANEL PWR	12 68 71 72
72 71 63 12	EDP IG BKL ON	==	EDP IG BKL ON	12 63 71 72
71 63 12	EDP IG BKL PWM	==	EDP IG BKL PWM	12 63 71
75 68 5	DP INT ML C P<3..0>	==	TP DP IG A MLP<3..0>	
75 68 5	DP INT ML C N<3..0>	==	TP DP IG A MLN<3..0>	
75 71 68 5	DP INT AUXCH C P	==	DP INT AUXCH C P	5 68 71 75
75 71 68 5	DP INT AUXCH C N	==	DP INT AUXCH C N	5 68 71 75
71 28 12	DP TBTSNK0 HPD	==	DP TBTSNK0 HPD	12 28 71
75 28 5	DP TBTSNK0 ML C P<3..0>	==	TP DP IG B MLP<3..0>	
75 28 5	DP TBTSNK0 ML C N<3..0>	==	TP DP IG B MLN<3..0>	
75 71 28 12	DP TBTSNK0 AUXCH C P	==	DP TBTSNK0 AUXCH C P	12 28 71 75
75 71 28 12	DP TBTSNK0 AUXCH C N	==	DP TBTSNK0 AUXCH C N	12 28 71 75
71 33 12	DP TBTSNK0 DDC DATA	==	DP TBTSNK0 DDC DATA	12 33 71
71 33 12	DP TBTSNK0 DDC CLK	==	DP TBTSNK0 DDC CLK	12 33 71
71 28 12	DP TBTSNK1 HPD	==	DP TBTSNK1 HPD	12 28 71
75 28 5	DP TBTSNK1 ML C P<3..0>	==	TP DP IG C MLP<3..0>	
75 28 5	DP TBTSNK1 ML C N<3..0>	==	TP DP IG C MLN<3..0>	
75 71 28 12	DP TBTSNK1 AUXCH C P	==	DP TBTSNK1 AUXCH C P	12 28 71 75
75 71 28 12	DP TBTSNK1 AUXCH C N	==	DP TBTSNK1 AUXCH C N	12 28 71 75
71 33 12	DP TBTSNK1 DDC DATA	==	DP TBTSNK1 DDC DATA	12 33 71
71 33 12	DP TBTSNK1 DDC CLK	==	DP TBTSNK1 DDC CLK	12 33 71
72 71 69 20 12	HDMI HPD	==	HDMI HPD	12 20 69 71 72
75 72 69	HDMI DATA P<0..2>	==	TP DP IG D MLP<2..0>	5
75 72 69	HDMI DATA N<0..2>	==	TP DP IG D MLN<2..0>	5
75 72 71 69 5	HDMI CLK P	==	HDMI CLK P	5 69 71 72 75
75 72 71 69 5	HDMI CLK N	==	HDMI CLK N	5 69 71 72 75
72 71 69 12	HDMI DDC CLK	==	HDMI DDC CLK	12 69 71 72
72 71 69 12	HDMI DDC DATA	==	HDMI DDC DATA	12 69 71 72

PP3V3 S4

Unused TPAD SPI

72 71 39	TPAD SPI BUS EN	==	TPAD SPI BUS EN	71 39
77 72 71 39	TPAD SPI CS L	==	TPAD SPI CS L	TRUE
72 71 39	TPAD SPI INT L	==	TPAD SPI INT L	TRUE
77 72 71 39	TPAD SPI MISO	==	TPAD SPI MISO	TRUE
77 72 71 39	TPAD SPI MOSI	==	TPAD SPI MOSI	TRUE
77 72 71 39	TPAD SPI SCLK	==	TPAD SPI SCLK	TRUE

78 75 71 24 23 22	PP0V75 S3 MEM VREFDQ A	==	0.675V	TRUE	PP0V75 S3 MEM VREFDQ A	22 23 24 71 75 78
75 71 26 25 22	PP0V75 S3 MEM VREFDQ B	==	0.675V	TRUE	PP0V75 S3 MEM VREFDQ B	22 25 26 71 75
78 75 71 26 25 24 23 22	PP0V75 S3 MEM VREFCA	==	0.675V	TRUE	PP0V75 S3 MEM VREFCA	22 23 24 25 26 71 75 78
78 75 71 26 25 24 23 22	PP0V75 S3 MEM VREFCA	==				
78 75 71 26 25 24 23 22	PP0V75 S3 MEM VREFCA	==				

EDP CABLE

72 71 68 63	I2C_BKLT_SCL	==	I2C_BKLT_SCL	63 68 71 72
72 71 68 63	I2C_BKLT_SDA	==	I2C_BKLT_SDA	63 68 71 72
81 72 71 68 48 44 41 37	SMBUS_SMC_0_S0_SCL	==	SMBUS_SMC_0_S0_SCL	37 41 44 48 68 71 72 81
81 72 71 68 48 44 41 37	SMBUS_SMC_0_S0_SDA	==	SMBUS_SMC_0_S0_SDA	37 41 44 48 68 71 72 81

CPU signals

71 60 21	MEMVTT_EN	==	MEMVTT_EN	21 60 71
----------	-----------	----	-----------	----------

J9510 RIO FLEX CONN

77 72 71 69 44 38 13	SMBUS_PCH_DATA	==	SMBUS_PCH_DATA	13 38 44 69 71 72 77
77 72 71 69 44 38 13	SMBUS_PCH_CLK	==	SMBUS_PCH_CLK	13 38 44 69 71 72 77

SSD Signals Through PEG

75 35 5	PCIE SSD D2R P<3..0>	==	=PEG D2R P<3..0>	
75 35 5	PCIE SSD D2R N<3..0>	==	=PEG D2R N<3..0>	
75 35 5	PCIE SSD R2D C P<3..0>	==	=PEG R2D C P<3..0>	
75 35 5	PCIE SSD R2D C N<3..0>	==	=PEG R2D C N<3..0>	

Thunderbolt Signals Through PEG

75 28 5	PCIE TBT D2R P<3..0>	==	=PEG D2R P<11..8>	
75 28 5	PCIE TBT D2R N<3..0>	==	=PEG D2R N<11..8>	
75 28 5	PCIE TBT R2D C P<3..0>	==	=PEG R2D C P<11..8>	
75 28 5	PCIE TBT R2D C N<3..0>	==	=PEG R2D C N<11..8>	

Unused PEG Lanes

TP PEG D2RP<7..4>	==	=PEG D2R P<7..4>	5
TP PEG D2RN<7..4>	==	=PEG D2R N<7..4>	5
TP PEG R2D CP<7..4>	==	=PEG R2D C P<7..4>	5
TP PEG R2D CN<7..4>	==	=PEG R2D C N<7..4>	5
TP PEG D2RP<15..12>	==	=PEG D2R P<15..12>	5
TP PEG D2RN<15..12>	==	=PEG D2R N<15..12>	5
TP PEG R2D CP<15..12>	==	=PEG R2D C P<15..12>	5
TP PEG R2D CN<15..12>	==	=PEG R2D C N<15..12>	5

Unused PCH PCIE Lanes

13	NC PCIE SSD D2RP<3..0>	MAKE_BASE	NO_TEST	==	=PCIE SSD D2R P<3..0>
13	NC PCIE SSD D2RN<3..0>	TRUE	TRUE	==	=PCIE SSD D2R N<3..0>
13	NC PCIE SSD R2D CP<3..0>	TRUE	TRUE	==	=PCIE SSD R2D C P<3..0>
13	NC PCIE SSD R2D CN<3..0>	TRUE	TRUE	==	=PCIE SSD R2D C N<3..0>

Unused signals

12	BT_PWRRST_L	
14	MEM_VDD_SEL_1V5_L	
14	FW_PWR_EN_PCH	
14	WOL_EN	
14	FW_PME_L	
11	DP_TBT_SEL	
11	ENET_MEDIA_SENSE_RDIV	
11	AUD_IPHS_SWITCH_EN_PCH	
12	AUD_IP_PERIPHERAL_DET	
12	AUD_I2C_INT_L	
14	TBT_G02SX_BIDIR	
14	DPMUX_UC_IRQ	
11	PEG_CLKREQ_L	
11	ENET_CLKREQ_L	
12	ENET_LOW_PWR_PCH	
28	HDMI_TBTMUX_SEL_TBT	
12	SDCONN_OC_L	
14	LPCPLUS_GPTIO	

SYNC MASTER=J15 MLB		SYNC DATE=10/31/2012	
PAGE TITLE		Signal Aliases	
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IV ALL RIGHTS RESERVED		SHEET	71 OF 82

Functional Test Points

D

D

C

C

B

B

A

A

FUNC_TEST J3501 - airport	
TRUE AP CLKREQ0 0 L	34
TRUE AP RESET CONN L	34
TRUE PCIE AP D2R PI N	34 77
TRUE PCIE AP D2R PI P	34 77
TRUE PCIE AP R2D N	34 77
TRUE PCIE AP R2D P	34 77
TRUE PCIE CLK100M AP CONN N	34 77
TRUE PCIE CLK100M AP CONN P	34 77
TRUE PCIE WAKE L	12 34 36 77
TRUE PP3V3 S3RS4 BT F	34
TRUE PP3V3 WLAN	34 42
TRUE USB BT CONN N	34 76
TRUE USB BT CONN P	34 76
TRUE WIFI EVENT L	34 41 42
TRUE GND	4X

J4002 - Camera	
TRUE MIPI CLK CONN N	37 88
TRUE MIPI CLK CONN P	37 88
TRUE CAM SENSOR WAKE L CONN	37
TRUE MIPI DATA CONN N	37 88
TRUE MIPI DATA CONN P	37 88
TRUE SMBUS SMC 0 S0 SDA	37 41 44 48 68 71 72 81
TRUE SMBUS SMC 0 S0 SCL	37 41 44 48 68 71 72 81
TRUE I2C CAM SCK	36 37
TRUE I2C CAM SDA	36 37
TRUE PP5V S3RS0 ALSCAM F	37
TRUE GND	

J9500 - rio coax	
TRUE HDMI CLK N	5 69 71 75
TRUE HDMI CLK P	5 69 71 75
TRUE HDMI DATA N<0>	69 71 75
TRUE HDMI DATA N<1>	69 71 75
TRUE HDMI DATA N<2>	69 71 75
TRUE HDMI DATA P<0>	69 71 75
TRUE HDMI DATA P<1>	69 71 75
TRUE HDMI DATA P<2>	69 71 75

TRUE USB3 SD D2R N	13 20 69 77
TRUE USB3 SD D2R P	13 20 69 77
TRUE USB3 SD R2D C N	13 20 69 77
TRUE USB3 SD R2D C P	13 20 69 77
TRUE USB3 EXTB D2R N	13 69 76
TRUE USB3 EXTB D2R P	13 69 76
TRUE USB3 EXTB R2D N	69 76
TRUE USB3 EXTB R2D P	69 76
TRUE USB EXTB N	13 69 76
TRUE USB EXTB P	13 69 76
TRUE GND	19X

J9510 - rio flex	
TRUE SD PWR EN	13 18 69
TRUE HDMI DDC CLK	12 69 71
TRUE HDMI DDC DATA	12 69 71
TRUE HDMI HPD	12 20 69 71
TRUE SMBUS PCH CLK	13 18 44 69 71 77
TRUE SMBUS PCH DATA	13 18 44 69 71 77
TRUE PM SLP S3 BUF L	61 66 67 69
TRUE PM SLP S4 L	12 21 34 38 41 67 69
TRUE PP3V3 S3	3X13 20 21 44 46 47 66 69 70
TRUE PP3V3 S4	20 34 39 40 42 43 46 47 65 66
TRUE PP5V S4	67 69 70 71 72
TRUE RIO SDCONN STATE CHANGE L	5X39 39 51 61 66 67 68 69 70
TRUE USB EXTB OC L	18 69
TRUE GND	10X

J5150 - hall effect	
TRUE PP3V42 G3H	19 35 38 39 41 42 43 44 50 56
TRUE SMC LID R	43
TRUE GND	

J6050 - left fan	
TRUE FAN LT PWM	49
TRUE FAN LT TACH	49
TRUE PP5V S0	3X18 19 37 49 58 59 62 63 66
TRUE GND	5X

J6060 - right fan	
TRUE FAN RT PWM	49
TRUE FAN RT TACH	49
TRUE PP5V S0	3X18 19 37 49 58 59 62 63 66
TRUE GND	5X

FUNC_TEST J6100 - spi	
TRUE PP3V42 G3H	42 43 44 50 56
TRUE SMC RESET L	41 42 50 57
TRUE SMC TCK	41 42 50
TRUE SMC TMS	41 42 50

TRUE SPIROM USE MLB	14 50
TRUE SPI ALT CLK	50 77
TRUE SPI ALT CS L	50 77
TRUE SPI ALT IO0 MOSI	50 77
TRUE SPI ALT IO1 MISO	50 77
TRUE SPI ALT IO2 WP L	50 77
TRUE SPI ALT IO1 HOLD L	50 77
TRUE GND	2X

J4801 - ipd flex	
TRUE TPAD SPI INT L	39 71
TRUE TPAD SPI CS L	39 71 77
TRUE TPAD SPI MOSI	39 71 77
TRUE TPAD SPI MISO	39 71 77
TRUE TPAD SPI SCLK	39 71 77
TRUE TPAD SPI BUS EN	39 71
TRUE USB TPAD N	13 39 76
TRUE USB TPAD P	13 39 76
TRUE IOXP2 INT L	39
TRUE I2C IOXP SCL	39
TRUE I2C IOXP SDA	39
TRUE SMC PME S4 WAKE L	34 39 41 43
TRUE TPAD ACTUATOR THRMTRIP L	39 65
TRUE TPAD VBUS EN	39
TRUE SMBUS SMC 2 S3 SCL	39 41 44 81
TRUE SMBUS SMC 2 S3 SDA	39 41 44 81
TRUE SMC LID	39 41 42 43
TRUE SMC ACTUATOR EN L	39 41
TRUE PPVIN S4 TPAD	4X39 46 70
TRUE GND ACTUATOR	4X39
TRUE PP3V3 S4	39 39 41 43
TRUE PP5V S4	39 39 51 61 66 67 68 69 70 72
TRUE GND	2X

J4813 - keyboard	
TRUE PP3V3 S4	20 34 39 40 42 43 46 47 65 66
TRUE PP3V42 G3H	42 43 44 50 56
TRUE WS CONTROL KBD	39
TRUE WS KBD1	39
TRUE WS KBD10	39
TRUE WS KBD11	39
TRUE WS KBD12	39
TRUE WS KBD13	39
TRUE WS KBD14	39
TRUE WS KBD15 CAP	39
TRUE WS KBD16 NUM	39
TRUE WS KBD17	39
TRUE WS KBD18	39
TRUE WS KBD19	39
TRUE WS KBD2	39
TRUE WS KBD20	39
TRUE WS KBD21	39
TRUE WS KBD22	39
TRUE WS KBD23	39
TRUE WS KBD3	39
TRUE WS KBD4	39
TRUE WS KBD5	39
TRUE WS KBD6	39
TRUE WS KBD7	39
TRUE WS KBD8	39
TRUE WS KBD9	39
TRUE WS KBD ONOFF L	39
TRUE WS LEFT OPTION KBD	39
TRUE WS LEFT SHIFT KBD	39
TRUE GND	2X

J4915 - kbd bklt	
TRUE KBDBKLT RETURN1	2X40 63
TRUE KBDBKLT RETURN2	2X40 63
TRUE PPVOUT S0 KBDBKLT	40 63
TRUE GND	4X

J6601 - mic	
TRUE DMIC CLK3	52 55
TRUE PP3V3 S0	55 67 68 70 72 82
TRUE DMIC SDA2	55
TRUE DMIC SDA3	52 55
TRUE GND	

J6602 - L speaker	
TRUE SPKRCONN L ID	52 55
TRUE SPKRCONN L OUT N	53 55 82
TRUE SPKRCONN L OUT P	53 55 82
TRUE SPKRCONN SL OUT N	53 55 82
TRUE SPKRCONN SL OUT P	53 55 82
TRUE GND	

J6603 - R speaker	
TRUE SPKRCONN R ID	52 55
TRUE SPKRCONN R OUT N	53 55 82
TRUE SPKRCONN R OUT P	53 55 82
TRUE SPKRCONN SR OUT N	53 55 82
TRUE SPKRCONN SR OUT P	53 55 82
TRUE GND	

J7000 - DC PWR	
TRUE ADAPTER SENSE	56
TRUE PP20V DCIN FUSE	2X56
TRUE GND	2X

J7050 - battery	
TRUE PPVBAT G3H CONN	8X56 57
TRUE SMBUS SMC 5 G3 SCL	41 44 56 57 81
TRUE SMBUS SMC 5 G3 SDA	41 44 56 57 81
TRUE SYS DETECT L	56
TRUE GND	8X

J8300 - edp	
TRUE DP INT AUX N	60 75
TRUE DP INT AUX P	60 75
TRUE DP INT ML N<0>	60 75
TRUE DP INT ML N<1>	60 75
TRUE DP INT ML N<2>	60 75
TRUE DP INT ML N<3>	60 75
TRUE DP INT ML P<0>	60 75
TRUE DP INT ML P<1>	60 75
TRUE DP INT ML P<2>	60 75
TRUE DP INT ML P<3>	60 75
TRUE LCD IRQ L	60
TRUE LCD HPD CONN	60
TRUE LCD BKLT PWM R	63 68
TRUE SMBUS SMC 0 S0 SDA	37 41 44 48 68 71 72 81
TRUE SMBUS SMC 0 S0 SCL	37 41 44 48 68 71 72 81
TRUE I2C BKLT SDA	63 68 71
TRUE I2C BKLT SCL	63 68 71
TRUE PP5VR3V3 SW LCD	3X68
TRUE PPVOUT S0 LCDBKLT	63 68
TRUE GND	16X

Power Rails	
TRUE PM SLP S3 L	12 21 41 67
TRUE PPVTT S0 DDR	21 27 60 70
TRUE PP3V3 S0	60 67 68 70 72 82
TRUE PP3V3 S3	12 12 13 14 15 17 19 20 29 33
TRUE PP3V3 S5	35 44 45 46 47 48 49 51 52 55
TRUE PP3V3 S5 AVREF SMC	13 20 21 44 46 47 66 69 70 72
TRUE PP3V42 G3H	31 32 34
TRUE PP5V S0	41 42
TRUE PP5V S3	19 20 35 38 41 42 43 44 50 56
TRUE PP5V S5	67 68 69 70 72
TRUE PPBUS G3H	21 37 60 66 67 70
TRUE PPDCIN G3H	61 66 70
TRUE PPVCC S0 CPU	39 45 56 57 63 65 70
TRUE PPVTIDDR S3	56 57 70
TRUE PP3V3 S0SW SSD	6 8 10 46 59 70
TRUE PP1V5 S0	60 70
TRUE PP1V35 S3	35 46 70
TRUE PP1V35 S3	12 13 15 17 19 52 64 67 69
TRUE PP1V35 S3	21 46 60 66 70

XDP	
TRUE XDP CPU TCK	6 18 75
TRUE XDP PCH TCK	11 18
TRUE XDP CPU TDI	6 18 75
TRUE XDP CPU TDO	6 18 75
TRUE XDP CPUPCH TRST L	6 18 75
TRUE XDP CPU TMS	6 18 75
TRUE XDP PCH TMS	11 18
TRUE XDP PCH TDI	11 18
TRUE XDP PCH TDO	11 18
TRUE XDP CPU PREQ L	6 18 75
TRUE XDP CPU PRDY L	6 18 75
TRUE PM RSMRST L	12 67 77
TRUE PM PCH PWROK	12 19 77
TRUE PM SYSRST L	12 19 41 77
TRUE CPU CFG<3>	6 18 75
TRUE PP1V05 S0	10 14 15 17 18 42 62 67 70
TRUE GND	2X GND

Power Sequence	
TRUE SMC ONOFF L	39 41 42
TRUE PM DSW PWRGD	12 41 77
TRUE ALL SYS PWRGD	10 19 41 58 67
TRUE PM PCH SYS PWROK	12 18 19 41 77
TRUE PLT RESET L	12 18 20 21
TRUE EDP IG PANEL PWR	12 68 71
TRUE EDP IG BKL ON	12 63 71

SYNC MASTER=J15 MLB		SYNC DATE=10/31/2012	
PAGE TITLE		DRAWING NUMBER	
Functional Test Points		<SCH_NUM> D	
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IV ALL RIGHTS RESERVED		72 OF 82	

[illegible]

79	31	28	TBT A D2R P<1>	1	TP	SM	BEAD-PROBE	BPA531	NO_XNET_CONNECTION=TRUE
79	31	28	TBT A D2R N<1>	1	TP	SM	BEAD-PROBE	BPA532	NO_XNET_CONNECTION=TRUE

73	12	NC DP IG D AUXCHN	—	TRUE	TRUE	NC DP IG D AUXCHN	12	73
73	12	NC DP IG D AUXCHP	—	TRUE	TRUE	NC DP IG D AUXCHP	12	73

76	76	NC	PCIE	CLK100M	GPUN	==	TRUE	TRUE	NC	PCIE	CLK100M	GPUN	11	79
76	76	NC	PCIE	CLK100M	GPUP	==	TRUE	TRUE	NC	PCIE	CLK100M	GPUP	11	79
76	76	NC	PCIE	CLK100M	PE5N	==	TRUE	TRUE	NC	PCIE	CLK100M	PE5N	11	79
76	76	NC	PCIE	CLK100M	PE5P	==	TRUE	TRUE	NC	PCIE	CLK100M	PE5P	11	79
76	76	NC	PCIE	CLK100M	ENETSDN	==	TRUE	TRUE	NC	PCIE	CLK100M	ENETSDN	11	79
76	76	NC	PCIE	CLK100M	ENETSDP	==	TRUE	TRUE	NC	PCIE	CLK100M	ENETSDP	11	79
76	76	NC	PCIE	CLK100M	ENETN	==	TRUE	TRUE	NC	PCIE	CLK100M	ENETN	11	79
76	76	NC	PCIE	CLK100M	ENETP	==	TRUE	TRUE	NC	PCIE	CLK100M	ENETP	11	79
76	76	NC	PCIE	CLK100M	PEGBN	==	TRUE	TRUE	NC	PCIE	CLK100M	PEGBN	11	79
76	76	NC	PCIE	CLK100M	PEGBP	==	TRUE	TRUE	NC	PCIE	CLK100M	PEGBP	11	79
76	76	NC	PCIE	CLK100M	SWN	==	TRUE	TRUE	NC	PCIE	CLK100M	SWN	11	79
76	76	NC	PCIE	CLK100M	SWP	==	TRUE	TRUE	NC	PCIE	CLK100M	SWP	11	79

73	11	NC	PCH	GPI064	CLKOUTFLEX0	—	TRUE	TRUE	NC	PCH	GPI064	CLKOUTFLEX0	11
73	11	NC	PCH	GPI065	CLKOUTFLEX1	—	TRUE	TRUE	NC	PCH	GPI065	CLKOUTFLEX1	11
73	11	NC	PCH	GPI066	CLKOUTFLEX2	—	TRUE	TRUE	NC	PCH	GPI066	CLKOUTFLEX2	11
73	11	NC	PCH	GPI067	CLKOUTFLEX3	—	TRUE	TRUE	NC	PCH	GPI067	CLKOUTFLEX3	11

76	13	NC USB 4N	==	TRUE	TRUE	NC USB 4N	13	73
76	13	NC USB 4P	==	TRUE	TRUE	NC USB 4P	13	73

TRUE	PCIE TBT R2D P<3..0>	28 75
TRUE	PCIE TBT R2D N<3..0>	28 75
TRUE	PCIE TBT D2R C P<3..0>	28 75
TRUE	PCIE TBT D2R C N<3..0>	28 75

R497	TRUE	DMI S2N P<3..1>	5 12 75
R497	TRUE	DMI S2N N<3..1>	5 12 75
R497	TRUE	DMI N2S P<3..1>	5 12 75
R497	TRUE	DMI N2S N<3..1>	5 12 75

SYNC MASTER=J15 MLB		SYNC DATE=10/31/2012	
PAGE TITLE		DRAWING NUMBER	
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		BRANCH	
		<BRANCH>	
		PAGE	
		105 OF 118	
		SHEET	
		73 OF 82	

X425 BOARD-SPECIFIC SPACING & PHYSICAL CONSTRAINTS

BOARD LAYERS	BOARD AREAS	BOARD UNITS (MIL or MM)	ALLEGRO VERSION
TOP, ISL2, ISL3, ISL4, ISL5, ISL6, ISL7, ISL8, ISL9, ISL10, ISL11, BOTTOM	NO_TYPE, BGA, P65BGA	MM	16.2

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	=45_OHM_SE	=45_OHM_SE	10 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	10 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	TOP,BOTTOM	Y	0.095 MM	0.095 MM			
50_OHM_SE	*	Y	0.066 MM	0.066 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
45_OHM_SE	TOP,BOTTOM	Y	0.116 MM	0.116 MM			
45_OHM_SE	*	Y	0.083 MM	0.083 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
40_OHM_SE	TOP,BOTTOM	Y	0.145 MM	0.095 MM			
40_OHM_SE	*	Y	0.102 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
37_OHM_SE	TOP,BOTTOM	Y	0.165 MM	0.095 MM			
37_OHM_SE	*	Y	0.118 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
27P4_OHM_SE	TOP,BOTTOM	Y	0.265 MM	0.095 MM			
27P4_OHM_SE	*	Y	0.186 MM	0.1 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
72_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
72_OHM_DIFF	ISL3, ISL4, ISL9, ISL10	Y	0.105 MM	0.105 MM		0.120 MM	0.120 MM
72_OHM_DIFF	ISL2, ISL11	Y	0.105 MM	0.105 MM		0.120 MM	0.120 MM
72_OHM_DIFF	TOP, BOTTOM	Y	0.146 MM	0.146 MM		0.120 MM	0.120 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
80_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
80_OHM_DIFF	ISL3, ISL4, ISL9, ISL10	Y	0.092 MM	0.092 MM		0.120 MM	0.120 MM
80_OHM_DIFF	ISL2, ISL11	Y	0.092 MM	0.092 MM		0.120 MM	0.120 MM
80_OHM_DIFF	TOP, BOTTOM	Y	0.125 MM	0.125 MM		0.155 MM	0.155 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
85_OHM_DIFF	ISL3, ISL4, ISL9, ISL10	Y	0.080 MM	0.080 MM		0.120 MM	0.120 MM
85_OHM_DIFF	ISL2, ISL11	Y	0.080 MM	0.080 MM		0.120 MM	0.120 MM
85_OHM_DIFF	TOP, BOTTOM	Y	0.105 MM	0.105 MM		0.125 MM	0.125 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
90_OHM_DIFF	1SL3, 1SL4, 1SL9, 1SL10	Y	0.078 MM	0.078 MM		0.200 MM	0.200 MM
90_OHM_DIFF	1SL2, 1SL11	Y	0.078 MM	0.078 MM		0.200 MM	0.200 MM
90_OHM_DIFF	TOP,BOTTOM	Y	0.101 MM	0.101 MM		0.180 MM	0.180 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA	P072_SPACE
*	*	P65BGA	P075_SPACE

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?
BGA_P1MM	*	0.1 MM	?
BGA_P2MM	*	0.2 MM	?
P072_SPACE	*	0.071 MM	?
P075_SPACE	*	0.075 MM	?

Stackup-Defined Spacing Rules

Note: Outer dielectric is 0.058 mm nominal,
Inner dielectric is 0.053 mm nominal.

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1:1_SPACING	*	0.1 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1X_DIELECTRIC	TOP, BOTTOM	0.058 MM	?
1X_DIELECTRIC	ISL3, ISL4, ISL9, ISL10	0.053 MM	?
1X_DIELECTRIC	ISL2, ISL5, ISL6, ISL7, ISL8, ISL11	0.101 MM	?

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
P65_BGA	*	Y	0.071MM	0.071MM		0.075MM	0.126MM

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
*	P65BGA	P65_BGA

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		BRANCH	
		<BRANCH>	
		PAGE	
		110	OF 118
		SHEET	
		74	OF 82

D

SATA Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
SATA_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
SATA_37SE	*	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE
SATA_45SE	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
SATA_2SAME	*	=3X_DIELECTRIC	?	SATA_2SAME	TOP,BOTTOM	=4X_DIELECTRIC	?
SATA_TXRX	*	=6X_DIELECTRIC	?	SATA_TXRX	TOP,BOTTOM	=10X_DIELECTRIC	?
SATA_20THER	*	=4X_DIELECTRIC	?	SATA_20THER	TOP,BOTTOM	=6X_DIELECTRIC	?
SATA_RCOMP	*	=6X_DIELECTRIC	?	SATA_RCOMP	TOP,BOTTOM	=10X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SATA_*	=SAME	*	SATA_2SAME
SATA_R2D	SATA_D2R	*	SATA_TXRX
SATA_*	*	*	SATA_20THER

USB 2.0 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
PCH_USB_RB1AS	*	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
USB_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
USB	*	=4X_DIELECTRIC	?	USB	TOP,BOTTOM	=6X_DIELECTRIC	?
USB_RB1AS	*	=6X_DIELECTRIC	?	USB_RB1AS	TOP,BOTTOM	=10X_DIELECTRIC	?
BT_WAKE	*	=4X_DIELECTRIC	?	BT_WAKE	TOP,BOTTOM	=6X_DIELECTRIC	?

USB 3.0 INTERFACE CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
USB3_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
USB3_2SAME	*	=3X_DIELECTRIC	?	USB3_2SAME	TOP,BOTTOM	=4X_DIELECTRIC	?
USB3_TXRX	*	=6X_DIELECTRIC	?	USB3_TXRX	TOP,BOTTOM	=10X_DIELECTRIC	?
USB3_20THER	*	=4X_DIELECTRIC	?	USB3_20THER	TOP,BOTTOM	=6X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
USB3_*	=SAME	*	USB3_2SAME
USB3_R2D	USB3_D2R	*	USB3_TXRX
USB3_*	*	*	USB3_20THER

System Clock Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
CLK_SLOW_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
CLK_25M_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
CLK_SLOW	*	=4X_DIELECTRIC	?
CLK_25M	*	=5X_DIELECTRIC	?

NOTE: 25MHz system clocks very sensitive to noise.
NOTE: Latest Intel DG calls out 50ohms SE for sys clocks

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PCH Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
	SATA_85D	SATA_R2D		NC SATA A R2D_CP	11 73
	SATA_85D	SATA_R2D		NC SATA A R2D_CN	11 73
	SATA_85D	SATA_D2R		NC SATA A D2RP	11 73
	SATA_85D	SATA_D2R		NC SATA A D2RN	11 73
	SATA_85D	SATA_R2D		NC SATA B R2D_CP	11 73
	SATA_85D	SATA_R2D		NC SATA B R2D_CN	11 73
	SATA_85D	SATA_D2R		NC SATA B D2RP	11 73
	SATA_85D	SATA_D2R		NC SATA B D2RN	11 73
	PCH_SATA_RCOMP	SATA_45SE	SATA_RCOMP	PCH_SATA_RCOMP	11
	USB_EXT_A	USB_85D	USB	USB_EXT_A_P	13 38
	USB_EXT_A	USB_85D	USB	USB_EXT_A_N	13 38
	USB_EXT_A	USB_85D	USB	USB_EXT_A_MUXED_P	38
	USB_EXT_A	USB_85D	USB	USB_EXT_A_MUXED_N	38
	USB_EXT_A	USB_85D	USB	USB_LT1_P	38
	USB_EXT_A	USB_85D	USB	USB_LT1_N	38
	USB_NC	USB_85D	USB	NC_USB_EXTCP	13 73
	USB_NC	USB_85D	USB	NC_USB_EXTCN	13 73
	USB_NC	USB_85D	USB	NC_USB_SDP	13 73
	USB_NC	USB_85D	USB	NC_USB_SDN	13 73
	CPU_45S	CPU_ITP		SMC_DEBUGPRT_RX_L	38 41 42
	CPU_45S	CPU_ITP		SMC_DEBUGPRT_TX_L	38 41 42
	USB_SMC	USB_85D	USB	NC_USB_SMCP	73
	USB_SMC	USB_85D	USB	NC_USB_SMCN	73
	USB_NC	USB_85D	USB	NC_USB_6P	13 73
	USB_NC	USB_85D	USB	NC_USB_6N	13 73
	USB_NC	USB_85D	USB	NC_USB_7P	13 73
	USB_NC	USB_85D	USB	NC_USB_7N	13 73
	USB_EXTB	USB_85D	USB	USB_EXTB_P	13 69 72
	USB_EXTB	USB_85D	USB	USB_EXTB_N	13 69 72
	USB_NC	USB_85D	USB	NC_USB_EXTDP	13 73
	USB_NC	USB_85D	USB	NC_USB_EXTDN	13 73
	USB_BT	USB_85D	USB	USB_BT_P	13 34
	USB_BT	USB_85D	USB	USB_BT_N	13 34
	USB_BT	USB_85D	USB	USB_BT_CONN_P	34 72
	USB_BT	USB_85D	USB	USB_BT_CONN_N	34 72
	USB_NC	USB_85D	USB	NC_USB_IRP	13 73
	USB_NC	USB_85D	USB	NC_USB_IRN	13 73
	USB_TPAD	USB_85D	USB	USB_TPAD_P	13 39 72
	USB_TPAD	USB_85D	USB	USB_TPAD_N	13 39 72
	USB_TPAD	USB_85D	USB	USB_TPAD_R_P	
	USB_TPAD	USB_85D	USB	USB_TPAD_R_N	
	PCH_USB_RB1AS	PCH_USB_RB1AS	USB_RB1AS	PCH_USB_RB1AS	13
	USB3_EXT_A_RX	USB_85D	USB3_D2R	USB3_EXT_A_D2R_P	13 38
	USB3_EXT_A_RX	USB_85D	USB3_D2R	USB3_EXT_A_D2R_N	13 38
	USB_85D	USB3_D2R		USB3_EXT_A_D2R_C_P	
	USB_85D	USB3_D2R		USB3_EXT_A_D2R_C_N	
	USB3_EXT_A_TX	USB_85D	USB3_R2D	USB3_EXT_A_R2D_P	38
	USB3_EXT_A_TX	USB_85D	USB3_R2D	USB3_EXT_A_R2D_N	38
	USB_85D	USB3_R2D		USB3_EXT_A_R2D_C_P	13 38
	USB_85D	USB3_R2D		USB3_EXT_A_R2D_C_N	13 38
	USB3_EXTB_RX	USB_85D	USB3_D2R	USB3_EXTB_D2R_P	13 69 72
	USB3_EXTB_RX	USB_85D	USB3_D2R	USB3_EXTB_D2R_N	13 69 72
	USB_85D	USB3_D2R		USB3_EXTB_D2R_C_P	
	USB_85D	USB3_D2R		USB3_EXTB_D2R_C_N	
	USB3_EXTB_TX	USB_85D	USB3_R2D	USB3_EXTB_R2D_P	69 72
	USB3_EXTB_TX	USB_85D	USB3_R2D	USB3_EXTB_R2D_N	69 72
	USB_85D	USB3_R2D		USB3_EXTB_R2D_C_P	13 69
	USB_85D	USB3_R2D		USB3_EXTB_R2D_C_N	13 69
	NC_USB3	USB_85D	USB3_D2R	NC_USB3_EXTC_D2RP	13 73
	NC_USB3	USB_85D	USB3_D2R	NC_USB3_EXTC_D2RN	13 73
	USB_85D	USB3_R2D		NC_USB3_EXTC_R2D_CP	13 73
	USB_85D	USB3_R2D		NC_USB3_EXTC_R2D_CN	13 73
	NC_USB3	USB_85D	USB3_D2R	NC_USB3_EXTD_D2RP	13 73
	NC_USB3	USB_85D	USB3_D2R	NC_USB3_EXTD_D2RN	13 73
	USB_85D	USB3_R2D		NC_USB3_EXTD_R2D_CP	13 73
	USB_85D	USB3_R2D		NC_USB3_EXTD_R2D_CN	13 73

Clock Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
	SYSCLK_CLK32K_RTC	CLK_SLOW_45S	CLK_SLOW	SYSCLK_CLK32K_RTC	11 19
	SYSCLK_CLK25M_SB	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_SB	11 19
	SYSCLK_CLK25M_CAM	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_CAMERA	11 37
	SYSCLK_CLK25M_TBT	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_TBT	19 28
		CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_TBT_R	28

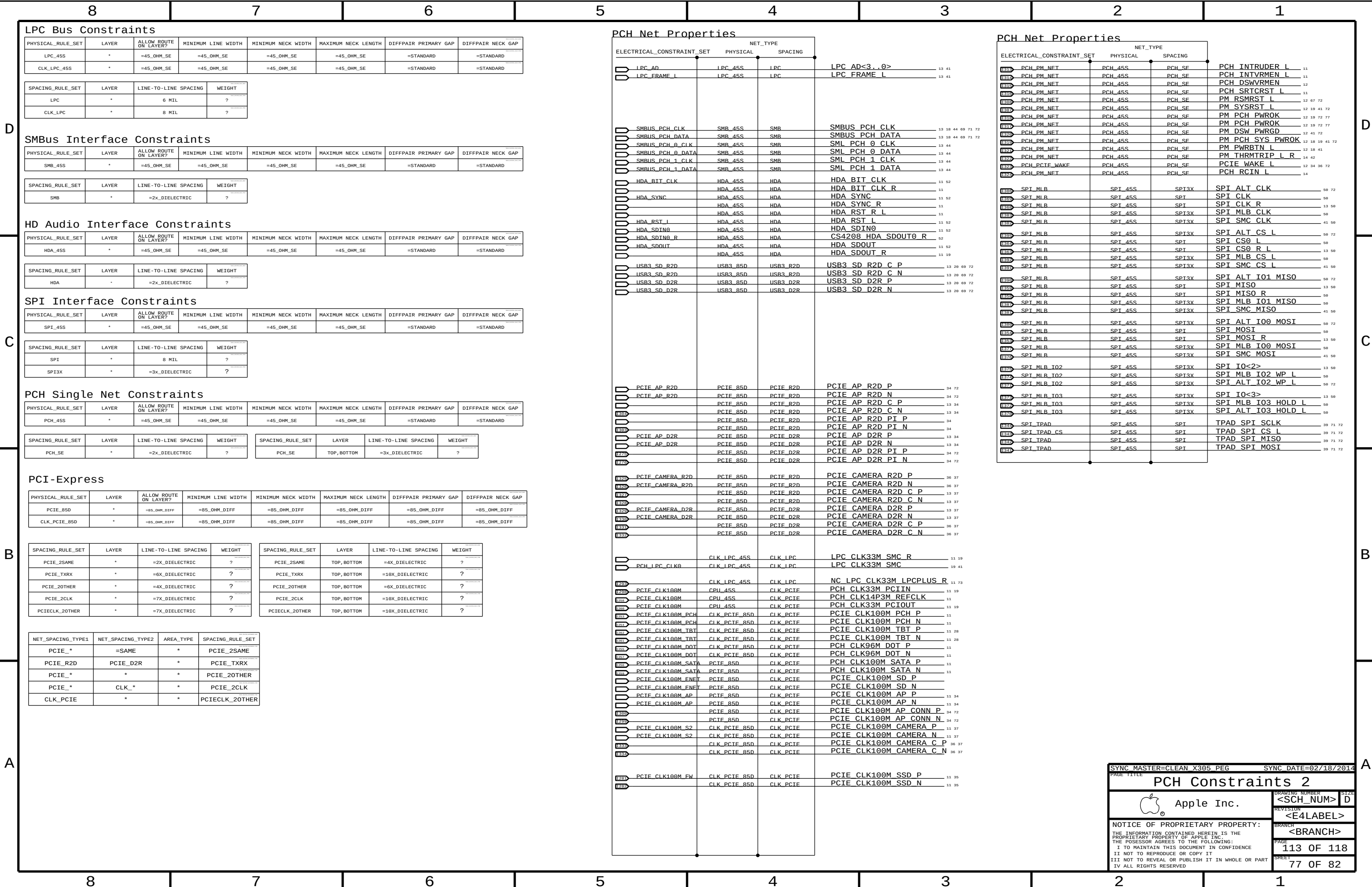
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		<E4LABEL>	
		BRANCH	<BRANCH>
		PAGE	112 OF 118
		SHEET	76 OF 82

D

C

B

A



Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_37S	*	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=STANDARD	=STANDARD
MEM_40S	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=STANDARD	=STANDARD
MEM_72D	*	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF
MEM_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
MEM_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

Spacing Rule Sets

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_DATA2SELF	*	=2x_DIELECTRIC	?
MEM_DQS2OWNDATA	*	=2x_DIELECTRIC	?
MEM_CMD2CMD	*	=2x_DIELECTRIC	?
MEM_CMD2CTRL	*	=2x_DIELECTRIC	?
MEM_CTRL2CTRL	*	=2x_DIELECTRIC	?
MEM_CLK2CLK	*	=4x_DIELECTRIC	?
MEM_2OTHERMEM	*	=4x_DIELECTRIC	?
MEM_2PWR	*	=2x_DIELECTRIC	?
MEM_2GND	*	=2x_DIELECTRIC	?
MEM_2OTHER	*	=6x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_DATA2SELF	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_DQS2OWNDATA	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CMD2CMD	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CMD2CTRL	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CTRL2CTRL	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CLK2CLK	TOP,BOTTOM	=8x_DIELECTRIC	?
MEM_2OTHERMEM	TOP,BOTTOM	=8x_DIELECTRIC	?
MEM_2PWR	TOP,BOTTOM	=4x_DIELECTRIC	?
MEM_2GND	TOP,BOTTOM	=4x_DIELECTRIC	?
MEM_2OTHER	TOP,BOTTOM	=10x_DIELECTRIC	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*_DATA_*	*	*	MEM_20THER
MEM_*_DQS_*	*	*	MEM_20THER
MEM_CMD	*	*	MEM_20THER
MEM_CTRL	*	*	MEM_20THER
MEM_CLK	*	*	MEM_20THER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*_DATA_*	=SAME	*	MEM_DATA2SELF

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_CMD	*	MEM_CMD2CMD
MEM_CMD	MEM_CTRL	*	MEM_CMD2CTRL
MEM_CTRL	MEM_CTRL	*	MEM_CTRL2CTRL
MEM_CLK	MEM_CLK	*	MEM_CLK2CLK
MEM_*	MEM_*	*	MEM_2OTHERMEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_A_DQS_0	MEM_A_DATA_0	*	MEM_DQS2OWNDATA
MEM_A_DQS_1	MEM_A_DATA_1	*	MEM_DQS2OWNDATA
MEM_A_DQS_2	MEM_A_DATA_2	*	MEM_DQS2OWNDATA
MEM_A_DQS_3	MEM_A_DATA_3	*	MEM_DQS2OWNDATA
MEM_A_DQS_4	MEM_A_DATA_4	*	MEM_DQS2OWNDATA
MEM_A_DQS_5	MEM_A_DATA_5	*	MEM_DQS2OWNDATA
MEM_A_DQS_6	MEM_A_DATA_6	*	MEM_DQS2OWNDATA
MEM_A_DQS_7	MEM_A_DATA_7	*	MEM_DQS2OWNDATA
MEM_B_DQS_0	MEM_B_DATA_0	*	MEM_DQS2OWNDATA
MEM_B_DQS_1	MEM_B_DATA_1	*	MEM_DQS2OWNDATA
MEM_B_DQS_2	MEM_B_DATA_2	*	MEM_DQS2OWNDATA
MEM_B_DQS_3	MEM_B_DATA_3	*	MEM_DQS2OWNDATA
MEM_B_DQS_4	MEM_B_DATA_4	*	MEM_DQS2OWNDATA
MEM_B_DQS_5	MEM_B_DATA_5	*	MEM_DQS2OWNDATA
MEM_B_DQS_6	MEM_B_DATA_6	*	MEM_DQS2OWNDATA
MEM_B_DQS_7	MEM_B_DATA_7	*	MEM_DQS2OWNDATA

DDR3 (Memory Down):

DQ signals should be matched within 0.508mm of associated DQS pair
DQS intra-pair matching should be within 0.127mm, no inter-pair matching requirement.
DQS to cClock matching should be within [CLK-139.73mm] and [CLK-30.48mm].
CLK intra-pair matching should be within 0.127mm, inter-pair matching should be within 0.508mm.
CONTROL signals should be matched within [CLK-2.54mm] to [CLK+0mm] of CLK pairs.
A/BA/CMD signals should be matched within [CLK-2.54mm] to [CLK+2.54mm] of CLK pairs.
DQ/DQS/A/BA/cmd signal spacing is 4x dielectric, CLK is 5x dielectric.
Maximum length of any signal from die pad to first DRAM device is 139.7mm max, to last DRAM device is 194.31mm max.

SOURCE: Double checked with Doc#486985 Chief River SFF Platform DG: Memory Down
SOURCE: Need to re-confirm CRW DG for memory down (Intel not yet provided)

Memory to Power Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_PWR	MEM_*	*	MEM_2PWR
MEM_PWR	*	*	DEFAULT

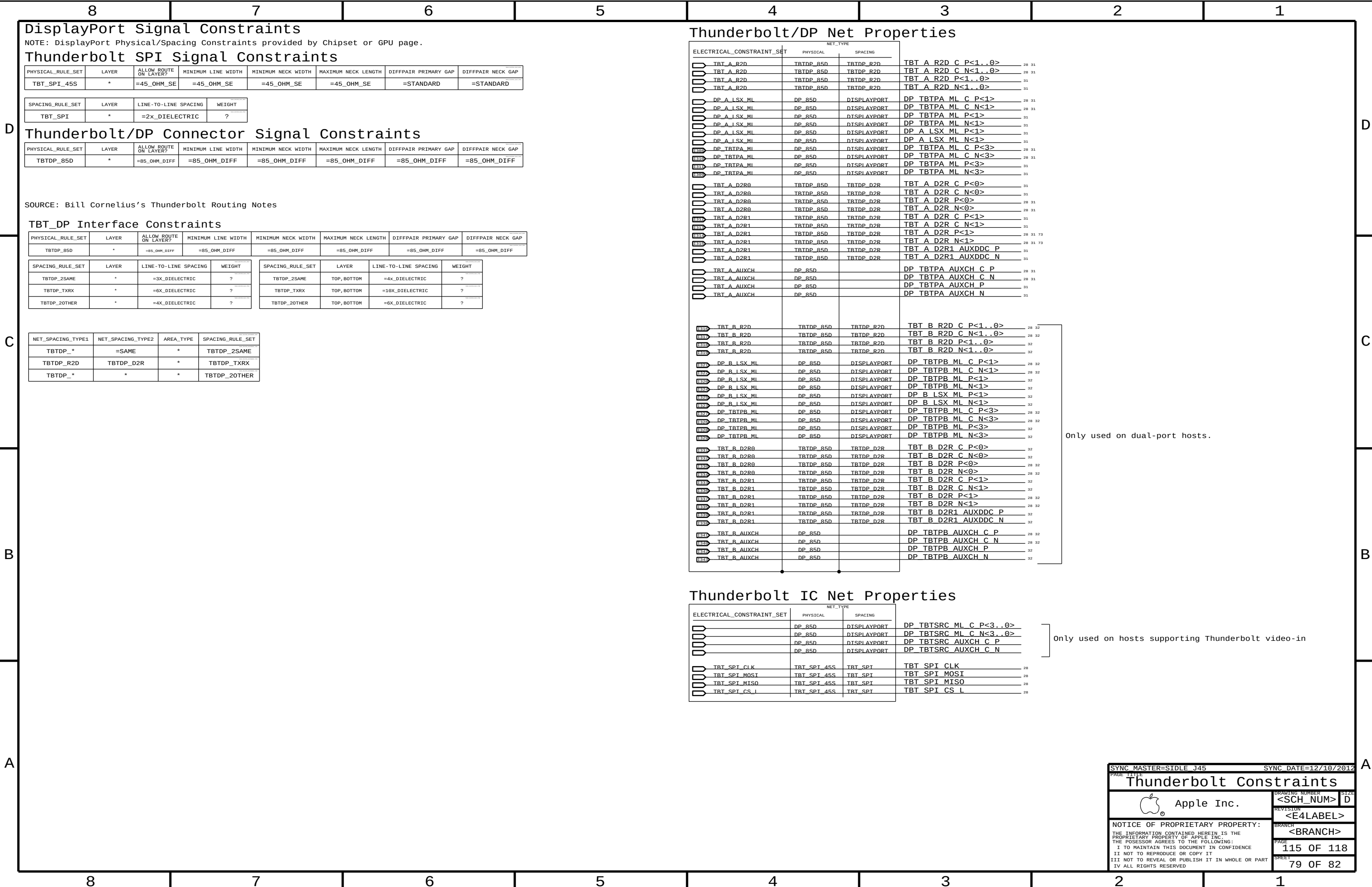
Memory to GND Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	MEM_*	*	MEM_2GND

Memory Net Properties

NET_TYPE			
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	
MEM_A_CLK0	MEM_72D	MEM_CLK	MEM A CLK P<0> 7 23 27
MEM_A_CLK0	MEM_72D	MEM_CLK	MEM A CLK N<0> 7 23 27
MEM_A_CLK1	MEM_72D	MEM_CLK	MEM A CLK P<1> 7 24 27
MEM_A_CLK1	MEM_72D	MEM_CLK	MEM A CLK N<1> 7 24 27
MEM_A_CNTRL0	MEM_40S	MEM_CTRL	MEM A CKE<0> 7 23 27
MEM_A_CNTRL1	MEM_40S	MEM_CTRL	MEM A CKE<1> 7 24 27
MEM_A_CNTRL0	MEM_40S	MEM_CTRL	MEM A CS L<0> 7 23 27
MEM_A_CNTRL1	MEM_40S	MEM_CTRL	MEM A CS L<1> 7 24 27
MEM_A_CNTRL0	MEM_40S	MEM_CTRL	MEM A ODT<0> 7 23 27
MEM_A_CNTRL1	MEM_40S	MEM_CTRL	MEM A ODT<1> 7 24 27
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A A<15..0> 7 23 24 27
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A BA<2..0> 7 23 24 27
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A RAS L 7 23 24 27
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A CAS L 7 23 24 27
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A WE L 7 23 24 27
MEM_A_DATA_0	MEM_45S	MEM_A_DATA_0	MEM A DQ<7..0> 7 23 24
MEM_A_DATA_1	MEM_45S	MEM_A_DATA_1	MEM A DQ<15..8> 7 23 24
MEM_A_DATA_2	MEM_45S	MEM_A_DATA_2	MEM A DQ<23..16> 7 23 24
MEM_A_DATA_3	MEM_45S	MEM_A_DATA_3	MEM A DQ<31..24> 7 23 24
MEM_A_DATA_4	MEM_45S	MEM_A_DATA_4	MEM A DQ<39..32> 7 23 24
MEM_A_DATA_5	MEM_45S	MEM_A_DATA_5	MEM A DQ<47..40> 7 23 24
MEM_A_DATA_6	MEM_45S	MEM_A_DATA_6	MEM A DQ<55..48> 7 23 24
MEM_A_DATA_7	MEM_45S	MEM_A_DATA_7	MEM A DQ<63..56> 7 23 24
MEM_A_DQS0	MEM_85D	MEM_A_DQS_0	MEM A DQS P<0> 7 23 24
MEM_A_DQS0	MEM_85D	MEM_A_DQS_0	MEM A DQS N<0> 7 23 24
MEM_A_DQS1	MEM_85D	MEM_A_DQS_1	MEM A DQS P<1> 7 23 24
MEM_A_DQS1	MEM_85D	MEM_A_DQS_1	MEM A DQS N<1> 7 23 24
MEM_A_DQS2	MEM_85D	MEM_A_DQS_2	MEM A DQS P<2> 7 23 24
MEM_A_DQS2	MEM_85D	MEM_A_DQS_2	MEM A DQS N<2> 7 23 24
MEM_A_DQS3	MEM_85D	MEM_A_DQS_3	MEM A DQS P<3> 7 23 24
MEM_A_DQS3	MEM_85D	MEM_A_DQS_3	MEM A DQS N<3> 7 23 24
MEM_A_DQS4	MEM_85D	MEM_A_DQS_4	MEM A DQS P<4> 7 23 24
MEM_A_DQS4	MEM_85D	MEM_A_DQS_4	MEM A DQS N<4> 7 23 24
MEM_A_DQS5	MEM_85D	MEM_A_DQS_5	MEM A DQS P<5> 7 23 24
MEM_A_DQS5	MEM_85D	MEM_A_DQS_5	MEM A DQS N<5> 7 23 24
MEM_A_DQS6	MEM_85D	MEM_A_DQS_6	MEM A DQS P<6> 7 23 24
MEM_A_DQS6	MEM_85D	MEM_A_DQS_6	MEM A DQS N<6> 7 23 24
MEM_A_DQS7	MEM_85D	MEM_A_DQS_7	MEM A DQS P<7> 7 23 24
MEM_A_DQS7	MEM_85D	MEM_A_DQS_7	MEM A DQS N<7> 7 23 24
MEM_B_CLK0	MEM_72D	MEM_CLK	MEM B CLK P<0> 7 25 27
MEM_B_CLK0	MEM_72D	MEM_CLK	MEM B CLK N<0> 7 25 27
MEM_B_CLK1	MEM_72D	MEM_CLK	MEM B CLK P<1> 7 26 27
MEM_B_CLK1	MEM_72D	MEM_CLK	MEM B CLK N<1> 7 26 27
MEM_B_CNTRL0	MEM_40S	MEM_CTRL	MEM B CKE<0> 7 25 27
MEM_B_CNTRL1	MEM_40S	MEM_CTRL	MEM B CKE<1> 7 26 27
MEM_B_CNTRL0	MEM_40S	MEM_CTRL	MEM B CS L<0> 7 25 27
MEM_B_CNTRL1	MEM_40S	MEM_CTRL	MEM B CS L<1> 7 26 27
MEM_B_CNTRL0	MEM_40S	MEM_CTRL	MEM B ODT<0> 7 25 27
MEM_B_CNTRL1	MEM_40S	MEM_CTRL	MEM B ODT<1> 7 26 27
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B A<15..0> 7 25 26 27
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B BA<2..0> 7 25 26 27
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B RAS L 7 25 26 27
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B CAS L 7 25 26 27
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B WE L 7 25 26 27
MEM_B_DATA_0	MEM_45S	MEM_B_DATA_0	MEM B DQ<7..0> 7 25 26
MEM_B_DATA_1	MEM_45S	MEM_B_DATA_1	MEM B DQ<15..8> 7 25 26
MEM_B_DATA_2	MEM_45S	MEM_B_DATA_2	MEM B DQ<23..16> 7 25 26
MEM_B_DATA_3	MEM_45S	MEM_B_DATA_3	MEM B DQ<31..24> 7 25 26
MEM_B_DATA_4	MEM_45S	MEM_B_DATA_4	MEM B DQ<39..32> 7 25 26
MEM_B_DATA_5	MEM_45S	MEM_B_DATA_5	MEM B DQ<47..40> 7 25 26
MEM_B_DATA_6	MEM_45S	MEM_B_DATA_6	MEM B DQ<55..48> 7 25 26
MEM_B_DATA_7	MEM_45S	MEM_B_DATA_7	MEM B DQ<63..56> 7 25 26
MEM_B_DQS0	MEM_85D	MEM_B_DQS_0	MEM B DQS P<0> 7 25 26
MEM_B_DQS0	MEM_85D	MEM_B_DQS_0	MEM B DQS N<0> 7 25 26
MEM_B_DQS1	MEM_85D	MEM_B_DQS_1	MEM B DQS P<1> 7 25 26
MEM_B_DQS1	MEM_85D	MEM_B_DQS_1	MEM B DQS N<1> 7 25 26
MEM_B_DQS2	MEM_85D	MEM_B_DQS_2	MEM B DQS P<2> 7 25 26
MEM_B_DQS2	MEM_85D	MEM_B_DQS_2	MEM B DQS N<2> 7 25 26
MEM_B_DQS3	MEM_85D	MEM_B_DQS_3	MEM B DQS P<3> 7 25 26
MEM_B_DQS3	MEM_85D	MEM_B_DQS_3	MEM B DQS N<3> 7 25 26
MEM_B_DQS4	MEM_85D	MEM_B_DQS_4	MEM B DQS P<4> 7 25 26
MEM_B_DQS4	MEM_85D	MEM_B_DQS_4	MEM B DQS N<4> 7 25 26
MEM_B_DQS5	MEM_85D	MEM_B_DQS_5	MEM B DQS P<5> 7 25 26
MEM_B_DQS5	MEM_85D	MEM_B_DQS_5	MEM B DQS N<5> 7 25 26
MEM_B_DQS6	MEM_85D	MEM_B_DQS_6	MEM B DQS P<6> 7 25 26
MEM_B_DQS6	MEM_85D	MEM_B_DQS_6	MEM B DQS N<6> 7 25 26
MEM_B_DQS7	MEM_85D	MEM_B_DQS_7	MEM B DQS P<7> 7 25 26
MEM_B_DQS7	MEM_85D	MEM_B_DQS_7	MEM B DQS N<7> 7 25 26
		MEM_PWR	PP0V75_S3 MEM VREFD0 A 22 23 24 71 75
		MEM_PWR	PP0V75_S3 MEM VREFCA 22 23 24 25 26 71 75
		MEM_PWR	PP1V35_S3 MEM 22 23 24 25 26 46 70

SYNC MASTER=SIDLE J45		SYNC DATE=12/10/2012	
PAGE TITLE			
Memory Constraints			
 Apple Inc.		DRAWING NUMBER	SIZE
		<SCH_NUM>	D
		REVISION	
		<E4LABEL>	
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		<BRANCH>	
		PAGE	114 OF 118
		SHEET	78 OF 82



MIPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
MIPI_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
MIPI_20THER	*	=4X_DIELECTRIC	?
MIPI_2CLK	*	=6X_DIELECTRIC	?
MIPICKL_20THER	*	=7X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MIPI_DATA	*	*	MIPI_20THER
MIPI_DATA	CLK_MIPI	*	MIPI_2CLK
CLK_MIPI	*	*	MIPICKL_20THER

Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
S2_MEM_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
S2_MEM_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

Spacing Rule Sets

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
S2_DATA2SELF	*	=2X_DIELECTRIC	?
S2_DQS20WNDATA	*	=2X_DIELECTRIC	?
S2_CMD2CMD	*	=2X_DIELECTRIC	?
S2_CMD2CTRL	*	=2X_DIELECTRIC	?
S2_CTRL2CTRL	*	=2X_DIELECTRIC	?
S2_20THERMEM	*	=4X_DIELECTRIC	?
S2MEM_2PWR	*	=2X_DIELECTRIC	?
S2MEM_2GND	*	=2X_DIELECTRIC	?
S2MEM_20THER	*	=6X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
S2_DATA2SELF	TOP,BOTTOM	=4X_DIELECTRIC	?
S2_DQS20WNDATA	TOP,BOTTOM	=4X_DIELECTRIC	?
S2_CMD2CMD	TOP,BOTTOM	=4X_DIELECTRIC	?
S2_CMD2CTRL	TOP,BOTTOM	=4X_DIELECTRIC	?
S2_CTRL2CTRL	TOP,BOTTOM	=4X_DIELECTRIC	?
S2_20THERMEM	TOP,BOTTOM	=6X_DIELECTRIC	?
S2MEM_2PWR	TOP,BOTTOM	=4X_DIELECTRIC	?
S2MEM_2GND	TOP,BOTTOM	=4X_DIELECTRIC	?
S2MEM_20THER	TOP,BOTTOM	=10X_DIELECTRIC	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_DATA*	*	*	S2MEM_20THER
S2_MEM_DQS*	*	*	S2MEM_20THER
S2_MEM_CMD	*	*	S2MEM_20THER
S2_MEM_CTRL	*	*	S2MEM_20THER
S2_MEM_CLK	*	*	S2MEM_20THER
S2_MEM_DATA*	=SAME	*	S2_DATA2SELF
S2_MEM_CMD	S2_MEM_CMD	*	S2_CMD2CMD
S2_MEM_CMD	S2_MEM_CTRL	*	S2_CMD2CTRL
S2_MEM_CTRL	S2_MEM_CTRL	*	S2_CTRL2CTRL
S2_MEM_*	S2_MEM_*	*	S2_20THERMEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_DQS1	S2_MEM_DATA1	*	S2_DQS20WNDATA
S2_MEM_DQS0	S2_MEM_DATA0	*	S2_DQS20WNDATA

Memory to Power Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_PWR	S2_MEM_*	*	S2MEM_2PWR
S2_MEM_PWR	*	*	DEFAULT

Memory to GND Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	S2_MEM_*	*	S2MEM_2GND

Camera Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE		
	PHYSICAL	SPACING	
S2_MEM_CLK	S2_MEM_85D	S2_MEM_CLK	MEM_CAM_CLK_P
S2_MEM_CLK	S2_MEM_85D	S2_MEM_CLK	MEM_CAM_CLK_N
S2_MEM_CTRL	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_CKE
S2_MEM_CTRL	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_CS_L
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_ODT
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_CAS_L
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_RAS_L
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_WE_L
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_BA<0>
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_BA<1>
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_BA<2>
S2_MEM_DQS0	S2_MEM_85D	S2_MEM_DQS0	MEM_CAM_DQS_P<0>
S2_MEM_DQS0	S2_MEM_85D	S2_MEM_DQS0	MEM_CAM_DQS_N<0>
S2_MEM_DQS1	S2_MEM_85D	S2_MEM_DQS1	MEM_CAM_DQS_P<1>
S2_MEM_DQS1	S2_MEM_85D	S2_MEM_DQS1	MEM_CAM_DQS_N<1>
S2_MEM_DATA_0	S2_MEM_45S	S2_MEM_DATA0	MEM_CAM_DM<0>
S2_MEM_DATA_1	S2_MEM_45S	S2_MEM_DATA1	MEM_CAM_DM<1>
S2_MEM_A	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_A<14..0>
S2_MEM_DATA_0	S2_MEM_45S	S2_MEM_DATA0	MEM_CAM_DQ<7..0>
S2_MEM_DATA_1	S2_MEM_45S	S2_MEM_DATA1	MEM_CAM_DQ<15..8>
MIPI_DATA_S2	MIPI_85D	MIPI_DATA	MIPI_DATA_P
MIPI_DATA_S2	MIPI_85D	MIPI_DATA	MIPI_DATA_N
	MIPI_85D	MIPI_DATA	MIPI_DATA_CONN_P
	MIPI_85D	MIPI_DATA	MIPI_DATA_CONN_N
MIPI_CLK_S2	MIPI_85D	CLK_MIPI	MIPI_CLK_P
MIPI_CLK_S2	MIPI_85D	CLK_MIPI	MIPI_CLK_N
	MIPI_85D	CLK_MIPI	MIPI_CLK_CONN_P
	MIPI_85D	CLK_MIPI	MIPI_CLK_CONN_N
		S2_MEM_PWR	PP1V35_CAM
		S2_MEM_PWR	PP0V675_CAM_VREF
		S2_MEM_PWR	PP0V675_MEM_CAM_VREFCA
		S2_MEM_PWR	PP0V675_MEM_CAM_VREFDQ

SYNC_MASTER=SIDLE J45		SYNC_DATE=12/10/2012	
PAGE TITLE			
Camera Constraints			
 Apple Inc.		DRAWING_NUMBER	SIZE
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		REVISION	
		<E4LABEL>	
		BRANCH	<BRANCH>
		PAGE	116 OF 118
		SHEET	80 OF 82

8

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SMC SMBus Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
	SMBUS_SMC_2_S3_SCL	SMB_45S	SMB	SMBUS_SMC_2_S3_SCL	39 41 44 72
	SMBUS_SMC_2_S3_SDA	SMB_45S	SMB	SMBUS_SMC_2_S3_SDA	39 41 44 72
	SMBUS_SMC_1_S0_SCL	SMB_45S	SMB	SMBUS_SMC_1_S0_SCL	41 44 48
	SMBUS_SMC_1_S0_SDA	SMB_45S	SMB	SMBUS_SMC_1_S0_SDA	41 44 48
	SMBUS_SMC_0_S0_SCL	SMB_45S	SMB	SMBUS_SMC_0_S0_SCL	37 41 44 48 68 71 72
	SMBUS_SMC_0_S0_SDA	SMB_45S	SMB	SMBUS_SMC_0_S0_SDA	37 41 44 48 68 71 72
	SMBUS_SMC_5_SCL	SMB_45S	SMB	SMBUS_SMC_5_S3_SCL	41 44 56 57 72
	SMBUS_SMC_5_SDA	SMB_45S	SMB	SMBUS_SMC_5_S3_SDA	41 44 56 57 72
	SMBUS_SMC_3_SCL	SMB_45S	SMB	NC_SMBUS_SMC_3_SCL	41 43
	SMBUS_SMC_3_SDA	SMB_45S	SMB	NC_SMBUS_SMC_3_SDA	41 43

SMBus Charger Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
	CHGR_CS1	1T01_DIEFPAIR		CHGR_CS1_P	57
		1T01_DIEFPAIR		CHGR_CS1_N	57
	CHGR_CS0	1T01_DIEFPAIR		CHGR_CS0_P	57
		1T01_DIEFPAIR		CHGR_CS0_N	57

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SYNC MASTER=SIDLE J45		SYNC DATE=12/10/2012	
PAGE TITLE			
SMC Constraints			
 Apple Inc.	DRAWING NUMBER		SIZE
	<SCH_NUM>		D
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BRANCH			
<BRANCH>			
PAGE			
117		OF 118	
SHEET			
81		OF 82	

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SENSE_1T01_50S	"	=+1.1_DIFFPAIR	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
THERM_1T01_50S	"	=+1.1_DIFFPAIR	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
DIFFPAIR	"	=+1.1_DIFFPAIR			=1:1_DIFFPAIR	=1:1_DIFFPAIR	=1:1_DIFFPAIR
AUDIODIFF	"	=+1.1_DIFFPAIR	0.1 MM	0.1 MM	16 MM	0.1 MM	0.1 MM
THERM_45S_CPUVISMS1	"	=+1.1_DIFFPAIR	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	0.2 MM	0.2 MM
THERM_1T01_45S	"	=+1.1_DIFFPAIR	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
SENSE_1T01_45S	"	=+1.1_DIFFPAIR	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SENSE	*	=2X_DIELECTRIC	?
THERM	*	=2X_DIELECTRIC	?
AUDIO	*	=2X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_P2MM	*	0.20 MM	1000
PWR_P2MM	*	0.20 MM	1000

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_COMP	GND	*	GND_P2MM
CPU_VCCSENSE	GND	*	GND_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	GND	*	GND_P2Mm https://www.mouser.com/datasheet/2/678/930-107.pdf
GND	PCIE_*	*	GND_P2Mm https://www.mouser.com/datasheet/2/678/930-107.pdf
GND	SATA_*	*	GND_P2Mm https://www.mouser.com/datasheet/2/678/930-107.pdf
USB	GND	*	GND_P2Mm https://www.mouser.com/datasheet/2/678/930-107.pdf
CLK_PCIE	SB_POWER	*	PwR_P2Mm https://www.mouser.com/datasheet/2/678/930-107.pdf
SB_POWER	SATA_*	*	PwR_P2Mm https://www.mouser.com/datasheet/2/678/930-107.pdf
USB	SB_POWER	*	PwR_P2Mm https://www.mouser.com/datasheet/2/678/930-107.pdf

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_40S OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_72D OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_37S OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_85D OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
PCIE_85D OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	10 MM OVERRIDE	OVERRIDE	OVERRIDE
USB_85D	TOP			0.1 MM	500 MIL		
CPU_27P4S	BOTTOM			0.23 MM	100 MIL		
USB3_85D	TOP			0.1 MM	500 MIL		
USB3_85D	ISL10			0.075 MM			0.090 MM
DP_85D	ISL9			0.075 MM			0.090 MM
PCIE_85D	ISL10			0.075 MM			0.090 MM

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
1T01_DIFFPAIR	*	1:1_DIFFPAIR

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X425 Specific Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
E490	SENSE_DIEFFPATR	THERM_1T01_45S	THERM	ISNS CPUDDR P
E491	SENSE_DIEFFPAIR	THERM_1T01_45S	THERM	ISNS CPUDDR N
E492		THERM_1T01_45S	THERM	ISNS CPU DDR R P
E493		THERM_1T01_45S	THERM	ISNS CPU DDR R N
	SENSE_DIEFFPAIR	THERM_1T01_45S	THERM	CPUTHMSNS D2 P
	SENSE_DIEFFPAIR	THERM_1T01_45S	THERM	CPUTHMSNS D2 N
E496	SENSE_DIEFFPAIR	THERM_1T01_45S	THERM	ISNS LCD PANEL P
E497	SENSE_DIEFFPAIR	SENSE_1T01_45S	SENSE	ISNS LCD PANEL N
E498	SENSE_DIEFFPAIR	SENSE_1T01_45S	SENSE	DDR3THMSNS D1 P
	SENSE_DIEFFPAIR	THERM_1T01_45S	THERM	DDR3THMSNS D1 N
	SENSE_DIEFFPAIR	THERM_1T01_45S	THERM	FINTHMSNS D P
	SENSE_DIEFFPAIR	THERM_1T01_45S	THERM	FINTHMSNS D N
	SENSE_DIEFFPATR	SENSE_1T01_45S	SENSE	ISNS 1V35 MEM P
	SENSE_DIEFFPATR	SENSE_1T01_45S	SENSE	ISNS 1V35 MEM N
E491		SENSE_1T01_45S	SENSE	ISNS 1V35 MEM R P
E492		SENSE_1T01_45S	SENSE	ISNS 1V35 MEM R N
	SENSE_DIEFFPAIR	SENSE_1T01_45S	SENSE	ISNS AIRPORT P
	SENSE_DIEFFPAIR	SENSE_1T01_45S	SENSE	ISNS AIRPORT N
		SENSE_1T01_45S	SENSE	ISNS AIRPORT R P
		SENSE_1T01_45S	SENSE	ISNS AIRPORT R N
	SENSE_DIEFFPAIR	SENSE_1T01_45S	SENSE	ISNS LCDBKLT N
	SENSE_DIEFFPAIR	SENSE_1T01_45S	SENSE	ISNS LCDBKLT P
E493	SENSE_DIEFFPAIR	SENSE_1T01_45S	SENSE	ISNS LCD PANEL N
E494	SENSE_DIEFFPAIR	SENSE_1T01_45S	SENSE	ISNS LCD PANEL P
E497	SENSE_DIEFFPAIR	SENSE_1T01_45S	SENSE	ISNS PCH P
E498	SENSE_DIEFFPAIR	SENSE_1T01_45S	SENSE	ISNS PCH N
E499		SENSE_1T01_45S	SENSE	ISNS PCH R P
E500		SENSE_1T01_45S	SENSE	ISNS PCH R N
E501	SENSE_DIEFFPATR	SENSE_1T01_45S	SENSE	ISNS TPAD P
E502	SENSE_DIEFFPAIR	SENSE_1T01_45S	SENSE	ISNS TPAD N
	SENSE_DIEFFPATR	SENSE_1T01_45S	SENSE	ISNS HS OTHER5V P
	SENSE_DIEFFPATR	SENSE_1T01_45S	SENSE	ISNS HS OTHER5V N
E504	SENSE_DIEFFPATR	SENSE_1T01_45S	SENSE	ISNS HS OTHER3V3 P
E505	SENSE_DIEFFPATR	SENSE_1T01_45S	SENSE	ISNS HS OTHER3V3 N
	SENSE_DIEFFPATR	SENSE_1T01_45S	SENSE	ISNS HS COMPUTING P
	SENSE_DIEFFPATR	SENSE_1T01_45S	SENSE	ISNS HS COMPUTING N
		SENSE_1T01_45S	SENSE	CPUVR ISNS P
		SENSE_1T01_45S	SENSE	CPUVR ISNS N
E508	SENSE_DIEFFPAIR	THERM_1T01_45S	THERM	P1V05 GPU PEX IOVDD SNS P
E509	SENSE_DIEFFPAIR	THERM_1T01_45S	THERM	P1V05 GPU PEX IOVDD SNS N
DIFFERENTIAL_PAIR				
E510	SENSE_DIEFFPATR	THERM_45S_CPUVRISNS1	THERM	CPUVR ISNS1 P
E511	SENSE_DIEFFPAIR	THERM_45S_CPUVRISNS1	THERM	CPUVR ISNS1 N
E512	SENSE_DIEFFPAIR	THERM_45S_CPUVRISNS1	THERM	CPUVR ISNS2 P
E513	SENSE_DIEFFPAIR	THERM_45S_CPUVRISNS1	THERM	CPUVR ISNS2 N
E514	SENSE_DIEFFPAIR	THERM_45S_CPUVRISNS1	THERM	CPUVR ISNS3 P
E515	SENSE_DIEFFPAIR	THERM_45S_CPUVRISNS1	THERM	CPUVR ISNS3 N
E516	SENSE_DIEFFPATR	THERM_1T01_45S	THERM	CPUVR ISUM R P
E517	SENSE_DIEFFPAIR	THERM_1T01_45S	THERM	CPUVR ISUM R N
E490	SENSE_DIEFFPATR	THERM_1T01_45S	THERM	GFXIMVP ISNS1 P
E491	SENSE_DIEFFPAIR	THERM_1T01_45S	THERM	GFXIMVP ISNS1 N
E500	SENSE_DIEFFPATR	THERM_1T01_45S	THERM	GFXIMVP ISNS1 P
E501	SENSE_DIEFFPAIR	THERM_1T01_45S	THERM	GFXIMVP ISNS1 N
E490	AUDIO_DIEFFPATR	AUDIODIFF	AUDIO	ISNS TBT N
E491	AUDIO_DIEFFPAIR	AUDIODIFF	AUDIO	ISNS TBT P
E492		AUDIODIFF	AUDIO	ISNS TBT R N
E493		AUDIODIFF	AUDIO	ISNS TBT R P
E496	SENSE_DIEFFPAIR	THERM_1T01_45S	THERM	ISNS SSD P
E497		THERM_1T01_45S	THERM	ISNS SSD N
E498	SENSE_DIEFFPAIR	THERM_1T01_45S	THERM	ISNS SSD R P
E499		THERM_1T01_45S	THERM	ISNS SSD R N
E500	SENSE_DIEFFPAIR	THERM_1T01_45S	THERM	P1V05S0 CS P
E501		THERM_1T01_45S	THERM	P1V05S0 CS N
DIFFERENTIAL_PAIR				
E490	SENSE_DIEFFPATR	THERM_1T01_45S	THERM	P1V05S0 SENSE P1V05S0 SENSE P
E491		THERM_1T01_45S	THERM	P1V05S0 SENSE P1V05S0 SENSE N
E492	SENSE_DIEFFPAIR	THERM_1T01_45S	THERM	TBT THERMDP
E493		THERM_1T01_45S	THERM	TBT THERMDN
E490	AUDIO_DIEFFPAIR	AUDIODIFF	AUDIO	CHGR CSI R P
E491		AUDIODIFF	AUDIO	CHGR CSI R N
E492	AUDIO_DIEFFPATR	AUDIODIFF	AUDIO	CHGR CS0 R P
E493		AUDIODIFF	AUDIO	CHGR CS0 R N
E494	SENSE_DIEFFPATR	SENSE_1T01_45S	SENSE	ISNS S2 P
E495	SENSE_DIEFFPAIR	SENSE_1T01_45S	SENSE	ISNS S2 N
E496	SENSE_DIEFFPAIR	SENSE_1T01_45S	SENSE	ISNS S2 R P
E497	SENSE_DIEFFPAIR	SENSE_1T01_45S	SENSE	ISNS S2 R N

X425 Specific Net Properties

ELECTRICAL_CONSTRAINT_SET		NET TYPE	
	PHYSICAL	SPACING	
	AUDIODIFF	AUDIO	AUD_SPKRAMP_RSUBIN_P
	AUDIODIFF	AUDIO	AUD_SPKRAMP_RSUBIN_N
	AUDIODIFF	AUDIO	AUD_SPKRAMP_LSUBIN_P
	AUDIODIFF	AUDIO	AUD_SPKRAMP_LSUBIN_N
	AUDIODIFF	AUDIO	RSUBIN_P
	AUDIODIFF	AUDIO	RSUBIN_N
	AUDIODIFF	AUDIO	LSUBIN_P
	AUDIODIFF	AUDIO	LSUBIN_N
AUDIO_DIEFPAIR	AUDIODIFF	AUDIO	AUD_L02_R_P
AUDIO_DIEFPAIR	AUDIODIFF	AUDIO	AUD_L02_R_N
AUDIO_DIEFPAIR	AUDIODIFF	AUDIO	AUD_L02_L_P
AUDIO_DIEFPAIR	AUDIODIFF	AUDIO	AUD_L02_L_N
	AUDIODIFF	AUDIO	AUD_SPKRAMP_RIN_P
	AUDIODIFF	AUDIO	AUD_SPKRAMP_RIN_N
	AUDIODIFF	AUDIO	AUD_SPKRAMP_LIN_P
	AUDIODIFF	AUDIO	AUD_SPKRAMP_LIN_N
	AUDIODIFF	AUDIO	SPKRAMP_RIN_P
	AUDIODIFF	AUDIO	SPKRAMP_RIN_N
	AUDIODIFF	AUDIO	SPKRAMP_LIN_P
	AUDIODIFF	AUDIO	SPKRAMP_LIN_N
AUDIO_DIEFPAIR	DIEFPAIR	AUDIO	SPKRCONN_SL_OUT_P
AUDIO_DIEFPAIR	DIEFPAIR	AUDIO	SPKRCONN_SL_OUT_N
AUDIO_DIEFPAIR	DIEFPAIR	AUDIO	SPKRCONN_SR_OUT_P
AUDIO_DIEFPAIR	DIEFPAIR	AUDIO	SPKRCONN_SR_OUT_N
AUDIO_DIEFPAIR	DIEFPAIR	AUDIO	SPKRCONN_L_OUT_P
AUDIO_DIEFPAIR	DIEFPAIR	AUDIO	SPKRCONN_L_OUT_N
AUDIO_DIEFPAIR	DIEFPAIR	AUDIO	SPKRCONN_R_OUT_P
AUDIO_DIEFPAIR	DIEFPAIR	AUDIO	SPKRCONN_R_OUT_N
AUDIO_DIEFPAIR	DIEFPAIR	AUDIO	AUD_MIC_IN1_R_P
AUDIO_DIEFPAIR	DIEFPAIR	AUDIO	AUD_MIC_IN1_R_N
AUDIO_DIEFPAIR	DIEFPAIR	AUDIO	CODEC_HS_MIC_P
AUDIO_DIEFPAIR	DIEFPAIR	AUDIO	CODEC_HS_MIC_N
AUDIO_DIEFPAIR	DIEFPAIR	AUDIO	AUD_MIC_IN1_L_P
	DIEFPAIR	AUDIO	AUD_MIC_IN1_L_N
	DIEFPAIR	AUDIO	AUD_HS_MIC_P
	DIEFPAIR	AUDIO	AUD_HS_MIC_N
	DIEFPAIR	AUDIO	HS_MIC_P
	DIEFPAIR	AUDIO	HS_MIC_N
	DIEFPAIR	AUDIO	AUD_CONN_HS_MIC_P
	DIEFPAIR	AUDIO	AUD_CONN_HS_MIC_N
AUDIO_DIEFPAIR	AUDIODIFF	AUDIO	AUD_L03_R_P
AUDIO_DIEFPAIR	AUDIODIFF	AUDIO	AUD_L03_R_N
AUDIO_DIEFPAIR	AUDIODIFF	AUDIO	AUD_L03_L_P
AUDIO_DIEFPAIR	AUDIODIFF	AUDIO	AUD_L03_L_N
		SB_POWER	PP3V3_S5
		SB_POWER	PP3V3_S0
		SB_POWER	PP1V35_S3RS0_CPUDDR
		GND	GND

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		PAGE	
		118 OF 118	
		SHEET	
		82 OF 82	